Practical Quantum Simulation

A Scaffolded Handbook for Scientific Computing and Emerging Fault-Tolerant Hardware

Onri Jay Benally

July 21, 2026

Purpose and Scope

Quantum simulation is often introduced through qubits, gates, circuit diagrams, and asymptotic speedups. Those ideas define the logical calculation. A scientific result also passes through a physical processor, a classical control stack, finite sampling, numerical reference methods, calibration, and an uncertainty analysis. This handbook explains how those pieces fit together.

How can a quantum result continue to be credible after a full exact classical calculation exceeds available resources? The answer starts with a conclusion narrow enough to test. An energy density, a dynamical correlation function, a band energy, a benchmark score, or a success probability can each serve as the object of study. The record must also identify the circuit, processor, calibration interval, estimator, reference calculation, and acceptance criterion.

The following are three terms to keep in mind.

- A **model** is the mathematical object under calculation, such as a Hamiltonian, quantum channel, lattice equation, or compiled circuit.
- **Reference calculation** is a high-confidence analytical or numerical result inside that model. Its credibility comes from convergence tests, residuals, independent implementations, or exact limits.
- **Hardware estimate** is inferred from experimental data produced by a calibrated device and its measurement pipeline.

Verification connects these objects without merging them. It asks whether a bounded conclusion survives the known statistical, numerical, and physical sources of error. The result is conditional on the stated model and protocol.

Readers new to quantum computing can follow the chapters in order. Meanwhile readers focused on experiments can move from the primer to sampling, mitigation, benchmarking, and hardware validation. Those focused on scientific computing can use the dense-simulation, tensor-network, numerical-solver, accelerator-workflow, and resource-accounting chapters as a connected path.

The book concentrates on gate-based processors and sampled observables. Analog simulators, annealers, photonic systems, and other platforms enter through platform-specific benchmark contracts. The common requirement is the same across platforms: the reported quantity, physical implementation, uncertainty, and validation evidence must refer to one another explicitly.

Contents

List of Figures

List of Tables

Chapter 1

Scaffolded Primer on Quantum Systems

1.1 Core Terminology

Prerequisite

A previous quantum computing course is unnecessary. The reader needs elementary complex numbers, vectors, matrices, probabilities, and the idea that repeated measurements produce a statistical sample.

1.1.1 States, amplitudes, and probabilities

A classical bit records one of two symbols. A **qubit** is a two-level quantum system described relative to two basis states, conventionally written $|0\rangle$ and $|1\rangle$. Its pure state has the form

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

The complex numbers α and β are **probability amplitudes**. Their squared magnitudes give the computational-basis measurement probabilities,

$$P(0) = |\alpha|^2, \qquad P(1) = |\beta|^2.$$

Their relative phase affects interference in later operations. A single measurement still returns one classical label, 0 or 1.

For n qubits, the computational basis contains 2^n bitstrings. A pure state can be written

$$|\psi\rangle = \sum_{x\in\{0,1\}^n} \alpha_x\,|x\rangle\,, \qquad \sum_x |\alpha_x|^2 = 1.$$

This expression explains the dense state-vector memory wall developed in chapter 5. The number of amplitudes doubles with every added qubit.

1.1.2 Gates, circuits, measurements, and shots

An ideal **quantum gate** is represented by a unitary matrix U. Unitarity,

$$U^\dagger U = I,$$

preserves the norm of every state. An ordered collection of gates forms a **quantum circuit**. Compilation maps the logical circuit into the native operations, connectivity, timing, and control instructions supported by a processor.

A measurement converts quantum information into classical data. Measuring every qubit in the computational basis produces a **bitstring**, such as 00101. One circuit execution followed by measurement is a **shot**. Repeating the same setting produces a sample of bitstrings from which probabilities and expectation values are estimated.

An **observable** O is a Hermitian operator associated with a measurable quantity. For a state ρ, its model expectation is

$$\mu = \langle O \rangle = \mathrm{Tr}(\rho O).$$

The model assigns the value μ. A finite data set supplies an estimator $\widehat{\mu}$ with statistical uncertainty.

1.1.3 Noise, verification, mitigation, and correction

Quantum noise is a collective name for unwanted dynamics and measurement distortion. Useful categories include

- energy relaxation and dephasing;
- coherent overrotation, detuning, and pulse distortion;
- leakage outside the computational subspace;
- crosstalk and correlated errors;
- residual excitation and thermal population;
- measurement assignment error and classifier drift.

Each category leaves a different signature in data. One scalar error rate cannot describe all of them.

A **verification protocol** tests a stated conclusion using analytical checks, numerical references, calibrations, benchmarks, and uncertainty analysis. **Quantum error mitigation** (QEM) changes the experiment or estimator to reduce noise-induced bias without providing the encoded protection of fault-tolerant quantum error correction [1]. QEM can improve a central value and increase variance, circuit count, calibration demand, or sensitivity to model mismatch. **Quantum error correction** encodes logical information and uses repeated syndrome measurements to detect and correct physical faults.

Table 1.1: Symbols used throughout the primer.

Symbol	Mathematical meaning	Experimental interpretation
$\lvert\psi\rangle$	Pure state vector	Ideal state prepared by a circuit or pulse sequence
ρ	Density operator	State model that can include preparation uncertainty and decoherence
U	Unitary operation	Ideal gate or coherent time evolution
$\mathcal{E}$	Quantum channel	State transformation that can include noise
O	Observable	Quantity assigned to measurement outcomes
μ	Model expectation $\mathrm{Tr}(\rho O)$	Value predicted inside the stated model
$\widehat{\mu}$	Estimator computed from data	Reported finite-shot value
N	Number of shots	Recorded circuit executions used by the estimator

1.2 Supporting Theory and Physics

Prerequisite

Before reading this section, ensure that matrix multiplication, complex conjugation, and probability-weighted averages are familiar.

1.2.1 Pure states, mixed states, and reduced descriptions

A state vector describes one coherent preparation. A density operator is used when the preparation varies across trials or when part of a larger quantum system is excluded from the description. For one qubit,

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

gives

$$\rho_\psi = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

The diagonal entries give computational-basis measurement probabilities. The off-diagonal entries carry the phase coherence that supports interference.

An ensemble containing preparation $|\psi_j\rangle$ with probability p_j is represented by

$$\rho = \sum_j p_j\,|\psi_j\rangle\langle\psi_j|\,, \qquad p_j \geq 0, \qquad \sum_j p_j = 1.$$

A physical density operator is Hermitian, positive semidefinite, and normalized by $\mathrm{Tr}(\rho) = 1$. A pure state satisfies

$$\mathrm{Tr}(\rho^2) = 1,$$

whereas a mixed state satisfies

$$\mathrm{Tr}(\rho^2) < 1.$$

Purity measures how concentrated the state is in Hilbert space and leaves the physical source of mixing unresolved.

For a composite state ρ_{AB}, the description available to subsystem A is

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}).$$

This partial trace removes the degrees of freedom assigned to B. It appears whenever an environment, spectator qubit, unmeasured mode, or unobserved subsystem is excluded from the working model.

1.2.2 Unitary evolution and quantum channels

Closed-system evolution follows

$$\rho(t) = U(t)\rho(0)U^\dagger(t).$$

A general physical state transformation is a completely positive trace-preserving channel. One useful representation is

$$\mathcal{E}(\rho) = \sum_r K_r \rho K_r^\dagger, \qquad \sum_r K_r^\dagger K_r = I,$$

where the Kraus operators K_r encode the modeled transformation. Different Kraus representations can describe the same channel, so the channel is the physical object and the individual operators are a mathematical representation.

Many introductory device models assume weak coupling to a memoryless environment. Under Markovian and time-local assumptions, the density operator obeys the Gorini–Kossakowski–Sudarshan–Lindblad equation [2], [3], [4]

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right).$$

The Hamiltonian H generates coherent motion. The jump operators L_k describe modeled dissipative processes such as relaxation and dephasing. The equation gives a controlled model inside its assumptions. Memory effects, correlated noise, leakage, and drift can require a larger state space or a non-Markovian description. Verification therefore tests the numerical solution and the adequacy of the physical model.

1.3 Step-by-Step Mathematical Derivations

Prerequisite

Before reading this section, ensure that expectation, variance, independent sampling, and a Taylor expansion are familiar.

1.3.1 Finite-shot estimation for a two-outcome observable

Consider a Pauli observable, or any dichotomic observable satisfying $O^2 = I$, measured on N independently prepared states. Each outcome obeys $y_j \in \{-1, +1\}$ and has mean

$$\mu = \mathrm{Tr}(\rho O).$$

The sample mean is

$$\widehat{\mu} = \frac{1}{N} \sum_{j=1}^{N} y_j.$$

Linearity of expectation gives

$$\mathbb{E}[\widehat{\mu}] = \frac{1}{N} \sum_{j=1}^{N} \mathbb{E}[y_j] = \mu,$$

so the estimator is unbiased under the sampling model.

Since $y_j^2 = 1$, one outcome has variance

$$\mathrm{Var}(y) = \mathbb{E}[y^2] - \mathbb{E}[y]^2 = 1 - \mu^2.$$

Independence removes the covariance terms,

$$\mathrm{Var}(\widehat{\mu}) = \frac{1}{N^2} \sum_{j=1}^{N} \mathrm{Var}(y_j) = \frac{1-\mu^2}{N}.$$

The standard error scales as $N^{-1/2}$. Doubling the number of correct decimal places therefore requires far more than doubling the shot count.

Hoeffding's inequality supplies a distribution-free bound for outcomes in $[-1, 1]$,

$$\Pr(|\widehat{\mu} - \mu| \geq \delta) \leq 2 \exp\left(-\frac{N\delta^2}{2} \right).$$

Requiring failure probability at most α gives

$$N \geq \frac{2}{\delta^2} \ln\left(\frac{2}{\alpha}\right).$$

The inequality is conservative because it uses the outcome range and omits the actual value of μ. It is useful before the experiment, while the variance may still be unknown.

Temporal correlations and drift reduce the effective number of independent samples. Interleaved calibrations, randomized execution order, time-series diagnostics, and block-bootstrap intervals can expose this failure of the independent-shot model.

1.3.2 Bias, variance, and zero-noise extrapolation

An estimator is judged by its mean squared error,

$$\mathrm{MSE}(\widehat{\mu}) = \mathbb{E}[(\widehat{\mu} - \mu)^2] = \mathrm{Bias}(\widehat{\mu})^2 + \mathrm{Var}(\widehat{\mu}).$$

A lower bias helps only if the accompanying variance and model error remain controlled.

Zero-noise extrapolation illustrates the trade. Let ϵ denote the native noise strength and let $\lambda \geq 1$ scale that noise. In a smooth weak-noise regime,

$$\mu(\lambda\epsilon) = \mu(0) + a_1\lambda\epsilon + a_2(\lambda\epsilon)^2 + \mathcal{O}(\epsilon^3).$$

Two estimates at λ_1 and λ_2 can cancel the first-order term. The weights satisfy

$$w_1 + w_2 = 1, \qquad w_1\lambda_1 + w_2\lambda_2 = 0,$$

which gives

$$w_1 = \frac{\lambda_2}{\lambda_2 - \lambda_1}, \qquad w_2 = -\frac{\lambda_1}{\lambda_2 - \lambda_1}.$$

The extrapolated estimator is

$$\widehat{\mu}_{\mathrm{ZNE}} = w_1\widehat{\mu}(\lambda_1\epsilon) + w_2\widehat{\mu}(\lambda_2\epsilon).$$

For $\lambda_1 = 1$ and $\lambda_2 = 2$, the weights are 2 and -1. If both estimates are independent and have variance σ^2, then

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{ZNE}}) = (2^2 + (-1)^2)\sigma^2 = 5\sigma^2.$$

The extrapolation cancels the leading modeled bias and amplifies statistical noise at the same time [5], [6]. This simple calculation explains why every mitigation result needs raw data, uncertainty propagation, and a resource ledger.

Key takeaway

The primer leaves four ideas that support the simulation chapters.

- A pure state assigns one complex amplitude to each basis state in the chosen basis.
- The number of basis states grows as 2^n for n qubits.
- A density operator describes mixtures, reduced subsystems, and modeled open-system dynamics.
- An experiment returns finite data, so every reported observable carries statistical and systematic uncertainty.

The next question is computational. A dense simulator stores every amplitude or density-matrix entry. A structured simulator searches for smaller factors that reproduce the required state or

observable to a declared tolerance. Chapter 5 counts the cost of full storage. Chapter 6 explains how limited correlation between groups of qubits can support a compact factorization.
Keep one question in view as you read the intervening chapters:

> *Does the calculation require the full quantum object, or can the requested result be obtained from a smaller structured representation?*

Check your understanding

1. In one sentence, distinguish a model state ρ from an estimator $\widehat{\mu}$ obtained from shots.
2. A six-qubit pure state has $2^6 = 64$ basis amplitudes. State the representation question that must be asked before assuming that all 64 amplitudes need independent storage.

Practical rule of thumb

Keep the physical system, its mathematical representation, and the measured estimator separate. Solver choice concerns the representation. Experimental credibility concerns the estimator and its connection to the physical system. Before trusting a sampled expectation value, record its outcome range, shot count, confidence construction, and possible time correlation.

Chapter 2

The Practical Question

How the argument develops

The chapter turns the abstract idea of trust into a testable workflow. It first defines a bounded proposition, then assigns different diagnostics to statistical, numerical, experimental, and model errors. A controlled analogy with direct numerical simulation prepares the comparison developed in chapter 4.

2.1 Trust and Verification

Prerequisite

Before reading this chapter, ensure that you can distinguish a mathematical model from data produced by a physical instrument. Review estimator, bias, variance, and confidence interval in chapter 1.

How can a quantum result continue to be credible after a full exact classical calculation exceeds available resources? A defensible answer begins with a **bounded proposition**.

Suitable propositions include an energy density, a correlation function, a sampled distribution, a dynamical structure factor, or a benchmark score. The record should identify the compiled circuit, processor region, calibration interval, shot allocation, data processing, and accuracy target. A sentence such as "the processor reproduced the target observable within ± 0.02 over the registered circuit family" carries more scientific content than a general statement about processor accuracy.

Evidence accumulates across several scales.

- Exact simulators test small instances, shallow prefixes, and isolated patches.
- Analytical identities test normalization, symmetries, conserved quantities, and limiting cases.
- Tensor networks, circuit cutting, and observable propagation evaluate selected quantities where structure permits.
- Device diagnostics measure preparation error, gate behavior, leakage, crosstalk, readout assignment, and drift.
- Randomized and volumetric benchmarks probe compiled behavior that component metrics can miss.
- Application-specific checks compare the final result with domain residuals, reference data, or physical invariants.

Agreement across these tests narrows the plausible explanations for the data. The result remains empirical, with strength set by the diversity and sensitivity of the tests.

A useful uncertainty budget separates four contributions.

- **Statistical uncertainty** comes from finite and possibly correlated samples. Confidence intervals, bootstrap methods, and effective sample sizes address it.
- **Numerical uncertainty** comes from floating-point arithmetic, truncation, solver tolerances, and stochastic classical algorithms. Grid refinement, basis enlargement, residuals, and independent implementations address it.
- **Experimental systematic error** comes from calibration drift, leakage, crosstalk, pulse distortion, and measurement assignment. Sentinel circuits, repeated calibrations, and randomized acquisition order address it.
- **Model discrepancy** comes from differences between the mathematical model and the physical processor. Competing models, parameter sensitivity, and comparison with laboratory observables address it.

Increasing the shot count reduces ideal independent-sampling error. It leaves a drifting calibration or an incomplete Hamiltonian unchanged. A credible workflow therefore assigns a diagnostic to each uncertainty source before combining them into one conclusion.

2.2 Reasoning Analogies

Direct numerical simulation in computational fluid mechanics provides a controlled analogy. DNS resolves the dynamically relevant fluid scales inside a stated continuum model. Dense quantum simulation resolves every computational-basis amplitude or density-matrix element inside a stated circuit or open-system model. Both methods can provide high-confidence numerical references and both encounter severe resource limits.

The analogy concerns model-resolved computation. The governing equations and scaling laws are distinct. DNS convergence establishes fidelity to a discretized Navier–Stokes model. State-vector convergence establishes fidelity to a circuit or Hamiltonian model. Experimental validation then asks whether the model and its parameters represent the physical system closely enough for the stated proposition.

Check your understanding

1. Rewrite the statement “the processor accurately simulated the system” as a bounded proposition. Specify one observable, one circuit or input family, one numerical tolerance, and one confidence requirement.
2. Assign one diagnostic to each uncertainty category: statistical uncertainty, numerical uncertainty, experimental systematic error, and model discrepancy. Explain what evidence the diagnostic contributes.
3. A second run doubles the shot count. Its within-session interval narrows, and the session means continue to move over time. Identify the limiting uncertainty source and the next measurement or analysis step.

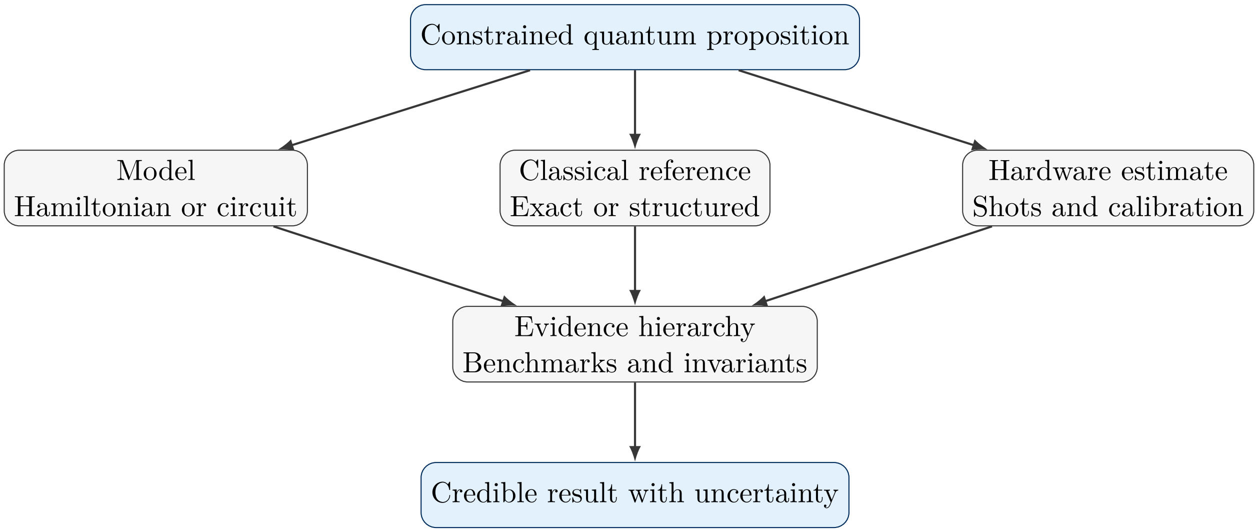

Figure 2.1: A bounded quantum conclusion gains credibility from convergent calculations, hardware estimates, benchmarks, and an explicit uncertainty budget.

Practical rule of thumb

State the narrowest conclusion that answers the scientific question. A local energy density with a declared uncertainty is easier to verify than a statement about an entire many-qubit state or output distribution.

Chapter 3

Models, Analytical Reasoning, and Numerical Reasoning

This chapter combines analytical constraints with numerical values and shows how both forms of evidence can be aligned with the same model, convention, and observable.

3.1 The Distinction Needed for Verification

Prerequisite

Before reading this chapter, ensure that matrix multiplication, a Taylor expansion, and the idea of a finite numerical representation are familiar.

Most verification problems need two anchors. Analytical reasoning supplies relations that every valid result must satisfy, including normalization, conservation laws, bounds, symmetries, and solvable limits. Numerical reasoning supplies values for the specific Hamiltonian, circuit, grid, boundary condition, or parameter set under study.

The two forms of evidence become comparable after they use the same mathematical object. Before interpreting a disagreement, align

- the Hamiltonian and boundary conditions;
- the initial state and basis ordering;
- the gate convention, phase convention, and endianness;
- the noise channel and parameter values;
- the measured observable and postprocessing rule.

Convention mismatches can create large apparent errors in otherwise consistent calculations.

3.2 Analytical and Numerical Anchors

An ideal single-qubit rotation about the x axis is

$$R_x(\theta) = \exp\left(-i\frac{\theta}{2}X\right) = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) X.$$

Applied to $|0\rangle$, it gives

$$R_x(\theta)\,|0\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle - i\sin\left(\frac{\theta}{2}\right)|1\rangle\,.$$

The probability of measuring 1 is therefore

$$P(1) = |\langle 1|\,R_x(\theta)\,|0\rangle|^2 = \sin^2\left(\frac{\theta}{2}\right).$$

This exact result supplies a calibration curve. A pulse experiment can test whether the measured estimator follows the curve after finite-shot uncertainty, readout assignment, pulse distortion, leakage, and drift are included.

Table 3.1: Analytical anchor points for an ideal $R_x(\theta)$ experiment.

Angle	Ideal $P(1)$	Primary use	Errors that can appear
0	0	Preparation and readout floor	Residual excitation, assignment error, spurious rotation
$\pi/2$	$1/2$	Amplitude calibration near maximum slope	Angle error, shot variance, pulse distortion
π	1	Population transfer	Underrotation, leakage, relaxation, assignment error
Ramsey phase scan	Sinusoidal fringe	Frame and detuning calibration	Frequency offset, phase error, dephasing

A two-qubit pure state introduces four complex amplitudes,

$$|\psi\rangle = \alpha_{00}\,|00\rangle + \alpha_{01}\,|01\rangle + \alpha_{10}\,|10\rangle + \alpha_{11}\,|11\rangle\,, \qquad \sum_x |\alpha_x|^2 = 1.$$

A state-vector simulator stores these amplitudes, applies gate updates, and calculates model probabilities. A processor implements a physical control sequence and returns empirical outcomes. Their comparison gains meaning after basis ordering, endianness, gate phases, measurement labels, and postprocessing have been aligned.

The calibration curve illustrates a reusable pattern. Start at points with known answers. Add tests whose sensitivities differ. Use the combined response to separate amplitude, phase, leakage, relaxation, and readout effects. The same logic scales to many-qubit verification through symmetries, conserved quantities, solvable limits, and classically tractable circuit fragments.

Check your understanding

1. Evaluate
$$P(1) = \sin^2\left(\frac{\theta}{2}\right)$$
at $\theta = 0$, $\pi/2$, and π. State which point is most sensitive to a small rotation-angle error and which points expose preparation or assignment floors most directly.

2. A simulator orders two-qubit basis states as $|q_1 q_0\rangle$, and the analysis code interprets them as $|q_0 q_1\rangle$. Describe the apparent discrepancy this can create and the convention check that resolves it.

3. For a many-body calculation with a conserved particle number, identify an analytical anchor, a numerical anchor, and a hardware diagnostic. Explain how the three tests respond to different failure modes.

Practical rule of thumb

Test analytical and numerical methods at zero coupling, one qubit, a symmetry point, and a known limit before using them in an unfamiliar regime. These checks expose convention and implementation errors at low cost.

Chapter 4

Direct Numerical Simulation as a Controlled Analogy

This chapter uses direct numerical simulation as a controlled analogy for separating convergence inside a mathematical model from validation against an experiment.

4.1 Meaning in Fluid Mechanics

Prerequisite

Before reading this chapter, ensure that a partial differential equation, spatial grid, timestep, and convergence test are familiar.

For an incompressible Newtonian fluid, the continuum model can be written

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\frac{1}{\rho_f}\nabla p + \nu \nabla^2 \boldsymbol{u} + \boldsymbol{f}, \qquad \nabla \cdot \boldsymbol{u} = 0.$$

Here $\boldsymbol{u}$ is the velocity field, p is pressure, ρ_f is mass density, ν is kinematic viscosity, and $\boldsymbol{f}$ is an applied body force. Direct numerical simulation resolves the dynamically relevant fluid scales on a spatial and temporal discretization. It avoids a turbulence closure at the resolved continuum level and still relies on constitutive assumptions, boundary data, grid resolution, time integration, and finite precision [7], [8].

A DNS result becomes a numerical reference after grid refinement, timestep refinement, conservation checks, and comparison with known limits. Those tests establish convergence to the stated continuum model.

4.2 Quantum Simulation Inside a Stated Model

A closed quantum system obeys

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle .$$

A dense simulator represents every basis amplitude of $|\psi\rangle$ and applies the specified Hamiltonian or circuit operations. An open-system calculation represents ρ and evolves a master equation or channel model.

Dense quantum simulation can remove hardware noise and finite-shot uncertainty from the circuit calculation. Its result is still conditional on the Hamiltonian, initial state, gate definitions, channel model, numerical precision, and implementation. Convergence establishes fidelity to those equations.

4.3 Limits of the Analogy

DNS and quantum state simulation have different physical variables and different scaling laws. Their shared lesson is epistemic. Numerical fidelity inside a model and fidelity of that model to an experiment are separate questions.

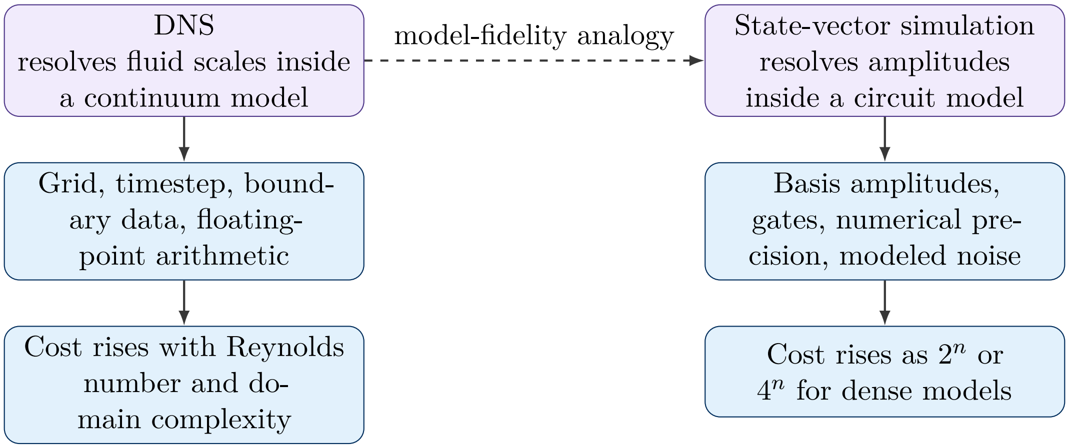

Figure 4.1: DNS and dense quantum simulation resolve their stated models directly. Their physical variables and computational scaling laws are different.

A dense pure-state calculation stores 2^n amplitudes. A dense density operator stores 4^n complex entries. Structured methods can avoid these arrays for favorable problem classes. Stabilizer simulation exploits Clifford algebra, tensor networks exploit factorization and limited entanglement, and hybrid Schrödinger–Feynman methods trade memory for repeated contractions. These methods move the accessible boundary without creating a universal classical shortcut.

Check your understanding

1. State what grid and timestep convergence establish for a DNS calculation. Then state the corresponding conclusion for a converged state-vector or density-matrix calculation.
2. Name two forms of evidence needed to connect a converged quantum model calculation to a physical experiment. Explain which part of the model each form of evidence tests.
3. Adding one qubit doubles the number of state-vector amplitudes and multiplies the number of density-matrix entries by four. Use this scaling to predict the storage change from n to $n + 3$ qubits for both representations.

Key takeaway

A converged simulation answers the equations that were supplied. Experimental validation tests whether those equations and parameters describe the physical system closely enough for the reported quantity.

Practical rule of thumb

Use the DNS analogy to separate two checks. First establish numerical convergence inside the model. Then test the model against calibration data, laboratory observables, or an independent experimental reference.

Interlude: From Quantum States to Simulation Strategies

The primer introduced state vectors, density operators, and observables. The next two chapters ask how a classical computer can represent those objects. The transition is easier after separating two questions:

1. How large is the full array?
2. Can smaller factors generate the part of the array needed by the scientific task?

A dense state vector answers the first question by listing every amplitude. A tensor network addresses the second question by testing whether correlations across chosen partitions can be carried by smaller internal indices. The symbol χ below denotes the bond dimension, which measures the correlation capacity assigned to one cut.

One idea that prevents a common misunderstanding

The number of nonzero amplitudes does not determine tensor-network compressibility. Factorization across bipartitions determines the required bond dimension.

State	Dense-basis view	Structure across a cut
$\lvert 0\rangle^{\otimes n}$	One nonzero amplitude	Product state; one paired pattern; exact $\chi = 1$
$\lvert +\rangle^{\otimes n}$	All 2^n amplitudes are nonzero	Product state; one paired pattern; exact $\chi = 1$
$\left(\lvert 0\rangle^{\otimes n} + \lvert 1\rangle^{\otimes n}\right)/\sqrt{2}$	Two nonzero amplitudes	Two paired patterns across every internal cut; exact $\chi = 2$
Generic strongly entangled state	Many amplitudes	Many independent paired patterns; χ can grow rapidly

The table supplies the conceptual bridge. Dense storage is controlled by the total basis size. MPS storage is controlled by the amount of left–right correlation that must cross each cut in the chosen qubit ordering. Chapter 5 develops the first cost. Chapter 6 develops the second.

Key takeaway

Amplitude count sets the size of a dense state vector. Cross-cut rank sets the exact bond dimension of an MPS. A state can have many nonzero amplitudes and still possess a compact product structure.

Chapter 5

Dense Simulation and the Memory Wall

How the argument develops

The interlude separated full storage from structured storage. This chapter develops the full-storage side first. It counts amplitudes, converts the count into bytes, and then identifies the additional features that determine whether a structured solver may be more useful.

5.1 State Vectors and Density Matrices

Prerequisite

Before reading this chapter, ensure that binary basis states, complex numbers, and the distinction between a pure state and a density operator are familiar. No tensor-network background is required.

5.1.1 Count entries before counting bytes

One qubit has two computational-basis amplitudes. Two qubits have four. Three qubits have eight. The pattern is

$$2,\ 4,\ 8,\ 16,\ \ldots,\ 2^n.$$

Each added qubit doubles the length of the dense state vector.

For n qubits,

$$|\psi\rangle = \sum_{x\in\{0,1\}^n} \alpha_x\,|x\rangle\,, \qquad \sum_x |\alpha_x|^2 = 1, \qquad \alpha_x \in \mathbb{C}.$$

Here x is an n-bit basis label and α_x is the complex amplitude stored for that label. A dense simulator keeps the full list, including entries that may be small or zero for a particular state.

A complex128 amplitude contains two 64-bit floating-point numbers and occupies 16 B. The raw state array therefore requires

$$M_{\mathrm{sv}} = 16 \cdot 2^n \text{ bytes}.$$

If a machine offers M usable bytes for one array, the raw storage bound is

$$n_{\max} = \left\lfloor \log_2\!\left(\frac{M}{16}\right) \right\rfloor.$$

This count is a first feasibility check. A working simulator also needs temporary arrays, communication buffers, metadata, and a safety margin.

An open-system density matrix contains $2^n \times 2^n$ complex entries,

$$M_\rho = 16 \cdot 4^n \text{ bytes.}$$

The extra factor of 2^n makes dense density-matrix calculations much smaller in width than dense pure-state calculations.

Practical simulators also need

- work arrays for gate application, transposition, and reduction;
- communication buffers and replicated metadata in distributed runs;
- checkpoint storage and fault recovery;
- memory for observables, random-number state, and noise trajectories;
- capacity lost to allocator fragmentation and runtime overhead.

Gate application can also become limited by memory bandwidth. A state that fits in memory can still carry an impractical time-to-solution.

Table 5.1: Raw dense state-vector storage using complex128 amplitudes. Workspace and communication memory are excluded.

Qubits	Complex amplitudes	Memory in GiB	Raw-storage interpretation
20	1.05×10^6	1.56×10^{-2}	Small local reference calculation
30	1.07×10^9	16	Single large-memory node or accelerator
35	3.44×10^{10}	512	Large-memory node or distributed calculation
40	1.10×10^{12}	1.64×10^4	Multi-node supercomputer scale
45	3.52×10^{13}	5.24×10^5	Approximately 0.5 PB of raw storage
50	1.13×10^{15}	1.68×10^7	Dense-storage barrier for ordinary computing systems

A distributed 45-qubit simulation using about 0.5 PB supplies a useful historical reference [9]. It illustrates the machine-scale cost of an uncompressed state vector and establishes no universal limit on classical simulation.

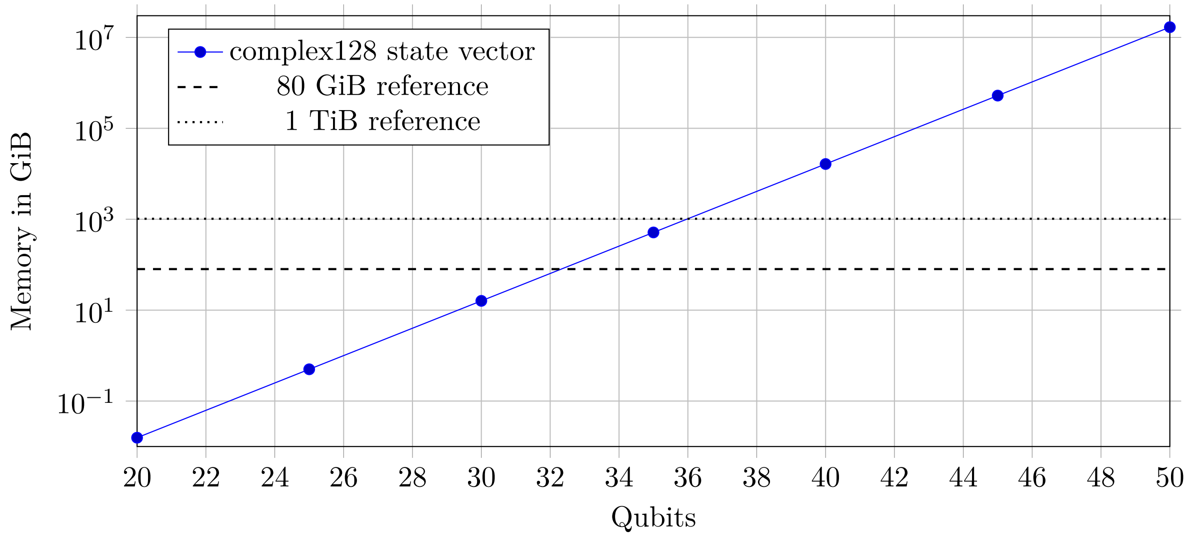

Figure 5.1: Raw dense state-vector storage grows as 2^n. The horizontal lines show capacity before simulator overhead is included.

5.2 Width Is Only One Complexity Coordinate

A fixed "classical qubit limit" hides the structure that determines cost. Clifford circuits can be simulated efficiently with stabilizer methods. Low-entanglement states can admit compact matrix-product-state representations. Shallow local circuits can favor tensor contraction, lightcone restriction, or circuit cutting. A request for one local observable can be much cheaper than a request for every amplitude.

Google qsim provides Schrödinger and hybrid qsimh approaches [10]. Qiskit Aer provides state-vector, density-matrix, stabilizer, extended-stabilizer, and matrix-product-state methods [11], [12]. A simulation report should therefore record

- qubit count, depth, connectivity, and gate family;
- entanglement structure and non-Clifford content;
- the requested output, such as a state, amplitude, sample, or observable;
- noise model, precision, approximation tolerance, and convergence checks;
- processor, accelerator, memory, interconnect, runtime, and software version.

The solver choice follows from this full description. A circuit above 50 qubits with strong volume-law entanglement can exceed dense state-vector memory and defeat low-bond-dimension MPS compression. General tensor contraction, hybrid partitioning, targeted observables, or a narrower scientific question may then provide the viable path.

Check your understanding

1. Complete the sequence for one through five qubits:

$$2,\ 4,\ 8,\ 16,\ \ldots$$

   State the general rule for n qubits.

2. The product state $|+\rangle^{\otimes n}$ has all 2^n computational-basis amplitudes nonzero. Explain why this fact alone does not make the state difficult for an MPS.

3. After a dense state vector exceeds available memory, what additional physical or mathematical property should be examined before declaring the calculation inaccessible?

Optional calculation. Verify that a 33-qubit complex128 state vector requires 128 GiB and that a dense 16-qubit complex128 density matrix requires approximately 64 GiB. Then identify two sources of workspace that lower the practical limit.

Key takeaway

Qubit count sets the dense storage scale. Circuit geometry, entanglement, gate content, target output, and accepted approximation error determine whether structure can reduce the cost.

Practical rule of thumb

Estimate raw storage and temporary workspace before launching a dense simulation. If projected arrays consume more than about half of available memory, select a distributed or structured method before allocation failures determine the outcome.

Chapter 6

Tensor Networks and Structured Simulation

How the argument develops

Chapter 5 counted every amplitude. This chapter asks whether those amplitudes can be generated from smaller factors. The controlling quantity is the amount of correlation that crosses each cut in the chosen qubit ordering. A small cross-cut description gives a small bond dimension. Entanglement growth enlarges the required description.

6.1 Compression by Entanglement Structure

Prerequisite

The reader needs product states and the idea of dividing qubits into left and right groups. Familiarity with the singular-value decomposition is helpful. The factorization and its physical meaning are developed below.

6.1.1 Begin with one cut

Place the qubits in a line and draw one cut between neighboring sites. The cut divides the system into a left block L and a right block R. The central question is simple:

How many independent left–right patterns are needed to reconstruct the state across this cut?

A product state needs one pattern. For example,

$$|+\rangle^{\otimes n} = \bigotimes_{k=1}^{n} \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

has 2^n nonzero amplitudes in the computational basis and still separates into one left factor and one right factor across every cut. Its Schmidt rank is one, so an exact MPS can use $\chi = 1$.

Entanglement increases the number of paired patterns. The Bell state

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

requires two left–right pairs across the cut between its qubits. Its Schmidt rank is two, so an exact representation requires $\chi \geq 2$ at that cut.

The required dimension can change from one cut to another. For

$$\left|\Phi^+\right\rangle_{12} \otimes |0\rangle_3 ,$$

the cut $1 \mid 23$ crosses the Bell pair and requires rank two. The cut $12 \mid 3$ separates the Bell pair from the third qubit and requires rank one. An MPS therefore carries a bond dimension for each cut, and the largest value usually controls memory and update cost.

6.1.2 From the coefficient table to the Schmidt decomposition

For a general bipartition, group the computational-basis labels into a left index ℓ and a right index r,

$$|\psi\rangle = \sum_{\ell,r} C_{\ell r} |\ell_L\rangle |r_R\rangle .$$

The coefficients form a matrix C. Its singular-value decomposition,

$$C = U\Lambda V^\dagger,$$

rewrites the state as

$$|\psi\rangle = \sum_{a=1}^{r} \lambda_a |a_L\rangle |a_R\rangle , \qquad \lambda_a \geq 0, \qquad \sum_a \lambda_a^2 = 1.$$

This is the Schmidt decomposition. The number of nonzero coefficients, r, is the Schmidt rank. It gives the exact dimension required across the cut. The entropy

$$S = -\sum_a \lambda_a^2 \log \lambda_a^2$$

describes how broadly the state weight is distributed across the paired patterns.

Plain-language interpretation

The Schmidt rank counts the exact number of cross-cut patterns. The bond dimension χ is the capacity assigned to carry those patterns in an MPS. Truncation keeps the largest-weight patterns and discards smaller ones, which introduces a numerical approximation that must be checked against the target observable.

6.1.3 From one cut to a matrix product state

Applying the factorization successively across the qubit chain gives a matrix product state,

$$\alpha_{i_1 i_2 \cdots i_n} = \sum_{a_1,\ldots,a_{n-1}} A^{[1]i_1}_{a_1} A^{[2]i_2}_{a_1 a_2} \cdots A^{[n]i_n}_{a_{n-1}}.$$

The physical index $i_k \in \{0, 1\}$ selects the basis state of qubit k. The bond index a_k carries the left–right correlation across the cut after that qubit. The largest bond-index range is the bond dimension χ.

For local dimension $d = 2$, storage scales near $\mathcal{O}(nd\chi^2)$ and a two-site update often scales near $\mathcal{O}(d^3\chi^3)$, with implementation-dependent constants [13]. The savings are large when the required χ remains moderate. The savings decline as entanglement spreads and the Schmidt ranks grow.

Check your understanding

1. Why can $|+\rangle^{\otimes n}$ have 2^n nonzero amplitudes and still require only $\chi = 1$?
2. For $|\Phi^+\rangle_{12} \otimes |0\rangle_3$, which cut requires the larger bond dimension, and what physical correlation crosses that cut?

This structure is useful for one-dimensional quantum magnets, shallow local circuits, and time windows in which entanglement remains concentrated across local cuts. In a verification workflow, an MPS can supply converged local observables for selected system sizes and evolution times.

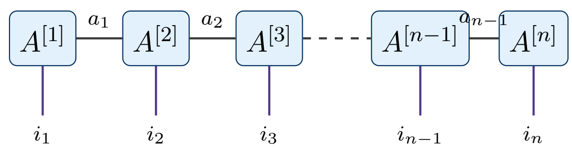

Figure 6.1: An open-boundary MPS. Physical indices label qubit states and internal bonds carry correlations between neighboring tensor blocks.

Low-entanglement states can remain compact. Deep random circuits and long real-time evolution can drive χ toward an exponential value. The representation then loses its advantage.

6.2 Truncation and Observable Convergence

After a two-site gate, an MPS algorithm performs a singular-value decomposition and can discard small Schmidt coefficients. If the retained set is $\mathcal{K}$, the discarded weight is

$$w_{\text{disc}} = \sum_{a \notin \mathcal{K}} \lambda_a^2.$$

This quantity measures the local norm loss at one truncation. The final observable error can accumulate across many steps and requires a direct convergence study.

A practical study repeats the calculation with increasing χ or decreasing truncation threshold. For a target observable O,

$$\Delta_O(\chi) = \left| \langle O \rangle_\chi - \langle O \rangle_{\chi_{\text{ref}}} \right|.$$

The reference bond dimension χ_{ref} should be large enough that a further increase changes the observable by less than the allocated numerical tolerance.

For an open-boundary qubit MPS with uniform internal bond dimension χ, the approximate number of stored complex entries is

$$N_{\text{MPS}} = 4\chi + 2(n-2)\chi^2, \qquad M_{\text{MPS}} = 16 N_{\text{MPS}} \text{ bytes}$$

for complex128 storage. Temporary singular-value-decomposition workspace and metadata can dominate the raw count.

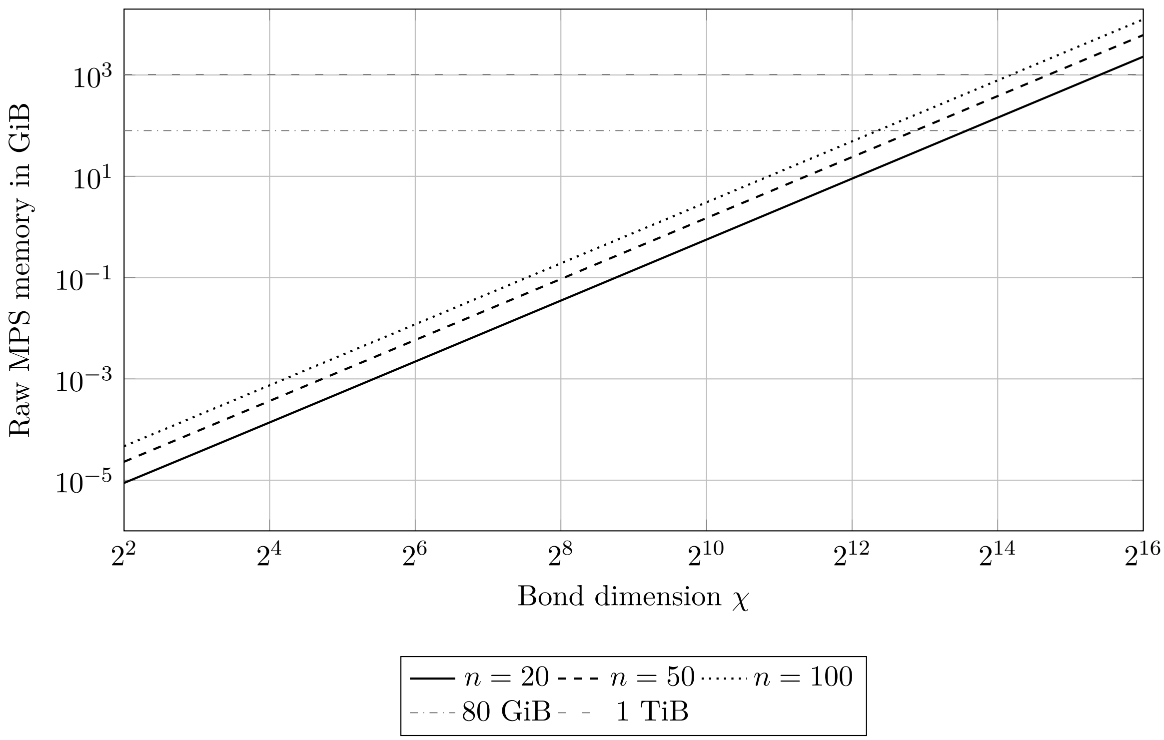

Figure 6.2: Raw complex128 memory for an open-boundary qubit MPS as the bond dimension increases. The three curves use the uniform estimate $N_{\mathrm{MPS}} = 4\chi + 2(n-2)\chi^2$. Runtime, singular-value-decomposition workspace, and metadata can impose stronger limits.

6.3 General Tensor Contraction and Related Methods

A general tensor-network simulator contracts a graph of tensors. The arithmetic cost and peak memory depend on the contraction order. Heuristic path optimizers seek a sequence with manageable intermediate tensors [14], [15]. The relevant structural quantity is the width of the chosen contraction path; qubit count alone is secondary.

The following methods serve different verification roles.

Table 6.1: Classical reference and estimation tools used in quantum verification.

Method	Strong regime	Verification utility
Dense state vector	Generic pure-state circuits at feasible widths	Supplies amplitudes, probabilities, and exact small-instance references
Density matrix	Small open systems and channel models	Tests noise models, channels, and readout correction
Stabilizer simulation	Clifford circuits with Pauli preparation and measurement	Supports very large exact references inside the stabilizer model
Extended stabilizer	Circuits with limited non-Clifford content	Supplies approximate references for near-Clifford workloads
Matrix product state	One-dimensional or weakly entangled systems	Tracks states and local observables with controlled bond truncation
General tensor networks	Geometries with favorable contraction width	Computes selected amplitudes, marginals, or observables

Method	Strong regime	Verification utility
Schrödinger–Feynman hybrid	Circuits that admit useful partitions	Trades repeated path summation for lower peak memory
Circuit cutting	Circuits with a manageable number of cut interactions	Reconstructs selected quantities from smaller fragment experiments [16]
Classical shadows	Many observables estimated from randomized measurements	Reduces measurement demand for compatible observable families [17]

Classical shadows are estimation tools. They support selected observable families and leave arbitrary amplitudes and the full output distribution outside their scope.

Check your understanding

1. Determine the Schmidt rank of $|00\rangle$ and of
$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$
across the cut between the two qubits. State the minimum exact bond dimension for each state.
2. Suppose the largest Schmidt rank encountered during an evolution is 32. What minimum bond dimension is required for an exact MPS representation at that cut? Describe what changes after truncation to $\chi = 16$.
3. Explain why a small discarded weight supports an MPS calculation yet leaves the target-observable error unresolved. State the convergence test that completes the verification record.
4. Compare an MPS calculation for a one-dimensional local Hamiltonian with a general tensor-network contraction for a two-dimensional circuit. Identify the structural quantity that controls cost in each case.

Key takeaway

Tensor networks replace a fixed qubit threshold with a structure-dependent cost. A credible calculation reports the representation, contraction strategy, maximum bond dimension, truncation rule, observable convergence, and requested output.

Practical rule of thumb

Increase χ until the reported observable changes by less than its numerical tolerance. A small discarded weight supports the calculation, and direct observable convergence supplies the decisive check.

Chapter 7

Sampling, Noise, and Estimation

How the argument develops

The chapter follows one result from analog readout to a reported estimator. It introduces the measurement model, separates physical noise from assignment error, derives the effect of correlated shots, and clarifies the difference between single-shot classification and multiple-shot precision.

7.1 Hardware Outputs as Estimators

Prerequisite

Before reading this chapter, ensure that a density matrix ρ, observable O, conditional probability, sample mean, and variance are familiar.

Gate-based processors usually return digitized bitstrings. The physical readout begins as an analog signal. Filtering, demodulation, integration, and classification convert that signal into a classical label. The estimator therefore depends on quantum dynamics and a classical measurement pipeline.

For a Pauli observable with outcomes $y_j \in \{-1, +1\}$,

$$\widehat{\mu} = \frac{1}{N}\sum_{j=1}^{N} y_j, \qquad \mathrm{Var}(\widehat{\mu}) = \frac{1-\mu^2}{N}$$

under the independent-shot model. Real data can include

- preparation error and residual excitation;
- relaxation, dephasing, coherent control error, and leakage;
- crosstalk and correlations across qubits or circuit layers;
- readout assignment error and classifier drift;
- slow calibration drift and batch-to-batch variation.

Randomized circuit order and timestamped records help distinguish shot noise from slow systematics.

7.1.1 Noise channels and measurement operators

Open-system noise is represented by a channel $\mathcal{E}$. A common single-qubit depolarizing convention is

$$\mathcal{E}_p(\rho) = (1-p)\rho + p\frac{I}{2}.$$

A global n-qubit version replaces $I/2$ with $I/2^n$. Local channels usually provide a more informative device model. Amplitude damping, dephasing, coherent overrotation, leakage, correlated errors, and readout confusion produce distinct data patterns. A single depolarizing parameter compresses those patterns into one limited approximation.

The most general measurement description uses a positive-operator-valued measure (POVM). Its elements E_x satisfy

$$E_x \succeq 0, \qquad \sum_x E_x = I, \qquad p(x) = \mathrm{Tr}(\rho E_x).$$

The label x is the reported classical outcome. For an ideal computational-basis measurement,

$$E_x = |x\rangle \langle x| .$$

A real readout chain changes the effective E_x through analog overlap, amplifier noise, classifier boundaries, and state transitions during integration.

Suppose an estimator assigns a real value $f(x)$ to each outcome. Its expectation and single-shot variance are

$$\mu_f = \sum_x f(x)\,\mathrm{Tr}(\rho E_x),$$
$$\sigma_f^2 = \sum_x f(x)^2\,\mathrm{Tr}(\rho E_x) - \mu_f^2.$$

The physical model determines the outcome probabilities. The estimator determines how those outcomes contribute to the final number.

7.1.2 Correlated shots and effective sample size

Shot correlations alter the variance of the sample mean. For a stationary sequence with variance σ^2 and lag-k autocorrelation ρ_k,

$$\mathrm{Var}(\widehat{\mu}) \approx \frac{\sigma^2}{N}\left(1 + 2\sum_{k=1}^{\infty}\rho_k\right) = \frac{\sigma^2}{N_{\mathrm{eff}}}.$$

Define the integrated autocorrelation factor

$$\tau_{\mathrm{int}} = 1 + 2\sum_{k=1}^{\infty}\rho_k, \qquad N_{\mathrm{eff}} = \frac{N}{\tau_{\mathrm{int}}}.$$

Positive slow drift gives $N_{\mathrm{eff}} < N$. Block resampling, session-level models, and repeated calibrations estimate uncertainty without treating adjacent shots as independent.

QEM constructs an adjusted estimator intended to reduce bias. Negative extrapolation weights, quasiprobability weights, matrix inversion, and postselection can increase variance or reduce effective sample size. A comparison should report

- raw and mitigated estimates with confidence intervals;
- total and accepted shots;
- circuit variants and calibration circuits;
- time ordering and drift diagnostics;
- sensitivity to the mitigation model and tuning choices.

7.2 Single-Shot and Multiple-Shot Verification

Single-shot readout means that one acquisition record contains enough information to assign an outcome. Its performance is described by conditional probabilities such as

$$P(\widehat{0} \mid 0), \qquad P(\widehat{1} \mid 1),$$

with leakage or erasure labels added where the classifier supports them.

Longer integration improves signal averaging until relaxation during measurement begins to erase information. Resonator response, amplifier noise, classifier design, and state-preparation error shape the observed confusion matrix. Assignment fidelity therefore belongs to a defined preparation-and-measurement protocol.

Multiple shots estimate probabilities and expectation values. Independent random error falls as $N^{-1/2}$. A fixed assignment bias persists. The required readout quality is set by the observable, circuit, mitigation method, accuracy target, and shot budget.

The following validation procedure is recommended

1. prepare known basis states and retain the analog records;
2. estimate the assignment matrix with uncertainty;
3. repeat the calibration across time and processor regions;
4. test whether estimator variance follows the expected $N^{-1/2}$ scaling;
5. compare raw and corrected estimates on held-out states.

A deviation from $N^{-1/2}$ can reveal drift or correlation. Correct scaling can coexist with a constant systematic bias, so both tests are needed.

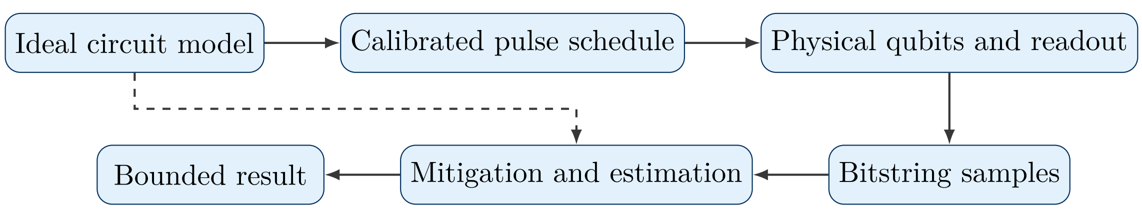

Figure 7.1: A hardware estimate joins the logical model, pulse implementation, physical measurement, calibration, mitigation, and statistical analysis.

Key takeaway

Single-shot fidelity describes classification under a specified protocol. Multiple-shot precision describes an estimator. Application accuracy requires both quantities plus a model for systematic error.

Practical rule of thumb

Plot the estimator against acquisition time and against $N^{-1/2}$ before assigning more shots. A plateau, hysteresis, or block-dependent mean directs attention to calibration stability.

Chapter 8

Quantum Error Mitigation on Near-Term Processors

How the argument develops

Quantum error mitigation changes an estimator by adding circuits, calibration data, or classical reconstruction. The chapter develops the governing relation for each major method, identifies its assumptions, and explains the evidence needed to show an improvement at fixed total resources.

8.1 Role and Limits

Prerequisite

Before reading this chapter, ensure that bias, variance, mean squared error, quantum channel, and calibration data are familiar.

QEM uses extra circuits, structural information, modified measurements, calibration data, or postprocessing to reduce noise-induced bias [1], [18]. Error suppression improves the physical implementation through pulse shaping, dynamical decoupling, randomized compiling, leakage reduction, or calibration. QEM acts on the reported estimate. Fault-tolerant quantum error correction encodes logical information and repeatedly detects faults so that logical error can fall with increasing code distance [19].

Let

$$\mu = \mathrm{Tr}(O\rho_{\mathrm{ideal}})$$

be the target expectation. Let $\widehat{\mu}_{\mathrm{raw}}$ be the hardware estimator and $\widehat{\mu}_{\mathrm{mit}}$ the mitigated estimator. A useful empirical gain is

$$G_{\mathrm{MSE}} = \frac{\mathbb{E}[(\widehat{\mu}_{\mathrm{raw}} - \mu)^2]}{\mathbb{E}[(\widehat{\mu}_{\mathrm{mit}} - \mu)^2]}.$$

A value above one indicates improvement on the validation distribution used to estimate the expectation. Generalization to a larger target circuit still requires held-out circuits whose noise, depth, and observable resemble the target.

8.2 Zero-Noise Extrapolation

8.2.1 Governing relation

Zero-noise extrapolation evaluates related circuits at several amplified noise scales and extrapolates to a zero-noise intercept [5], [6]. Experimental demonstrations have used pulse stretching and gate folding in chemistry and spin-model calculations [20].

Let ϵ denote the native noise strength and $\lambda_k \geq 1$ the scale factors. A Richardson estimator has the form

$$\widehat{\mu}_{\mathrm{ZNE}} = \sum_k w_k \widehat{\mu}(\lambda_k \epsilon),$$

with constraints

$$\sum_k w_k = 1, \qquad \sum_k w_k \lambda_k^r = 0 \quad (1 \leq r \leq p).$$

These equations cancel selected powers of ϵ in a smooth expansion. They place no guarantee on statistical efficiency. The variance contains the squared weights,

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{ZNE}}) = \sum_k w_k^2 \,\mathrm{Var}[\widehat{\mu}(\lambda_k \epsilon)]$$

for independent estimates.

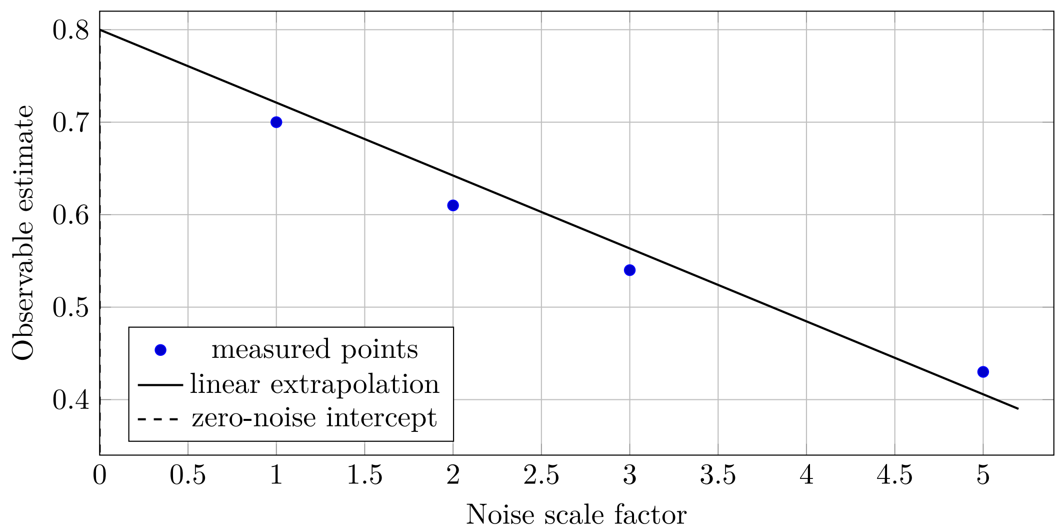

Figure 8.1: Illustrative ZNE data. The intercept is an extrapolated quantity whose uncertainty follows from the scaling procedure, regression model, and shot allocation.

8.2.2 Physical interpretation and experimental checks

Global or local unitary folding, identity insertion, and pulse stretching create the scaled circuits. Full global folding naturally gives odd factors $1, 3, 5, \ldots$. Partial folding and pulse-level methods can produce intermediate effective scales.

A complete ZNE record includes

- every raw point and its uncertainty;
- scale factors, fold construction, and shot allocation;
- regression equation, residuals, and confidence interval;

- sensitivity to removing one scale point;
- sentinel data that track drift during the sweep.

The method can fail if scaling changes coherent dynamics, leakage, temporal drift, or the relative composition of noise channels. Experimental evidence therefore tests the scaling law instead of assuming it.

8.3 Probabilistic Error Cancellation

8.3.1 Quasiprobability representation

PEC represents an ideal operation as a signed combination of implementable noisy operations [5],

$$\mathcal{U} = \sum_a \eta_a \mathcal{O}_a, \qquad \gamma = \sum_a |\eta_a|.$$

Sampling operation a with probability

$$p_a = \frac{|\eta_a|}{\gamma}$$

and multiplying the measured outcome by

$$\gamma \operatorname{sgn}(\eta_a)$$

produces an unbiased estimator if the learned representation is exact. The single-sample magnitude grows by γ, so the variance bound grows as γ^2.

Sparse Pauli–Lindblad models provide an experimentally demonstrated route to learning implementable decompositions [21]. For a circuit with layer norms γ_ℓ,

$$\Gamma = \prod_{\ell=1}^{L} \gamma_\ell.$$

The shot demand for fixed additive precision can scale as $\mathcal{O}(\Gamma^2)$. Small per-layer overheads therefore accumulate rapidly with noisy circuit volume. Imperfect noise learning adds residual bias, and model violation can be bounded separately under stated assumptions [22].

Experimentally, PEC is credible after the learned noise model predicts held-out circuits, the weighted estimator remains stable under recalibration, and the full sampling overhead is included in the resource comparison.

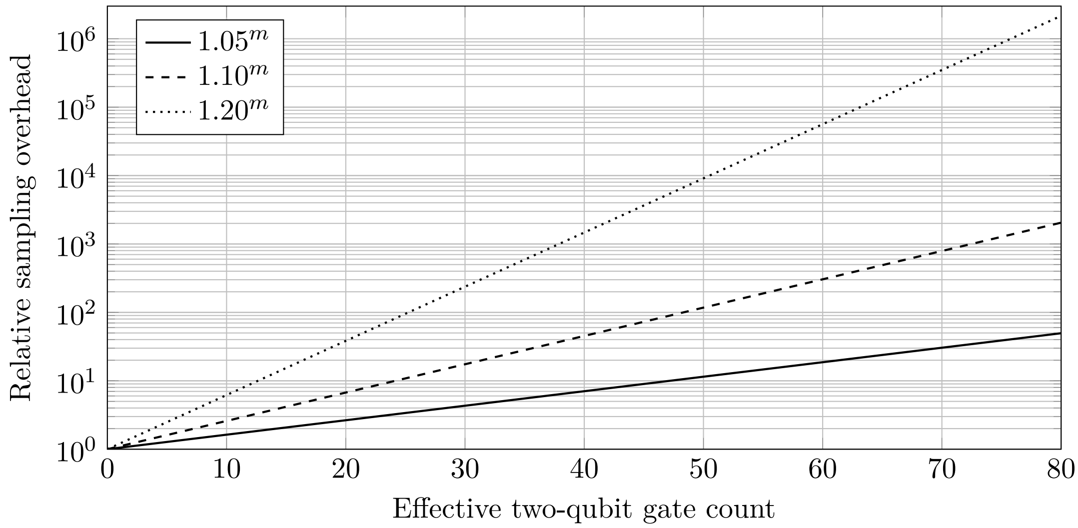

Figure 8.2: Illustrative PEC overhead multipliers for an effective noisy gate count m. The curves show generic exponential sensitivity and use no hardware fit.

8.4 Measurement Error Mitigation and Virtual Distillation

8.4.1 Measurement error mitigation

Measurement error mitigation models the observed probability vector as

$$\boldsymbol{p}_{\text{obs}} = A\boldsymbol{p}_{\text{true}},$$

where

$$A_{yx} = P(y \mid x)$$

is an assignment matrix estimated from prepared basis states. Direct inversion gives

$$\widehat{\boldsymbol{p}}_{\text{true}} = A^{-1}\widehat{\boldsymbol{p}}_{\text{obs}}.$$

An ill-conditioned inverse amplifies calibration and shot noise and can produce negative components. Constrained least squares, Bayesian methods, tensor-product approximations, and sparse correlated models trade calibration cost against model expressiveness. Scalable correlated measurement models have been demonstrated on multiqubit hardware [23].

A useful report gives the condition number of A, calibration uncertainty, time separation between calibration and target circuits, and a held-out basis-state test.

8.4.2 Virtual distillation

Virtual distillation uses powers of a noisy state,

$$\mu_M = \frac{\mathrm{Tr}(O\rho^M)}{\mathrm{Tr}(\rho^M)}.$$

If

$$\rho = \sum_j p_j \left|j\right\rangle \left\langle j\right|, \qquad p_0 > p_1 \geq \cdots,$$

then the relative weight of component j falls as

$$\left(\frac{p_j}{p_0}\right)^M .$$

Implementations require multiple state copies or equivalent permutation measurements, extra gates, and ratio estimation. State-preparation mismatch, coherent error in the dominant eigenvector, and a small denominator can limit the improvement [24], [25].

8.5 Symmetry Verification and Postselection

Known symmetries identify outcomes outside the physical sector of interest. For a symmetry operator S with eigenvalue $s \in \{-1, +1\}$,

$$\Pi_s = \frac{I + sS}{2}.$$

For an observable commuting with S,

$$\langle O \rangle_s = \frac{\mathrm{Tr}(O\Pi_s\rho)}{\mathrm{Tr}(\Pi_s\rho)}.$$

Postselection removes detected sector-changing errors and reduces the accepted sample count. Errors that preserve the symmetry remain hidden. An inaccurate sector assumption biases the result. Symmetry verification has been demonstrated in variational eigensolver experiments [26], [27].

The accepted fraction

$$a_s = \mathrm{Tr}(\Pi_s\rho)$$

must accompany the corrected estimate. A small accepted fraction can dominate the variance and execution cost.

8.6 Learning-Based Mitigation Approaches

Clifford data regression fits a map from noisy hardware values to exact values using classically tractable surrogate circuits near the target [28]. Variable-noise CDR adds noise-scaled training data and combines regression with ZNE [29]. Performance depends on how well the training circuits represent the target circuit, observable, and device noise.

Validation should separate training and test circuits, include an unmitigated baseline, and evaluate drift or distribution shift. Mitiq implements several QEM workflows, including ZNE, PEC, and CDR [30], [31].

Table 8.1: Assumptions and common failure modes of representative QEM methods.

Method	Main resource	Central assumption	Common failure mode
ZNE	Noise-scaled circuits and regression	Scaled circuits share a smooth zero-noise limit	Noise composition changes with scale
PEC	Characterized operations and weighted samples	Quasiprobability model represents the implemented noise	Norm overhead or model mismatch dominates

Method	Main resource	Central assumption	Common failure mode
Readout mitigation	Assignment calibration and inversion	Calibration remains valid for the target measurement	Drift, correlations, or ill conditioning amplify error
Virtual distillation	Multiple copies or permutation measurements	Dominant eigenvector approximates the target state	Coherent eigenvector error persists
Symmetry verification	Sector checks and rejected samples	Target remains in a known symmetry sector	Symmetry-preserving errors remain hidden
CDR and vnCDR	Tractable training circuits	Training and target errors share a learnable relation	Distribution shift invalidates the fitted map

Key takeaway

QEM is an estimator-design problem with assumptions, bias, variance, calibration cost, circuit overhead, and model risk. A mitigated point without raw data, uncertainty propagation, and held-out validation supplies incomplete evidence.

Practical rule of thumb

Adopt a mitigation method after it lowers held-out-circuit mean squared error at the same total execution budget. Improvement on the tuning circuits measures fit quality.

Chapter 9

Lightcones, Shaded Lightcones, and Classically Accelerated QEM

How the argument develops

A local observable is influenced by only part of an ideal circuit. This chapter shows how causal support can focus mitigation, how influence bounds produce shaded lightcones, and how operator propagation exchanges quantum depth for classical Pauli growth.

9.1 Causal Lightcones and Influence Bounds

Prerequisite

Before reading this chapter, ensure that you can conjugate an operator by a unitary and identify the qubits on which a Pauli operator acts nontrivially.

Consider a final measurement of Z on the target qubit of a CNOT gate. Backward conjugation gives

$$\mathrm{CNOT}^\dagger (I \otimes Z)\, \mathrm{CNOT} = Z \otimes Z.$$

The target measurement therefore depends on the control qubit one layer earlier. Repeating this conjugation gate by gate traces the backward causal lightcone of the observable.

For a circuit

$$U = U_L \cdots U_2 U_1$$

and terminal observable O, define

$$O_L = O, \qquad O_{t-1} = U_t^\dagger O_t U_t.$$

The support of O_t identifies the qubits that can influence the final expectation at circuit time t. A local gate with support disjoint from O_t leaves the operator unchanged and lies outside the ideal backward causal lightcone.

This statement refers to the specified circuit decomposition. Crosstalk, correlated noise, leakage, residual couplings, and control spillover can enlarge the physical region of influence. A useful experiment compares the ideal lightcone with sentinel operations placed just outside it. Any measurable response reveals a physical coupling that the ideal circuit graph omits.

Locality can reduce mitigation cost for local observables. Local PEC applies inverse noise models inside the relevant lightcone. Local structure also yields bounds on the remainder of ZNE [32]. The gain follows from circuit geometry, depth, observable support, and the spatial range of the noise.

Fundamental results show that broad QEM classes can require sampling overhead that grows rapidly with noise strength and circuit size [33], [34], [35], [36]. These statements are conditional on a noise model, accessible operations, estimator class, precision target, and asymptotic regime. Finite local observables can occupy easier regions inside those general bounds.

9.1.1 Shaded lightcones

A binary lightcone marks possible influence. A shaded lightcone assigns an upper bound to the contribution of each error channel. The channels with the largest bounds receive mitigation resources first.

A tight influence bound can itself be expensive to compute. Practical methods truncate Pauli paths, approximate operator growth, or restrict the classical propagation. Reported examples showed about two orders of magnitude of runtime reduction for a structured 127-qubit Trotter circuit, with conclusions tied to that circuit family and approximation [37].

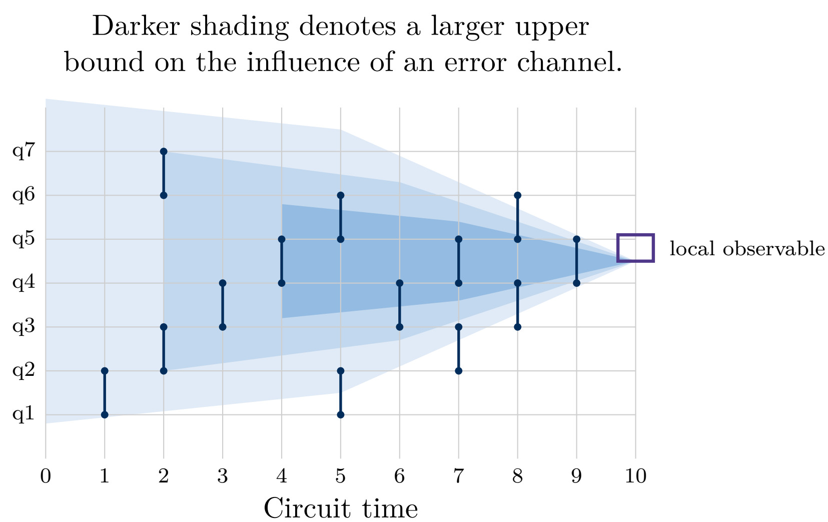

Figure 9.1: Conceptual shaded lightcone. The causal cone marks possible influence and the shading ranks modeled error channels by an influence bound.

9.2 Operator Backpropagation and Noise Absorption

9.2.1 Operator backpropagation

Operator backpropagation removes a suffix of gates from the hardware execution and conjugates the observable by that suffix on a classical computer. If

$$U = U_B U_A,$$

then

$$\langle O \rangle = \langle \psi_0 | U_A^\dagger \underbrace{\left(U_B^\dagger O U_B \right)}_{O_B} U_A | \psi_0 \rangle .$$

The processor executes the shorter circuit U_A and measures the Pauli terms in O_B.

Conjugation can produce an exponentially large Pauli expansion,

$$O_B = \sum_{\ell} c_\ell P_\ell.$$

Coefficient, Pauli-weight, and path truncation control memory and measurement cost at the price of approximation bias. OBP therefore trades quantum depth for classical propagation, Pauli storage, measurement grouping, and shots [38], [39].

9.2.2 Propagated noise absorption

Propagated noise absorption starts from a learned Pauli–Lindblad model and propagates inverse-noise contributions into a modified observable. Measuring that observable on the noisy processor can cancel the modeled gate noise.

The Pauli term count can approach 4^n. Truncation and noise-model error therefore form central parts of the uncertainty budget. A 2026 formulation reported numerical studies and a 56-qubit superconducting experiment [40], [41].

Table 9.1: Classically accelerated observable methods.

Method	Classical object	Quantum benefit	Principal cost
Lightcone PEC	Causal support of the observable	Avoids mitigating irrelevant local channels	Cone growth and nonlocal noise
Shaded lightcone	Influence bound for each error channel	Allocates PEC to influential channels	Bound computation and truncation
OBP	Backpropagated Pauli observable	Shortens the executed circuit	Pauli growth, measurements, and approximation bias
PNA	Noise-canceling modified observable	Absorbs a learned inverse-noise model classically	Noise-model error and Pauli growth

A complete record tracks the Pauli term count, total absolute coefficient weight,

$$\|c\|_1 = \sum_{\ell} |c_\ell|,$$

measurement groups, truncation rule, classical runtime, quantum depth removed, and held-out accuracy.

Key takeaway

Locality and operator propagation move part of the work from quantum depth or PEC sampling into classical computation. The exchange is useful only after Pauli growth, truncation bias, measurement groups, noise-model error, and wall-clock cost are included.

Practical rule of thumb

Track the number and total absolute weight of propagated Pauli terms after every layer. Stop extending the classical boundary once truncation bias or measurement growth exceeds the gain from removing another quantum layer.

Chapter 10

Benchmarking and Processor Verification

How the argument develops

A benchmark is a protocol whose score has meaning inside that protocol. The chapter explains what several widely used protocols measure, which assumptions connect their fitted parameters to physical error, and how to select a benchmark whose circuit ensemble resembles the intended workload.

10.1 Benchmark Viewpoints and Methodologies

Prerequisite

Before reading this chapter, ensure that exponential decay fitting, confidence intervals, and state-preparation and measurement error are familiar.

A processor benchmark specifies an input ensemble, circuit family, compilation rule, measurement, scoring function, sampling plan, and acceptance criterion. Changing one of these elements changes the quantity under test. A portfolio of benchmarks is therefore more informative than one scalar score.

10.1.1 Randomized benchmarking

Standard randomized benchmarking samples Clifford sequences of length m, appends an inverse, and fits a survival probability

$$P(m) = Ap^m + B.$$

The constants A and B absorb state-preparation and measurement offsets in the idealized model. For Hilbert-space dimension d, the fitted decay is often converted to an average error per Clifford,

$$r = \frac{d-1}{d}(1-p).$$

This conversion relies on assumptions about Markovianity, gate dependence, leakage, sequence sampling, and the fit model [42].

Interleaved RB alternates a target gate with random Cliffords and compares the reference and interleaved decays. The inferred target-gate error also relies on assumptions about how target and reference errors compose [43].

10.1.2 Wide, deep, and application-linked tests

Layer fidelity applies simultaneous direct RB across disjoint two-qubit layers to estimate connected wide-layer performance and sensitivity to concurrent operation [44]. Quantum volume runs square random model circuits and reports the largest width that passes a heavy-output test under a prescribed statistical rule [45].

Mirror circuits pair a randomized circuit with a compiled inverse or quasi-inverse, which creates a known ideal output for deeper tests [46]. Cross-entropy benchmarking compares experimental samples with ideal random-circuit probabilities. The linear score

$$F_{\text{XEB}} = 2^n \frac{1}{N} \sum_{j=1}^{N} p_U(x_j) - 1$$

acts as a fidelity proxy under specified random-circuit and noise assumptions. It also requires ideal probabilities for the sampled bitstrings [47], [48].

Volumetric benchmarking distinctly preserves width and depth as independent evaluation axes [49], whereas application invariants supply direct scientific relevance by utilizing symmetries, conserved sectors, identifiable trends, and solvable limits.

Table 10.1: Representative processor benchmarks and their evidentiary limits.

Method	Measured quantity	Primary strength	Evidentiary gap
Randomized benchmarking	Average Clifford sequence decay	Reduces sensitivity to preparation and measurement offsets	Limited noise specificity and model-dependent error conversion
Interleaved RB	Relative decay with a target gate inserted	Focuses on one implemented gate	Depends on reference errors and composition assumptions
Layer fidelity	Simultaneous layer performance over connected qubits	Probes scale and concurrent operation	Remains an indirect application predictor
Quantum volume	Largest square model-circuit width passing heavy-output generation	Compact full-stack test	Discrete score can hide topology and depth dependence
Mirror circuits	Success decay for circuits with known ideal outputs	Scales to deep compiled structures	Synthetic structure can differ from the application
Cross-entropy benchmarking	Correlation with ideal random-circuit probabilities	Sensitive random-circuit sampling score	Ideal-probability computation and fidelity interpretation are limiting
Application invariants	Symmetries, conserved sectors, trends, and solvable limits	Direct relevance to the target workload	Invariant-preserving errors can remain hidden

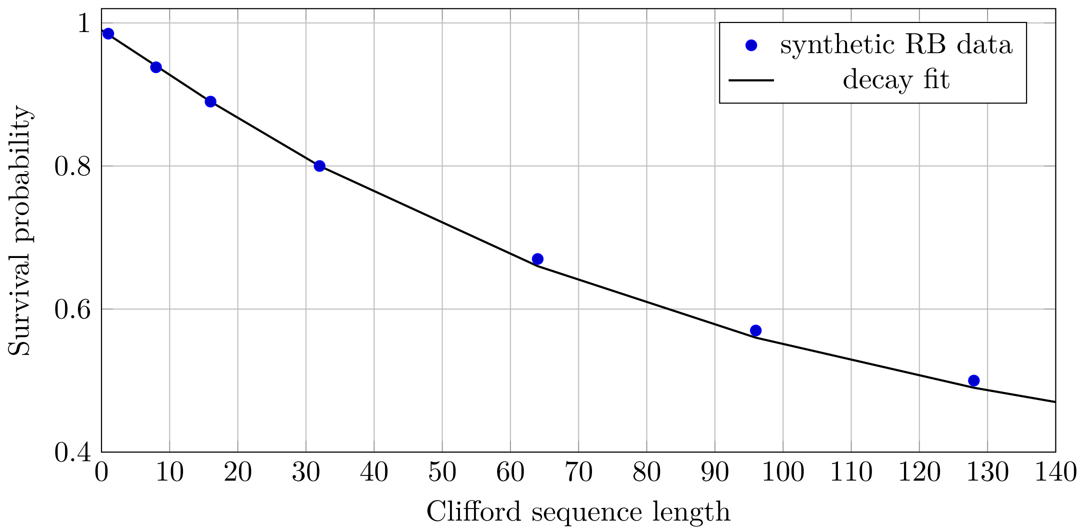

Figure 10.1: Synthetic RB data and an exponential fit. Residuals, sequence-to-sequence variation, leakage, and fit sensitivity should accompany the decay parameter.

10.2 Reporting and Benchmark Selection

A benchmark report should include

- confidence intervals and fit residuals;
- device region and calibration timestamps;
- circuit ensemble, depth convention, and sequence count;
- compilation settings and native-gate definitions;
- shot allocation, inclusion rules, and leakage treatment;
- protocol version and every pass or fail criterion.

Cross-device comparisons require matched protocol versions and success criteria. A higher score on one benchmark can provide little information about a different application family. Modern benchmarking methodologies emphasize this protocol dependence [50].

Key takeaway

RB measures an average sequence decay. Layer fidelity tests simultaneous connected layers. Quantum volume tests square random circuits. Mirror circuits supply known outputs at larger depth. XEB measures correlation with ideal random-circuit probabilities. The results overlap in purpose and remain distinct in interpretation.

Practical rule of thumb

Choose a benchmark whose circuits resemble the width, depth, connectivity, gate mix, and concurrency of the application. Add a second benchmark with a different sensitivity whenever the conclusion relies on one fitted score.

Chapter 11

Beyond DiVincenzo and the Qualification of Demonstrated Capability

How the argument develops

DiVincenzo's criteria describe the physical ingredients required for quantum computation. This chapter extends that foundation into a qualification record that separates the object under test, error-management stage, benchmark contract, reliability, scaling, application value, and resource cost.

11.1 From Physical Feasibility to Evidence

Prerequisite

Before reading this chapter, ensure that you are comfortable with the concepts of a density matrix ρ, a measurement shot, a confidence interval, and an error corrected logical qubit. The primer in chapter 1 supplies the required background.

DiVincenzo's criteria answer a foundational engineering question: what capabilities must a physical platform possess before universal quantum computation can be implemented? The original technique identifies five requirements for computation and two additional requirements for quantum communication [51]. In compact form, the platform needs

- a scalable collection of well characterized qubits whose computational states can be identified and controlled;
- preparation of a reproducible fiducial state, commonly $|0\rangle^{\otimes n}$, so that experiments begin from a defined quantum condition;
- coherence times that exceed the characteristic gate and measurement times by a margin compatible with the intended circuit;
- a universal family of controllable quantum operations;
- qubit resolved measurement with a declared assignment model;
- conversion between stationary and flying quantum information for communication architectures; and

- faithful transmission of flying qubits across the intended channel.

These requirements remain useful. They describe whether a proposed technology possesses the ingredients from which a quantum computer can be built. A mature experimental program also needs a second layer that asks what the assembled system has demonstrated, how strong the evidence is, how performance changes with scale, and whether an application gains scientific or economic value.

A useful progression is

$$\begin{aligned}\text{physical feasibility} &\longrightarrow \text{controlled implementation} \longrightarrow \text{benchmark conditioned reliability}\\ &\longrightarrow \text{scaling evidence} \longrightarrow \text{scientific utility} \longrightarrow \text{economic deployment}.\end{aligned}$$

The arrows represent additional evidence. A later stage preserves the earlier requirements and adds a more demanding acceptance test.

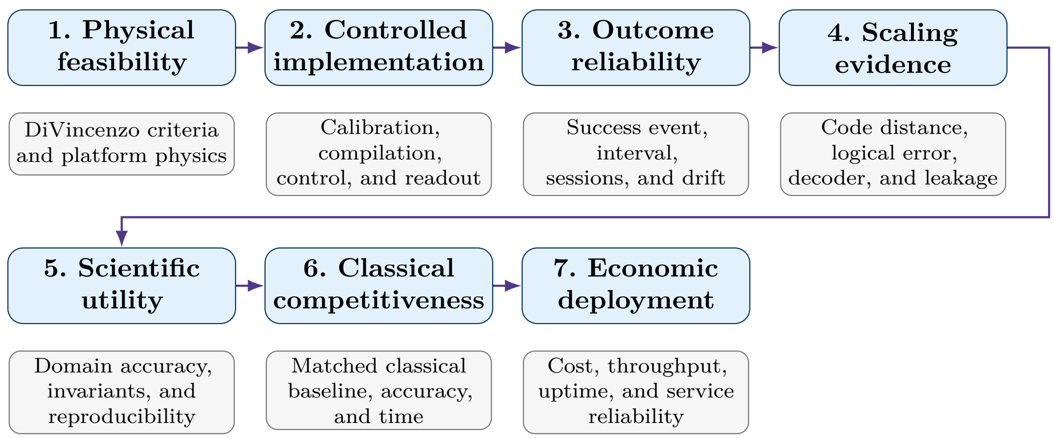

Figure 11.1: A post-DiVincenzo hierarchy of evidence.

The ladder prevents one numerical score from carrying several meanings. A gate fidelity describes a calibrated operation. A benchmark score describes performance on a prescribed circuit ensemble. A logical error rate describes an encoded process under a code, syndrome circuit, and decoder. An application result describes a scientific workflow. Economic deployment adds time, cost, uptime, and repeatability. Each quantity is useful after its object and protocol are stated.

11.2 The Object Under Test and the Qualification Record

The first decision identifies the object receiving the qualification. A physical qubit, encoded memory, logical gate, processor, and end to end workflow occupy different levels of the system. The notation

$$O \in \{\text{physical qubit},\ \text{encoded memory},\ \text{logical qubit},\ \text{logical gate},\ \text{processor},\ \text{workflow}\}$$

keeps that choice explicit.

A complete qualification record can be written as

$$\mathcal{Q}^{+} = (O,\ P,\ E,\ B,\ R,\ S,\ A,\ U)\,.$$

The coordinates separate concepts that often appear together in informal descriptions.

Table 11.1: Coordinates in a capability qualification record.

Symbol	Coordinate	Experimental meaning
O	Object under test	Identifies whether the record describes a physical qubit, encoded memory, logical operation, processor, or complete workflow.
P	Platform and protection modality	Records the hardware family, native interactions, connectivity, encoding mechanism, and any intrinsic protection such as a bosonic or topological degree of freedom.
E	Error management status	Distinguishes unencoded control, error suppression, mitigation, error detection, logical correction, below threshold scaling, and fault-tolerant logical operations.
B	Benchmark contract	Fixes the input ensemble, implemented operations, width, depth, duration, measurement rule, decoder, and postselection policy.
R	Reliability evidence	Reports the observed estimator, confidence interval, shot count, effective sample size, number of sessions, and between session variation.
S	Scaling evidence	Records how logical error, runtime, memory, or another target quantity changes as code distance, system size, depth, or precision is increased.
A	Application qualification	Locates the result between hardware execution, scientific validity, reproducible utility, classical competitiveness, and economic deployment.
U	Resource and utility ledger	Records wall clock time, QPU use, CPU or GPU work, calibration, queueing, energy, financial cost, and personnel effort.

This representation resolves an important classification problem. Terms such as *NISQ*, *near-term*, *logical*, *fault-tolerant*, *bosonic*, and *topological* describe different coordinates. A topologically protected object can also be logical and fault-tolerant. A superconducting, trapped ion, neutral atom, spin, or photonic platform can pass through several error management stages. A high success fraction on a shallow benchmark provides limited information about a deep logical computation. The vector preserves these distinctions without discarding familiar labels.

11.2.1 A benchmark contract gives the success probability meaning

Consider a repeated experiment that returns a binary result $Y \in \{0, 1\}$. The value $Y = 1$ indicates that the registered success criterion has been satisfied. The criterion acquires physical meaning from a benchmark contract

$$B = (\mathcal{I}, \mathcal{G}, w, d, t, \mathcal{M}, \mathcal{D}, \mathcal{P}_{\text{post}}) ,$$

where

- $\mathcal{I}$ is the input state or input instance ensemble;
- $\mathcal{G}$ is the allowed logical or native operation family;

- w and d are the compiled width and depth;
- t is the elapsed quantum evolution or memory duration;
- $\mathcal{M}$ is the measurement and scoring rule;
- $\mathcal{D}$ is the decoder or classical postprocessing map; and
- $\mathcal{P}_{\text{post}}$ is the postselection, discard, and erasure policy.

The benchmark conditioned success probability is

$$p_B = \Pr(Y = 1 \mid B)\,.$$

The subscript matters. A processor can have a large p_B for a shallow memory test and a small $p_{B'}$ for a deeper nonlocal observable. The probability belongs to the device under a specific contract.

A null model and an ideal model provide two useful reference points. Let $p_{\text{null},B}$ denote the success probability for random guessing or another declared baseline, and let $p_{\text{ideal},B}$ denote the ideal value. The normalized outcome score

$$S_B = \frac{\widehat{p}_B - p_{\text{null},B}}{p_{\text{ideal},B} - p_{\text{null},B}}$$

places the observed result relative to both references. Values below zero reveal performance below the null model. Values above one can occur from finite sampling or an imperfect idealization, so the raw value should remain visible.

Qualification should use an uncertainty bound in addition to the point estimate. For a required threshold τ_B, a one sided acceptance rule can be written

$$\underline{p}_{B,1-\alpha} \geq \tau_B,$$

where $\underline{p}_{B,1-\alpha}$ is a lower confidence bound. This rule asks whether the evidence supports the threshold at confidence $1 - \alpha$.

11.3 An Illustrative Finite-Shot Outcome Study

The following calculation isolates finite-shot evidence under one fictional benchmark contract. Each row uses a different underlying success probability and the same nominal shot count. The rows are statistical scenarios and carry no hardware classification. They illustrate how point estimates, Wilson intervals, shot requirements, and zero-failure bounds change as the success probability approaches one.

Hardware descriptions such as NISQ, logical, fault-tolerant, bosonic, and topological belong to the platform, error-management, and scaling coordinates of the qualification record. Those descriptions require evidence about architecture, encoding, repeated correction, logical operations, and scaling.

Key takeaway

Use these rows to study finite-shot uncertainty. Hardware maturity and protection mechanisms require separate qualification evidence.

Table 11.2: Illustrative benchmark conditioned outcomes from 10,000 shots. The interval uses the observed successes and the two sided 95% Wilson construction.

Model	Target p	Shots	Successes	Failures	Observed $\widehat{p}$	Wilson low	Wilson high	CI width	Cost (USD)
Toy working qubit	0.50	10,000	5,025	4,975	0.5025	0.4927	0.5123	0.0196	$0.60
NISQ qubit	0.60	10,000	6,010	3,990	0.6010	0.5914	0.6106	0.0192	$0.60
Near-term qubit	0.70	10,000	7,008	2,992	0.7008	0.6917	0.7097	0.0179	$0.60
Advanced near-term qubit	0.80	10,000	8,006	1,994	0.8006	0.7927	0.8083	0.0157	$0.60
Fault-tolerant logical qubit	0.90	10,000	9,013	987	0.9013	0.8953	0.9070	0.0117	$0.60
Topologically protected qubit	0.98	10,000	9,797	203	0.9797	0.9767	0.9823	0.0055	$0.60
Ideal qubit	1.00	10,000	10,000	0	1.0000	0.9996	1.0000	0.0004	$0.60

Table 11.3: Approximate shot planning for a two sided 95% interval with requested absolute half width ϵ. The idealized boundary row is axiomatic, resulting in singular deterministic shot planning for absolute certainty under theoretical conditions.

Model	Required shots for ±2%	Required shots for ±1%	Required shots for ±0.5%
Toy working qubit	2,401	9,604	38,415
NISQ qubit	2,305	9,220	36,879
Near-term qubit	2,017	8,068	32,269
Advanced near-term qubit	1,537	6,147	24,586
Fault-tolerant logical qubit	865	3,458	13,830
Topologically protected qubit	189	753	3,012
Ideal qubit	1	1	1

The tables show how the type of qubit used affects the number of measurements (shots) needed to reach a reliable result. As we move from basic "toy" qubits to advanced, error-corrected ones, the statistical uncertainty drops.

We categorize these models by their capability.

- **Toy and Noisy Intermediate-Scale Quantum (NISQ) qubits** are the starting point, plagued by high noise and low coherence.
- **Near-term and Advanced near-term qubits** introduce better calibration and early error-detection methods to improve reliability.
- **Fault-tolerant and Topologically protected qubits** represent the gold standard, using active error correction and intrinsic physical protections to keep success rates very high.
- **The Ideal qubit** is a theoretical perfect model used for comparison.

A key takeaway from the data is that the more reliable a qubit is, the fewer shots are needed to prove its performance. For example, a "topologically protected" qubit requires far fewer shots to hit a $\pm 1\%$ precision window than a "toy" qubit does.

Regarding cost, as detailed in table 11.2, we have listed a flat rate of \$0.60 per 10,000 shots. It is important to note that this is a simplified figure. In reality, advanced qubits require massive amounts of classical computing power for error decoding, which makes them much more expensive to operate than the raw "per-shot" price suggests. We analyze this conversion from cost to scientific value in the next section.

11.4 From Success Counts to a Wilson Interval

Let X be the number of successes in n independent Bernoulli trials,

$$X \sim \text{Binomial}(n, p), \qquad \widehat{p} = \frac{X}{n}.$$

The elementary normal interval replaces the unknown standard deviation with $\sqrt{\widehat{p}(1-\widehat{p})/n}$. That substitution performs poorly near $p = 0$ and $p = 1$ and can produce endpoints outside the physical interval $[0, 1]$. The Wilson construction inverts a score test and retains the parameter p inside the variance term [52].

For a two sided confidence level $1 - \alpha$, let $z = z_{1-\alpha/2}$. The score inequality is

$$\left| \frac{\widehat{p} - p}{\sqrt{p(1-p)/n}} \right| \le z.$$

Squaring and collecting powers of p gives

$$(n + z^2)p^2 - (2n\widehat{p} + z^2)p + n\widehat{p}^2 \le 0.$$

The accepted values of p lie between the roots of this quadratic. After division by n, the endpoints are

$$p_\pm = \frac{\widehat{p} + \dfrac{z^2}{2n} \pm z\sqrt{\dfrac{\widehat{p}(1-\widehat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}.$$

For a 95% interval, $z = z_{0.975} = 1.959964$. Substituting $X = 5{,}025$ and $n = 10{,}000$ gives

$$\widehat{p} = 0.5025, \qquad p_- = 0.4927, \qquad p_+ = 0.5123.$$

The interval width is $p_+ - p_- = 0.0196$. This width quantifies uncertainty under the independent Bernoulli model. Drift and serial correlation require an expanded model, as developed later in this chapter.

11.4.1 Planning the shot count

Far from the boundaries and for moderate n, the half width is approximately

$$\epsilon \approx z\sqrt{\frac{p(1-p)}{n}}.$$

Solving for n gives the planning expression

$$n = \left\lceil \frac{z^2 p(1-p)}{\epsilon^2} \right\rceil.$$

The factor $p(1-p)$ reaches its maximum at $p = 1/2$. A conservative plan that lacks a prior estimate therefore uses

$$n_{\max} = \left\lceil \frac{z^2}{4\epsilon^2} \right\rceil.$$

At 95% confidence, this gives 2,401 shots for $\epsilon = 0.02$, 9,604 shots for $\epsilon = 0.01$, and 38,415 shots for $\epsilon = 0.005$.

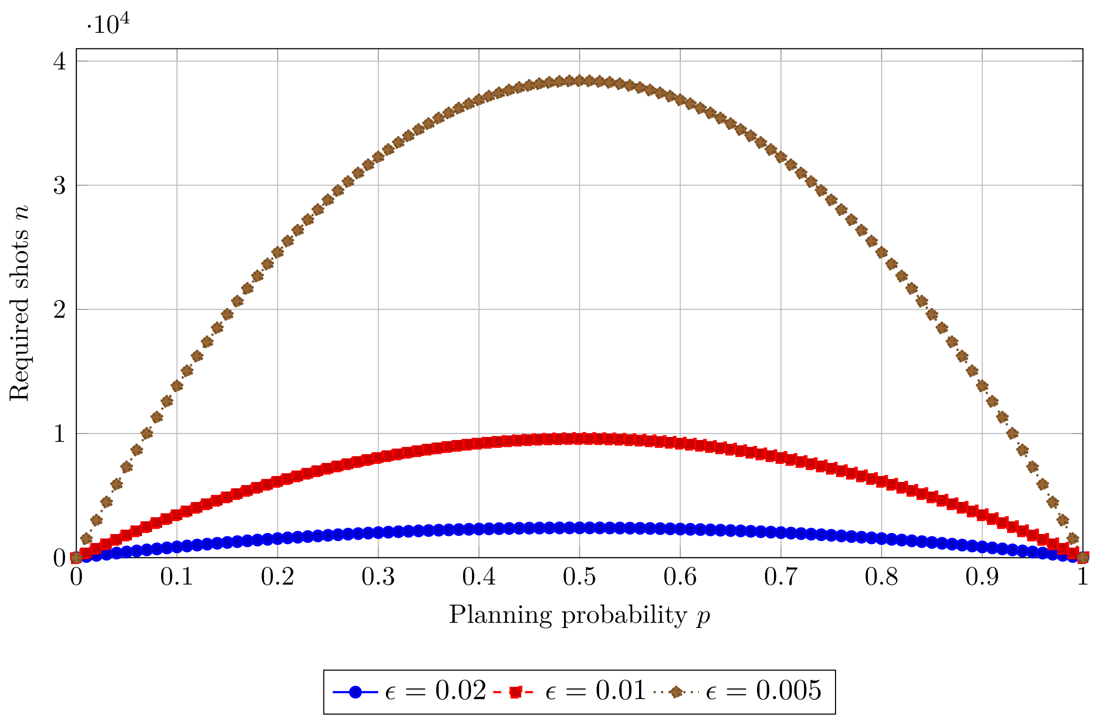

Figure 11.2: Approximate 95% shot planning as a function of the assumed success probability. The largest requirement occurs near $p = 1/2$, where Bernoulli variance is maximal.

11.4.2 Zero failures and evidence near the boundary

The planning expression uses a known design value of p. Setting $p = 1$ therefore yields zero variance and zero planned shots. The idealized boundary case can use $p = 1$ as an axiom. An empirical device has an unknown probability, so a run with zero observed failures still requires a finite lower bound.

Suppose n trials all succeed. Under a candidate probability p_0, the probability of that event is

$$\Pr(X = n \mid p_0) = p_0^n.$$

A one sided test at significance α rejects values whose all success probability is at most α. The lower confidence bound is therefore

$$p_{\mathrm{L}} = \alpha^{1/n}.$$

Equivalently, the number of zero failure trials required to support $p \geq p_0$ is

$$n \geq \left\lceil \frac{\ln \alpha}{\ln p_0} \right\rceil.$$

At 95% confidence, the resulting requirements are

- 299 consecutive successes for $p_0 = 0.99$;
- 2,995 consecutive successes for $p_0 = 0.999$;
- 29,956 consecutive successes for $p_0 = 0.9999$; and
- 299,572 consecutive successes for $p_0 = 0.99999$.

This boundary construction is closely related to the exact binomial interval of Clopper and Pearson [53]. It shows why a perfect observed fraction and perfect underlying reliability are different statements.

11.5 Drift, Correlation, and Reproducibility

The independent Bernoulli model assumes a fixed probability and independent trials. Quantum experiments can contain calibration drift, residual heating, leakage accumulation, low frequency noise, classifier drift, or batch effects. A large shot count collected inside one short calibration interval can therefore overstate long term evidence.

Let $Y_i \in \{0, 1\}$ be a stationary sequence with mean p, variance $\sigma^2 = p(1-p)$, and lag k correlation ρ_k. The variance of the sample mean is

$$\mathrm{Var}(\widehat{p}) = \frac{\sigma^2}{n} \left[1 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n} \right) \rho_k \right].$$

For a long sequence with decaying correlations, define the integrated autocorrelation factor

$$\tau_{\mathrm{int}} = 1 + 2 \sum_{k=1}^{\infty} \rho_k, \qquad n_{\mathrm{eff}} = \frac{n}{\tau_{\mathrm{int}}}.$$

Positive correlation gives $n_{\mathrm{eff}} < n$. The effective count communicates how much independent information remains after serial structure is considered. A block bootstrap, session level model, or hierarchical binomial analysis provides a more faithful interval than an iid formula applied to every raw shot.

Session to session variation can be represented by a beta binomial model,

$$p_j \sim \mathrm{Beta}(\alpha, \beta), \qquad X_j \mid p_j \sim \mathrm{Binomial}(n_j, p_j).$$

The mean success probability is

$$\mu = \frac{\alpha}{\alpha + \beta},$$

and the within session intraclass correlation is

$$\rho = \frac{1}{\alpha + \beta + 1}.$$

For equal session sizes m, the binomial variance is inflated by the design factor

$$D = 1 + (m - 1)\rho.$$

The model separates repeatability inside one calibration window from reproducibility across independent windows. Both quantities belong in a qualification record.

A strong acquisition plan therefore distributes measurements across

- several calibration blocks;
- independent sessions and experimental days;
- distinct processor regions or devices where generalization matters;
- randomized circuit order to decorrelate drift from problem difficulty; and
- archived raw records that permit post hoc correlation diagnostics.

11.6 Fault Tolerance Requires Scaling Evidence

A large success probability on one benchmark cannot by itself establish fault tolerance. Fault tolerance concerns the behavior of encoded information under repeated correction and logical operations as resources are increased. For a surface code memory, a common scaling form is

$$\epsilon_L(d) \approx A \left(\frac{p}{p_{\text{th}}}\right)^{(d+1)/2},$$

where d is code distance, p is a representative physical error parameter, p_{th} is the threshold, and $\epsilon_L(d)$ is the logical error per correction cycle. The relation is approximate and its parameters depend on the circuit level noise, decoder, leakage treatment, boundaries, and correlated errors.

The suppression factor for a distance increase of two is

$$\Lambda(d) = \frac{\epsilon_L(d)}{\epsilon_L(d+2)}.$$

Values $\Lambda > 1$ indicate logical error suppression as distance grows. Breakeven adds a second comparison: the encoded memory or operation must outperform the selected physical reference under matched conditions. Recent surface code experiments have demonstrated below threshold memory scaling, beyond breakeven lifetime, repeated syndrome extraction, and real time decoding on superconducting hardware [54]. Those achievements supply several coordinates in $\mathcal{Q}^+$; scalable logical computation still requires a fault-tolerant logical gate set, sustained multi logical qubit operation, leakage control, long duration stability, and manageable resource overhead.

The error management coordinate can use the following progression.

Table 11.4: Error management stages and their characteristic evidence.

Stage	Regime	Characteristic evidence
E0	Unencoded physical operation	Calibrated preparation, gates, measurement, coherence, and benchmark performance on physical qubits.

Stage	Regime	Characteristic evidence
E1	Error-suppressed operation	Pulse shaping, dynamical decoupling, leakage reduction, symmetry preservation, or architecture level suppression with measured improvement.
E2	Error-mitigated estimation	Reduced estimator bias under a declared QEM protocol, accompanied by variance, calibration, and resource overhead.
E3	Error-detected encoding	Repeated detection events, postselection or erasure handling, and a documented accepted fraction.
E4	Beyond breakeven logical memory	Encoded lifetime or logical error improves upon a matched physical reference.
E5	Below-threshold scaling	Logical error decreases with code distance or an equivalent protection parameter across repeated cycles.
E6	Fault-tolerant logical operations	State preparation, gates, measurement, and decoding preserve logical protection under the applicable fault tolerance construction.
E7	Utility scale logical computation	Multiple logical qubits execute an application relevant circuit at the required logical error, runtime, and resource overhead.

Topological protection belongs primarily to the platform and protection coordinate P. It can reduce selected local error processes, yet experimental qualification still needs control, measurement, poisoning, leakage, decoder, and scaling evidence. This placement allows topological, bosonic, and conventional stabilizer encodings to be compared without turning one mechanism into a universal maturity rank.

11.7 A Reusable Qualification Record

A publication or engineering report can summarize the full record in a compact sequence.

- Identify O, the exact object under test, and describe its platform and protection modality P.
- State the error management stage E and the evidence required for entry into that stage.
- Register the benchmark contract B, including inputs, compiled circuit, duration, measurement, decoding, and postselection.
- Report R with point estimates, intervals, raw and effective sample sizes, session count, and drift diagnostics.
- Report S with code distance or problem size scaling, convergence tests, and the resource range over which the trend holds.
- Locate the result on the application ladder A developed in chapter 15.
- Publish the resource ledger U with quantum, classical, calibration, queue, verification, energy, and financial contributions.

The resulting statement is concise and auditable. It describes what succeeded, under which contract, with which evidence, at what scale, and at what cost. The approach extends DiVincenzo's physical requirements into a capability record that can follow a system from the first qubit demonstration to deployed scientific computation.

Key takeaway

For a success probability to be scientifically meaningful, it must be supported by a clear object of study, a defined benchmark, a method for calculating confidence, proof of reproducibility, an analysis of scaling behavior, and a record of resources used. Additionally, the physical implementation and the maturity of the technology should be treated as two separate metrics.

Practical rule of thumb

Qualify a result with the lower confidence bound under a registered benchmark contract. Reserve the labels *below threshold* and *fault-tolerant* for experiments that include scaling evidence across encoded resources.

Chapter 12

A Precise Vocabulary for Quantum Systems, Workflows, and Performance

How the argument develops

The word *quantum* can describe a physical state, a design orientation, a role inside a larger workflow, or a measured comparison with a classical method. These meanings answer different questions. This chapter separates them, develops the governing relations behind the physical and performance terms, and then joins them in one reusable description record.

12.1 Primer: One Prefix, Several Scientific Questions

Prerequisite

Before reading this chapter, ensure that you are comfortable with a density matrix ρ, an observable O, a finite-shot estimator, and the distinction between a quantum circuit and a complete scientific workflow. Review chapters 1 and 7 if these ideas need reinforcement.

A phrase such as "quantum enhanced coherent native accelerator" can sound precise and still leave its scientific content unresolved. Each adjective belongs to a different coordinate.

- **Physical coordinate** describes the state or device physics. *Quantum coherent* and *quantum limited* belong here.
- **Design coordinate** describes how the problem, data, algorithm, or architecture uses quantum information. *Quantum inspired*, *quantum aided*, *quantum native*, *quantum assisted*, and *purely quantum* belong here.
- **Comparative coordinate** describes a measured change against a baseline. *Quantum enhanced* and *quantum accelerated* belong here.
- **Capability coordinate** describes how useful and competitive the complete result is. *Quantum utility*, *quantum advantage*, and the historical term *quantum supremacy* belong here.

Devices can be coherent and quantum-limited without producing a useful computation. A workflow can be quantum-native yet slower than its classical counterpart. Similarly, a quantum-assisted calculation can provide utility before a broad advantage study is complete. These combinations are consistent as long as these metrics are treated as distinct.

The following description record keeps the coordinates visible,

$$\mathcal{L} = (R,\, I,\, K,\, B,\, M,\, Q,\, \tau)\,.$$

Its entries have the following meanings.

- R identifies the quantum resource, such as coherence, squeezing, entanglement, interference, contextuality, a protected logical subspace, or quantum statistics.
- I identifies the integration class, such as inspired, aided, native, assisted, or purely quantum.
- K states the quantum kernel and its boundary with classical computation.
- B states the classical, conventional, null, or physical baseline.
- M states the metric, acceptance tolerance, confidence level, and resource boundary.
- Q states the supported capability term, such as enhancement, acceleration, utility, advantage, or a computational advantage milestone.
- τ records the comparison date and the reproducibility window.

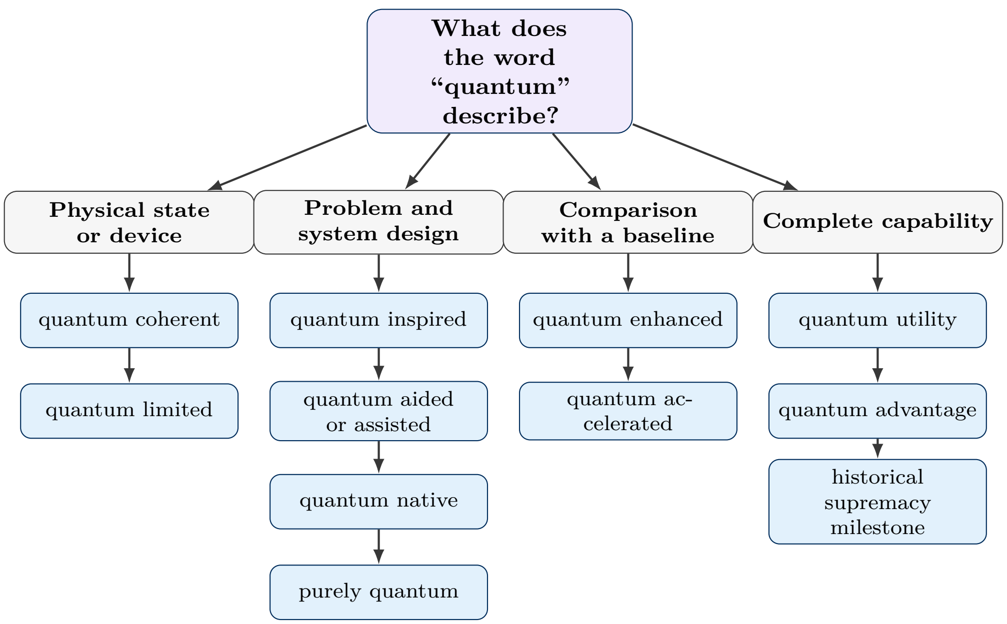

Figure 12.1: The principal quantum descriptors occupy separate coordinates. A complete application can carry one term from several branches at the same time.

12.2 Physical Descriptors: Coherence and Quantum Limits

12.2.1 Quantum coherent systems

Physical setting and terminology

Coherence is the capacity of a quantum state to retain phase relations in a declared basis. Those phase relations permit interference and support entangling operations, sensing sequences, state transfer, and algorithmic amplitude control. Coherence is basis dependent. A density matrix can be diagonal in one basis and contain off-diagonal entries in another. Resource theories of coherence make this basis dependence explicit [55].

For a qubit in the computational basis,

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}.$$

The populations ρ_{00} and ρ_{11} determine Z-basis probabilities. The off-diagonal entry ρ_{01} carries the relative phase information that produces Ramsey interference.

Step-by-step dephasing relation

Consider free evolution at angular frequency ω with exponential transverse decay time T_2. The off-diagonal entry obeys

$$\frac{d\rho_{01}}{dt} = -\left(\frac{1}{T_2} + i\omega\right)\rho_{01}.$$

Separating variables gives

$$\frac{d\rho_{01}}{\rho_{01}} = -\left(\frac{1}{T_2} + i\omega\right)dt.$$

Integration from 0 to t gives

$$\ln\left[\frac{\rho_{01}(t)}{\rho_{01}(0)}\right] = -\frac{t}{T_2} - i\omega t,$$

so

$$\rho_{01}(t) = \rho_{01}(0)\exp\left(-\frac{t}{T_2}\right)\exp(-i\omega t).$$

A normalized Ramsey contrast can be written

$$\mathcal{C}(t) = \frac{|\rho_{01}(t)|}{|\rho_{01}(0)|} = \exp\left(-\frac{t}{T_2}\right).$$

The phase factor produces oscillation. The exponential envelope records the loss of usable phase memory.

Interpretation and experimental consequence

The term *quantum coherent* is informative after the state basis, operation duration, measurement, and contrast criterion are stated. A useful task-level condition is

$$\mathcal{C}(t_{\text{task}}) \geq \mathcal{C}_{\text{min}}.$$

This condition connects a physical property to the time required by the experiment. A long quoted T_2 leaves a fast gate sequence well supported and may leave a long transport or memory protocol outside its accepted region. Coherence supports the resource. It leaves utility, acceleration, and advantage for separate tests.

12.2.2 Quantum limited devices and measurements

Physical setting and assumptions

A quantum limit is a lower or upper bound imposed by quantum mechanics for a defined device, observable, resource count, and measurement model. The phrase gains content from the bound that

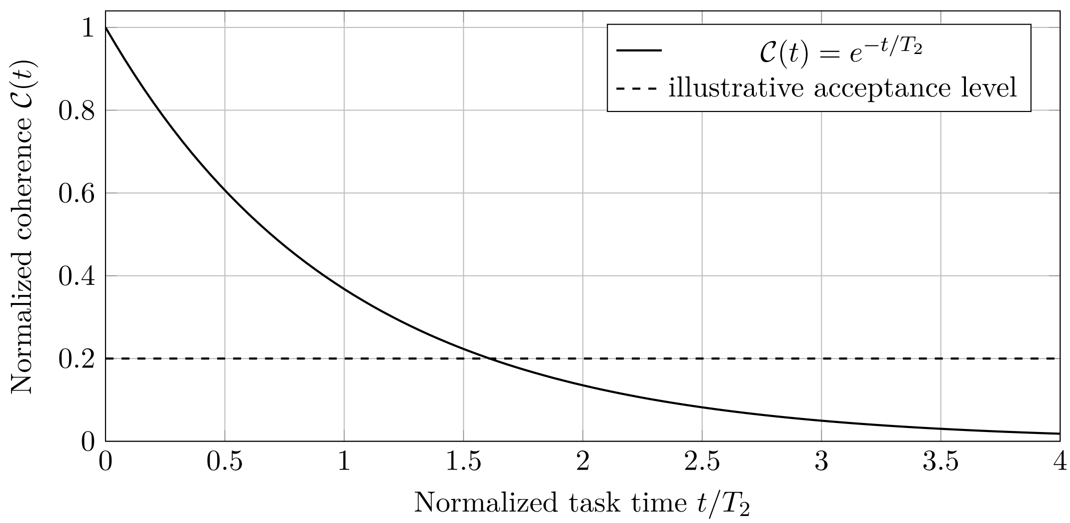

Figure 12.2: Exponential coherence envelope for a simple Markovian dephasing model. The accepted task duration follows from the declared contrast threshold.

follows it. Examples include a minimum estimator variance, minimum input-referred added noise, maximum discrimination fidelity, or minimum backaction compatible with a measurement protocol.

The qualifier *quantum limited* describes proximity to that bound. It carries no universal numerical threshold across amplifiers, interferometers, clocks, force sensors, or qubit readout chains. A selective single-mode and double-mode amplifier offers one device-level example of performance assessed near the applicable quantum noise limit [56].

Quantum Cramér–Rao bound

Let ρ_θ be a quantum state that depends on an unknown parameter θ. Define the symmetric logarithmic derivative L_θ by

$$\frac{\partial \rho_\theta}{\partial \theta} = \frac{1}{2}\left(L_\theta \rho_\theta + \rho_\theta L_\theta\right).$$

The quantum Fisher information is

$$F_Q(\theta) = \mathrm{Tr}\Big(\rho_\theta L_\theta^2\Big).$$

For ν independent preparations and a locally unbiased estimator $\widehat{\theta}$, the quantum Cramér–Rao bound is

$$\mathrm{Var}(\widehat{\theta}) \geq \frac{1}{\nu F_Q(\theta)}.$$

The derivation places the quantum state before the measurement. A chosen measurement has classical Fisher information $F_C \leq F_Q$. Saturation therefore requires a state, measurement, and estimator that approach the same bound under the stated assumptions [57].

A useful proximity ratio is

$$\eta_{\mathrm{QL}} = \frac{1}{\nu F_Q(\theta)\,\mathrm{Var}(\widehat{\theta})}, \qquad 0 < \eta_{\mathrm{QL}} \leq 1.$$

Values near one indicate approach to the selected quantum Cramér–Rao bound. Bias, finite-sample behavior, nuisance parameters, loss, and estimator regularization can require a different bound.

Phase-preserving amplifier limit

A phase-preserving linear amplifier with power gain G must add noise to preserve bosonic commutation relations. In symmetrized noise-quanta units, an input-referred lower bound can be written

$$N_{\mathrm{add}} \geq \frac{1}{2}\left|1-\frac{1}{G}\right|.$$

At large gain,

$$N_{\mathrm{add}} \geq \frac{1}{2}.$$

This half-quantum high-gain limit is the standard reference for an ideal phase-preserving amplifier [58], [59]. A phase-sensitive amplifier follows a different measurement model and can amplify one quadrature with a different noise constraint.

What the bound permits experimentally

A quantum-limit statement should identify

- the input and output variables;
- the gain, bandwidth, operating point, and reference plane;
- the noise convention and calibration chain;
- the resource count and estimator assumptions;
- the measured distance from the selected bound.

The resulting record separates an intrinsic quantum bound from technical noise. It can guide amplifier design, readout-chain allocation, sensing protocols, and metrological resource planning.

12.3 Design and Integration Terms

12.3.1 The common hybrid map

Many applications can be written as

$$\widehat{y} = \mathcal{C}_{\mathrm{post}} \circ \mathcal{M} \circ \mathcal{Q} \circ \mathcal{C}_{\mathrm{pre}}(x).$$

Here x is the original problem input, $\mathcal{C}_{\mathrm{pre}}$ performs classical preparation, $\mathcal{Q}$ is the quantum state preparation or evolution, $\mathcal{M}$ converts the quantum state into classical data, and $\mathcal{C}_{\mathrm{post}}$ reconstructs the requested output. The location and size of $\mathcal{Q}$ define the accelerator boundary.

The integration terms below are document-local conventions. Research communities use some of them with overlapping meanings. The local definitions preserve the distinctions needed by this handbook.

- **Quantum-inspired** describes a classical method whose mathematics or design draws from quantum theory and uses classical hardware during execution. Tensor networks, simulated annealing variants, quantum-inspired low-rank methods, and selected probabilistic models can occupy this class. Practical quantum-inspired linear-algebra studies illustrate the need to state rank, conditioning, sparsity, and data-access assumptions [60].
- **Quantum-aided** describes a hybrid process in which quantum hardware, quantum-generated data, or a quantum measurement improves part of a larger classical activity. The emphasis falls on the additional resource supplied by the quantum component.

- **Quantum assisted** describes a bounded quantum kernel that carries a specific task inside a larger classical workflow. The emphasis falls on workload partitioning and on the input-output contract of the kernel.
- **Quantum-native** describes a problem formulation, data representation, algorithm, or architecture designed from the outset around quantum states, measurements, information primitives, or hardware constraints. Recent database and reinforcement-learning proposals use the term for systems that keep quantum data structures and Hilbert-space operations central to the design [61], [62]. The label describes orientation. Acceleration, utility, and advantage require separate comparisons.
- **Purely quantum** describes a declared data-bearing path in which the input, processing, storage or communication, and output object remain quantum across the chosen system boundary.

12.3.2 Classification of quantum applications

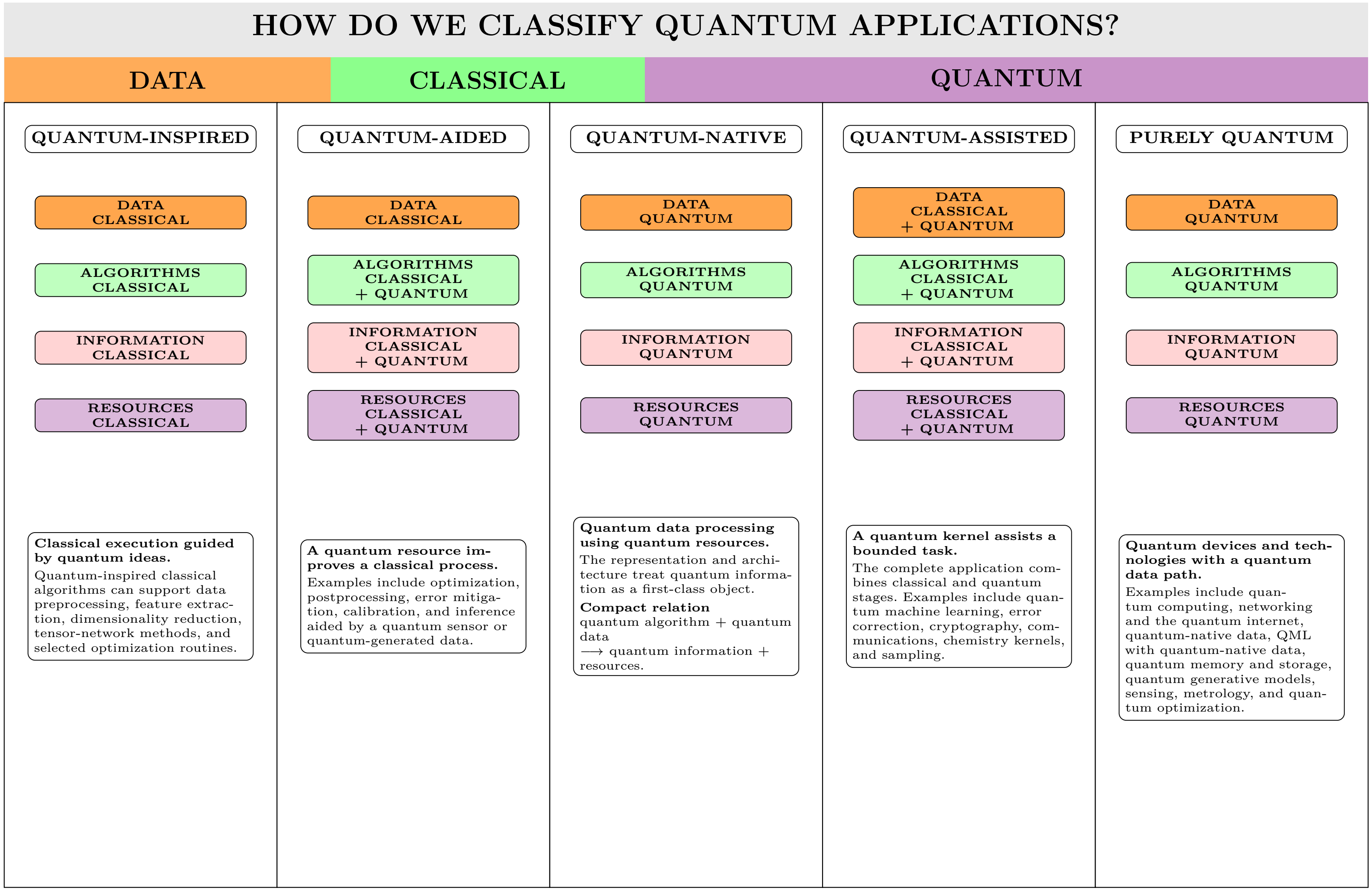

Figure 12.3: The four rows preserve its data, algorithm, information, and resource coordinates, and the text boxes retain its examples.

12.3.3 How the five classes differ

The four rows in figure 12.3 answer a sequence of questions about the data, algorithms, information, and physical resources carried by the application.

- Quantum-inspired applications execute on classical hardware and carries classical data across its stated boundary. Quantum theory supplies mathematical structure or design ideas.
- Quantum-aided applications retain a primarily classical purpose and receives an improvement from a quantum sensor, quantum-generated data set, hardware call, or mitigation resource.
- Quantum-native applications treat quantum data and Hilbert-space operations as first-class design objects from the start.
- Quantum-assisted applications assign a bounded task to a quantum kernel inside a complete classical and quantum workflow.
- Purely quantum applications keep the data-bearing object quantum across the chosen system boundary, such as a networked state, stored mode, processor register, or sensor state before final readout.

The categories can overlap as well. Quantum-native databases can route selected queries to a quantum-assisted kernel. Quantum-aided materials experiments can combine a quantum sensor with a classical inverse problem. Purely quantum networks can contain quantum-native protocols and classical supervisory control.

12.4 Comparative Performance Terms

12.4.1 Quantum-enhanced

A quantum enhanced method uses a declared quantum resource to improve a stated metric against a defined classical or conventional baseline. The resource can be coherence, squeezing, entanglement, interference, quantum statistics, or another physically identified ingredient. The metric can be sensitivity, precision, accuracy, security, accessible regime, or solution quality. Large Fock-state metrology supplies one experimental example in which the quantum state improves the metrological metric [63].

For a metric m that is larger when performance improves, define

$$G_m = \frac{m_Q}{m_C}.$$

For an error ϵ that is smaller when performance improves, define

$$G_\epsilon = \frac{\epsilon_C}{\epsilon_Q}.$$

Quantum enhancement corresponds to $G_m > 1$ or $G_\epsilon > 1$ under the registered comparison. The record should also show that removal of the quantum resource removes the gain or reduces it in the predicted direction.

In metrology, a Fisher-information gain can be written

$$G_F = \frac{F_Q^{(\text{resource})}}{F_C^{(\text{baseline})}}.$$

At matched repetitions ν, the Cramér–Rao scaling gives

$$\frac{\Delta\theta_C}{\Delta\theta_Q} \approx \sqrt{G_F}$$

when both estimators approach their respective bounds.

12.4.2 Quantum accelerated

A quantum accelerated workflow reduces computational complexity, runtime, or time to solution for a task or subroutine against a stated classical method. The acceleration can be mathematical, projected from a resource model, or measured on hardware. These three forms should be labeled separately.

Let

$$T_Q(\epsilon, \alpha) \quad \text{and} \quad T_C(\epsilon, \alpha)$$

be the total times required to meet error ϵ at confidence $1 - \alpha$. The end-to-end acceleration factor is

$$A_T(\epsilon, \alpha) = \frac{T_C(\epsilon, \alpha)}{T_Q(\epsilon, \alpha)}.$$

Measured acceleration requires $A_T > 1$ under matched inputs, outputs, accuracy, confidence, accounting boundary, and comparison date. A kernel-level speedup can coexist with a slower complete workflow if data loading, sampling, queueing, calibration, or reconstruction controls the critical path. Definitions of quantum speedup therefore require a stated classical comparator and problem ensemble [64].

For asymptotic acceleration, suppose

$$T_C(n) = \Theta[f(n)], \qquad T_Q(n) = \Theta[g(n)].$$

The asymptotic ratio is

$$A_{\text{asym}}(n) = \frac{f(n)}{g(n)}.$$

The result gains application meaning after oracle construction, state preparation, precision dependence, fault-tolerant overhead, and output extraction are included.

12.5 Capability Terms: Utility, Advantage, and the Historical Supremacy Milestone

12.5.1 Quantum utility

Quantum utility describes a result that is scientifically or practically useful at a scale and accuracy beyond straightforward brute-force classical treatment. The term permits useful hardware experiments before a comprehensive superiority study against every optimized classical method. The 127-qubit Ising experiment established one influential use of this term and also showed why later classical reassessment is important [65], [66].

Define the scientific acceptance event

$$\mathcal{S}_Q(W) = \{L(\widehat{y}_Q, y_{\text{ref}}) \leq \epsilon_{\text{req}}, \quad V_{\text{checks}} = 1\}.$$

A utility statement needs

$$\Pr[\mathcal{S}_Q(W)] \geq 1 - \alpha$$

and a clear account of how the accepted result changes a scientific inference, design decision, or experimental plan. The statement can precede a full A5 classical-competitiveness assessment from table 15.1.

12.5.2 Quantum advantage

Quantum advantage establishes superior quantum performance over a relevant classical alternative for a specified task and metric under a fair resource comparison. The comparison can concern time, cost, sample quality, accuracy, energy, communication, security, or accessible problem regime. Practical quantum advantage adds application usefulness to the superiority result. Programmable photonic sampling has supplied a prominent quantum computational advantage experiment under a sampling-specific comparison [67].

For a loss metric L and total resource R, two common matched comparisons are

$$L_Q(R) < L_C(R)$$

and

$$R_Q(\epsilon, \alpha) < R_C(\epsilon, \alpha).$$

The first compares quality at equal resource. The second compares resource at equal accepted quality. A strong advantage gives both curves across a range of problem sizes and shows the region in which the ordering stays stable.

12.5.3 Quantum supremacy and quantum computational advantage

Quantum supremacy is the historical term for a well-defined computation completed by a quantum processor at a scale considered practically infeasible for classical computers, with immediate application value left outside the requirement. Random-circuit sampling on a programmable superconducting processor supplied a landmark example [48]. *Quantum computational advantage* is now common for the same general milestone and avoids language that can distract from the scientific comparison.

The term belongs to a narrow computational task. It carries no automatic conclusion about broad economic deployment, general scientific usefulness, or performance on a different workload. The supporting record should include the task distribution, success metric, classical algorithms, hardware, extrapolation method, uncertainty, and comparison date.

Table 12.1: Meaning and required companion record for common quantum descriptors.

Term	What the term establishes	Companion record
Quantum-coherent	Phase relations survive over the declared task interval	Basis, preparation, task time, contrast, T_2 model, and interference measurement
Quantum-limited	Performance approaches a selected quantum bound	Bound, resource count, gain or measurement model, calibration, and distance from the bound
Quantum-inspired	Classical execution uses ideas drawn from quantum theory	Classical hardware, access model, approximation assumptions, and classical comparator
Quantum-aided	A quantum resource improves part of a larger activity	Resource identity, coupling to the classical process, and ablation study
Quantum-assisted	A bounded QPU kernel carries a task inside a classical workflow	Kernel input, output, circuit family, orchestration, and end-to-end ledger

Term	What the term establishes	Companion record
Quantum-native	The representation or architecture is designed around quantum information	Quantum data object, native operations, storage or communication assumptions, and device constraints
Purely quantum	The data-bearing path remains quantum across the chosen boundary	Input state, processing path, memory or network, output object, loss, and readout boundary
Quantum enhanced	A quantum resource improves a stated metric	Resource ablation, baseline, metric, uncertainty, and matched resource count
Quantum-accelerated	Quantum computation reduces time or scaling for a defined task	Classical comparator, equal-accuracy target, complete time ledger, and comparison date
Quantum-utility	The accepted result changes a scientific or practical decision	Scientific tolerance, repeated-job success, reference checks, and decision consequence
Quantum advantage	Quantum performance exceeds a relevant classical method	Matched task, metric, resources, confidence, strongest available baseline, and comparison date
Quantum supremacy	A narrow computation exceeds practical classical reach under the registered benchmark	Task distribution, classical extrapolation, benchmark score, uncertainty, and scope restriction

12.6 A Step-by-Step Qualification Calculation

12.6.1 Physical setting

Consider a hybrid materials workflow that estimates a spectral quantity $s(\omega)$ from a prepared many-body state. A classical stage constructs the Hamiltonian and selects the observable. A QPU prepares the state, evolves it, and estimates time correlations. Classical processing performs a Fourier transform and compares the spectrum with a reference calculation or laboratory measurement.

Let the complete record be

$$\mathcal{L}_{\text{spec}} = (R, I, K, B, M, Q, \tau).$$

Choose

- $R =$ coherent many-body evolution and entangling interactions;
- $I =$ quantum assisted, with a quantum native spin representation inside the kernel;
- $K =$ state preparation, time evolution, and correlation measurement;
- $B =$ a converged matrix-product-state calculation on the same finite model;
- $M =$ spectral loss, total time, shot count, confidence, and accepted frequency window;
- $Q =$ the strongest supported performance term after the calculations below;
- $\tau =$ the acquisition and classical-comparison date.

12.6.2 Coherence check

Suppose the longest useful correlation time is $t_{\max} = 0.6T_2$. The simple dephasing model gives

$$\mathcal{C}(t_{\max}) = e^{-0.6} \approx 0.549.$$

If the registered contrast threshold is 0.50, the task meets its coherence condition. This result supports the label *quantum coherent* for the stated interval.

12.6.3 Enhancement check

Suppose a classical sensor chain reconstructs the spectral peak with standard uncertainty $\sigma_C = 0.080$ and a quantum-resource-assisted chain gives $\sigma_Q = 0.050$ at the same acquisition time. The uncertainty gain is

$$G_\sigma = \frac{\sigma_C}{\sigma_Q} = 1.6.$$

If an ablation run removes the quantum resource and returns the uncertainty near 0.080, the result supports quantum enhancement for this metric.

12.6.4 Acceleration check

Suppose the complete quantum-assisted workflow requires

$$T_Q = 7.5 \text{ h},$$

and the matched classical workflow requires

$$T_C = 5.0 \text{ h}.$$

Then

$$A_T = \frac{5.0}{7.5} \approx 0.67.$$

The experiment supports enhancement and leaves measured acceleration unsupported. This combination is scientifically coherent.

12.6.5 Utility and advantage checks

Suppose the spectral loss satisfies

$$L_Q = 0.018 < \epsilon_{\text{req}} = 0.025$$

across repeated sessions and the result changes the selected material parameter range for the next neutron-scattering experiment. The workflow supports quantum utility under the registered scientific contract.

Suppose the classical workflow also satisfies the tolerance with lower cost and shorter time. The current record leaves quantum advantage unsupported. A later experiment can revisit the comparison after hardware, algorithms, and classical baselines change.

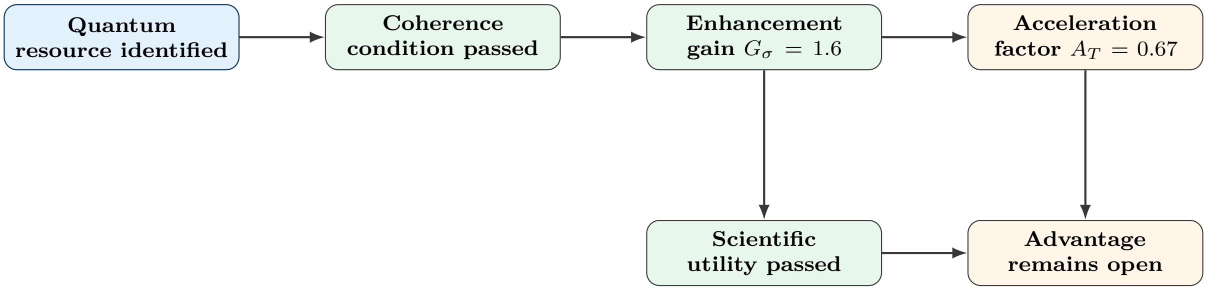

Figure 12.4: A workflow can pass coherence, enhancement, and utility tests and still leave acceleration and advantage open. The terms describe separate coordinates.

12.7 A Reusable Language Record

A publication, engineering review, or technology roadmap can use the following sequence.

1. Identify the physical quantum resource R and the experiment that measures it.
2. State the integration class I and draw the boundary of the quantum kernel K.
3. Register the baseline B, metric M, acceptance tolerance, confidence level, and resource boundary.
4. Report physical descriptors such as coherent or quantum limited with the corresponding bound or decay model.
5. Report comparative descriptors such as enhanced or accelerated with ratios and uncertainty.
6. Report capability descriptors such as utility or advantage after the complete workflow passes its registered test.
7. Record the comparison date τ to preserve the classical algorithms, quantum hardware, and service conditions used in the comparison.

Descriptor sentence template

For workload W, the system uses quantum resource R in integration class I. The quantum kernel K receives the stated input and returns the stated output. Against baseline B, the workflow achieved metric M with the reported uncertainty and resource ledger. These results support the descriptors listed in Q over the registered problem range and comparison date τ.

Check your understanding

1. A sensor reaches the quantum Cramér–Rao bound and has lower sensitivity than a competing classical sensor with more probes. Which descriptor is supported by the first fact, and which comparison is still required for quantum enhancement?
2. A quantum native algorithm runs on a simulator and returns a useful result. State which design term is supported and which hardware-performance terms are still open.
3. A QPU kernel is faster than its classical replacement and the complete workflow is slower. Calculate the two time ratios that should appear in the report and identify the supported acceleration statement.
4. Select one panel in figure 12.3. Redraw its data path for a real application and identify the point at which quantum information becomes classical data.

Key takeaway

Quantum coherent and quantum limited describe physical behavior. Quantum inspired, aided, native, assisted, and purely quantum describe design and integration. Quantum enhanced and quantum accelerated describe comparative gains. Quantum utility and quantum advantage describe the capability of a complete result under a registered scientific and resource contract. The terms can coexist and they should remain distinct.

Practical rule of thumb

Attach every quantum adjective to the physical resource, system boundary, baseline, and measured metric. Use the strongest term supported by the complete record and leave the remaining distinctions open.

Chapter 13

Hardware Measurement and Validation Chains

This chapter follows a microwave-controlled superconducting processor from digital instructions to classified readout and identifies the records needed to trace hardware faults into a reported estimator.

13.1 Control Electronics and Device Diagnostics

Prerequisite

Before reading this chapter, ensure that microwave amplitude and phase, digitization, conditional assignment probability, and the difference between a logical gate and its calibrated pulse implementation are familiar.

A logical circuit reaches a superconducting chip as timed microwave and flux controls. The return signal passes through cryogenic and room-temperature electronics before a classifier produces a bitstring. Other platforms use different transducers and detectors. The same systems principle applies across them: the reported quantum result includes the classical control and measurement path.

The forward path contains a controller, sequencer, digital-to-analog conversion, local oscillators, mixers, attenuators, filters, cables, and cryogenic components. The return path contains resonators or sensors, isolators, cryogenic amplification, room-temperature amplification, analog-to-digital conversion, demodulation, integration, and classification. Reviews of superconducting-qubit engineering and fabrication describe these coupled layers in detail [68], [69], [70].

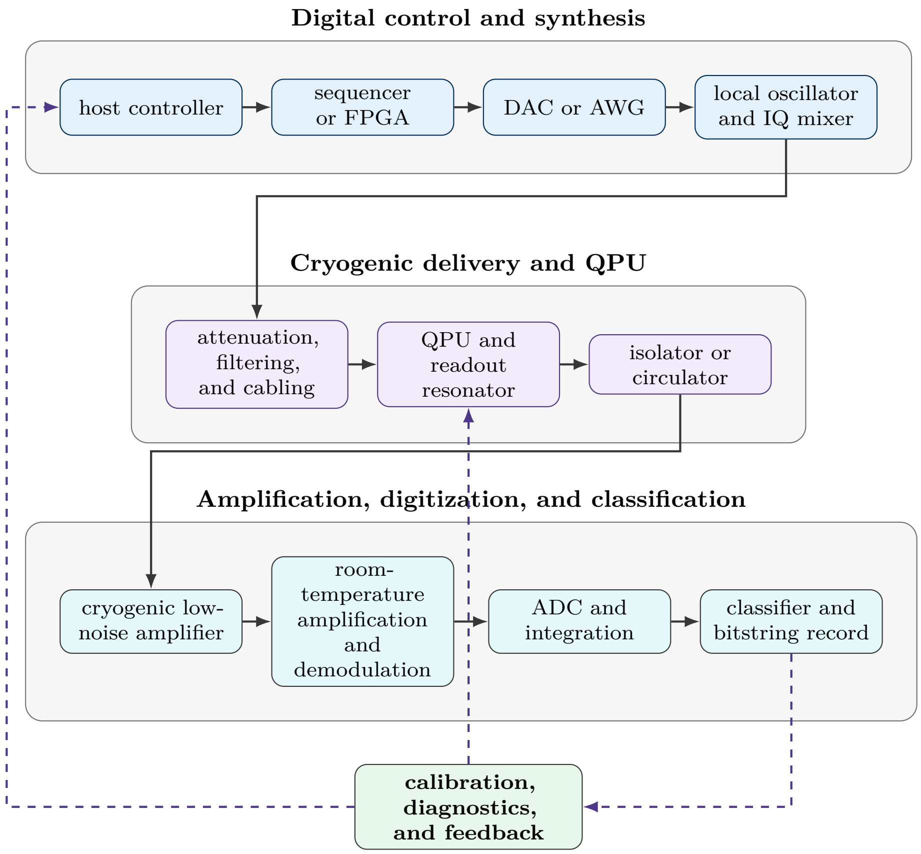

Figure 13.1: Grouped control and readout chain for a microwave-controlled superconducting processor. Digital synthesis drives the cryogenic delivery path, the return signal is amplified and digitized, and classified data feed calibration and control updates. Platform-specific hardware can differ.

13.1.1 How hardware faults enter the result

Several hardware effects can resemble a qubit error.

- Clock misalignment changes pulse phases and frame relationships.
- Converter nonlinearity and mixer imbalance create harmonics and image tones.
- Cable reflections and filter dispersion distort pulse envelopes.
- Thermal radiation raises residual excitation.
- Amplifier compression changes readout distributions.
- Classifier drift changes assignment probabilities without changing the quantum state.

Verification therefore records the control configuration and calibration state with the circuit data.

Routine validation tracks energy relaxation T_1, Ramsey and echo coherence, residual excitation, leakage, gate error, simultaneous-operation effects, readout assignment, resonator collisions, crosstalk, and drift. Each quantity has its own protocol and uncertainty. A quoted two-qubit fidelity can come from RB, interleaved RB, cycle benchmarking, process tomography, or another method, and the values can differ.

13.1.2 Minimum reproducibility record

A reproducible run should include

- processor identifier, qubit map, and native-gate definitions;
- compiler version, optimization settings, and pulse-calibration version;
- acquisition timestamps, shot order, and random seeds;
- measurement level, classifier version, and raw analog or count data;
- mitigation settings, calibration circuits, and discarded outcomes;
- sentinel-circuit time series that reveal nonstationarity.

Practical rule of thumb

Treat the calibration snapshot as part of the data set. Raw counts need their qubit map, compiler settings, pulse version, classifier, and acquisition time to support a reproducible verification result.

Chapter 14

Advanced Numerical Solvers, Ground Truth, and Open-System Dynamics

How the argument develops

This chapter develops a numerical ground-truth record from the model outward.

14.1 Primer: What Ground Truth Means in a Numerical Quantum Study

Prerequisite

Before reading this section, ensure that you are comfortable with a density matrix ρ, an eigenvalue equation, the variational principle, a confidence interval, and the distinction between a finite basis and the physical continuum.

A quantum processor and a classical solver should be compared inside the same scientific statement. That statement identifies the Hamiltonian, particle number, geometry, boundary conditions, basis or lattice, symmetry sector, observable, and requested tolerance. A numerical value acquires reference status after these choices and the unresolved errors are recorded.

Let $\mathcal{M}$ denote the declared mathematical model. Let $O_{\mathcal{M}}$ be the exact value of observable O inside that model, and let O_{phys} be the corresponding value for the physical system. A numerical solver returns $\widehat{O}_s$. The total difference can be written

$$\widehat{O}_s - O_{\text{phys}} = \underbrace{\widehat{O}_s - O_{\mathcal{M}}}_{\text{numerical difference}} + \underbrace{O_{\mathcal{M}} - O_{\text{phys}}}_{\text{model discrepancy}} .$$

The first term belongs to numerical analysis. The second term belongs to model construction and comparison with experiment. A solver can drive the first term below a chosen tolerance and leave the second term unchanged.

For bookkeeping, decompose the numerical contribution as

$$\delta_{\text{num}} = \delta_{\text{basis}} + \delta_{\text{size}} + \delta_{\text{time}} + \delta_{\text{trunc}} + \delta_{\text{stat}} + \delta_{\text{constr}} + \delta_{\text{fp}}.$$

The terms represent basis incompleteness, finite-size effects, timestep or imaginary-time discretization, representation truncation, Monte Carlo sampling, imposed sign or phase constraints, and floating-point effects. The sum is an accounting convention. Several contributions can couple, so a refined study may require joint sweeps or covariance terms.

A useful numerical ground-truth record is

$$\mathcal{G} = \left(\mathcal{M}, O, \boldsymbol{\lambda}, \widehat{O}_{\text{ref}}, \Delta_{\text{ref}}, \mathcal{P}\right).$$

Here $\boldsymbol{\lambda}$ contains every convergence control, $\widehat{O}_{\text{ref}}$ is the selected reference value, Δ_{ref} is its numerical uncertainty or bound, and $\mathcal{P}$ stores provenance such as code version, hardware, input files, random seeds, and solver settings. The pair

$$\mathcal{I}_{\text{ref}} = \left[\widehat{O}_{\text{ref}} - \Delta_{\text{ref}}, \widehat{O}_{\text{ref}} + \Delta_{\text{ref}}\right]$$

forms the reference interval used in the hardware comparison.

The phrase **ground truth** therefore means a converged and auditable answer for a declared model and observable. It carries no assertion of an exact description of nature. In this chapter, the term always carries the model, convergence controls, uncertainty, and provenance record with it.

14.1.1 Terminology used throughout the chapter

- **Model space** is the finite or discretized Hilbert space in which the solver works. Examples include a spin lattice, a Gaussian orbital basis, a plane-wave cutoff, an active orbital space, or an oscillator truncation.
- **Reference solver** is a method whose residual numerical difference is smaller than the comparison tolerance for the stated observable.
- **Convergence control** is an adjustable quantity whose refinement approaches a limiting calculation. Examples include basis size, lattice length, bond dimension, timestep, walker population, sample count, and coupled-cluster excitation rank.
- **Residual** measures failure to satisfy the governing equation. Eigenvalue, linear-equation, self-consistency, and steady-state residuals are common examples.
- **Constraint bias** is the change introduced by a stabilizing approximation such as a fixed-node, constrained-path, or phaseless condition.
- **Cross-method agreement** means that solvers with different representations and failure modes converge to compatible intervals for the same model and observable.

Key takeaway

A method name alone supplies no ground truth. The reference object is the full record $\mathcal{G}$, which joins the model, observable, convergence controls, reference interval, and computational provenance.

14.2 Universal Diagnostics for Convergence

Prerequisite

Before reading this section, ensure that vector norms, expectation values, variance, asymptotic error expansions, and autocorrelation are familiar.

The solvers in this chapter use different representations, yet several checks recur across all of them. These checks provide a common language for deciding when a result is ready to serve as a reference.

14.2.1 Eigenvalue residuals

Let H be Hermitian, let $\left|\widetilde{\psi}\right\rangle$ be normalized, and let $\widetilde{E}$ be an approximate eigenvalue. Define

$$|r\rangle = \left(H - \widetilde{E}\right)\left|\widetilde{\psi}\right\rangle.$$

The residual norm

$$\eta_r = \|r\|_2$$

measures direct failure of the eigenvalue equation. If $\widetilde{E} = \langle\widetilde{\psi}|H|\widetilde{\psi}\rangle$, then the distance from $\widetilde{E}$ to the spectrum of H obeys

$$\operatorname{dist}\left(\widetilde{E}, \operatorname{spec}(H)\right) \leq \eta_r.$$

A small residual therefore places the approximate energy near some exact eigenvalue of the finite matrix. Symmetry labels, overlap tracking, and state ordering identify which eigenvalue has been reached.

14.2.2 Energy variance

For a normalized state $|\psi\rangle$, define

$$E_\psi = \langle\psi|H|\psi\rangle$$

and

$$\sigma_H^2 = \langle\psi|H^2|\psi\rangle - E_\psi^2.$$

Since

$$\sigma_H^2 = \|(H - E_\psi)\,|\psi\rangle\|_2^2,$$

the energy variance is the squared residual evaluated at the expectation value. It vanishes for an exact eigenstate or a state contained inside a degenerate eigenspace. DMRG, variational Monte Carlo, variational quantum eigensolvers, and neural-state calculations can all use this diagnostic.

A sequence of approximate states can also support a variance extrapolation,

$$E(\sigma_H^2) = E_0 + a\sigma_H^2 + \mathcal{O}\Big((\sigma_H^2)^2\Big),$$

provided the sequence approaches the same eigenstate smoothly. The intercept supplies an additional reference estimate, and sensitivity to the fitted range becomes part of Δ_{ref}.

14.2.3 The variational upper bound

Expand a normalized trial state in exact eigenstates,

$$|\psi\rangle = \sum_n c_n\,|E_n\rangle\,, \qquad \sum_n |c_n|^2 = 1, \qquad E_0 \leq E_1 \leq \cdots.$$

Its energy is

$$E[\psi] = \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle} = \sum_n |c_n|^2 E_n \geq E_0.$$

This inequality gives a one-sided check for ground-state calculations that stay variational in the declared space. Full configuration interaction, variational MPS optimization, MCSCF, and variational Monte Carlo inherit this property. CCSD(T), phaseless AFQMC, and many corrected or extrapolated estimators lack a general upper-bound property, so they require different checks.

14.2.4 Refinement and extrapolation

Suppose a discretization control h follows

$$O(h) = O_0 + ah^p + \mathcal{O}(h^{p+1}).$$

For a refinement factor $s > 1$, two calculations at h and h/s give the Richardson estimate

$$O_0 \approx \frac{s^p O(h/s) - O(h)}{s^p - 1}.$$

The observed order can be estimated from three resolutions. For h, h/s, and h/s^2,

$$p_{\text{obs}} = \frac{\ln\left|\dfrac{O(h) - O(h/s)}{O(h/s) - O(h/s^2)}\right|}{\ln s}.$$

Agreement between p_{obs} and the theoretical order supports the asymptotic regime. An irregular trend signals an unresolved scale, state crossing, optimizer failure, sampling fluctuation, or coupling between convergence controls.

14.2.5 Stochastic uncertainty and effective sample size

Let O_k be a stationary Monte Carlo sequence with variance σ_O^2 and lag-ℓ autocorrelation ρ_ℓ. Define

$$\tau_{\text{int}} = 1 + 2\sum_{\ell=1}^{\infty} \rho_\ell, \qquad N_{\text{eff}} = \frac{N}{\tau_{\text{int}}}.$$

The standard uncertainty of the mean is approximately

$$u_{\text{stat}} = \sqrt{\frac{\sigma_O^2}{N_{\text{eff}}}}.$$

Blocking analysis, independent replicas, and chain-length studies estimate this quantity. Walker branching and population control can introduce additional correlation, so the raw sample count should never replace N_{eff} in the uncertainty record.

14.2.6 Agreement between independent methods

For independent reference estimates $\widehat{O}_A$ and $\widehat{O}_B$ with standard uncertainties u_A and u_B, define

$$z_{AB} = \frac{\widehat{O}_A - \widehat{O}_B}{\sqrt{u_A^2 + u_B^2}}.$$

Shared samples or common model parameters add a covariance term,

$$u_{A-B}^2 = u_A^2 + u_B^2 - 2\,\mathrm{Cov}\left(\widehat{O}_A, \widehat{O}_B\right).$$

A small standardized difference shows compatibility. Its scientific force is strongest when the methods fail in different ways, such as FCI versus AFQMC, DMRG versus ED, or CCSD(T) versus MCSCF near a geometry where the single-reference picture may weaken.

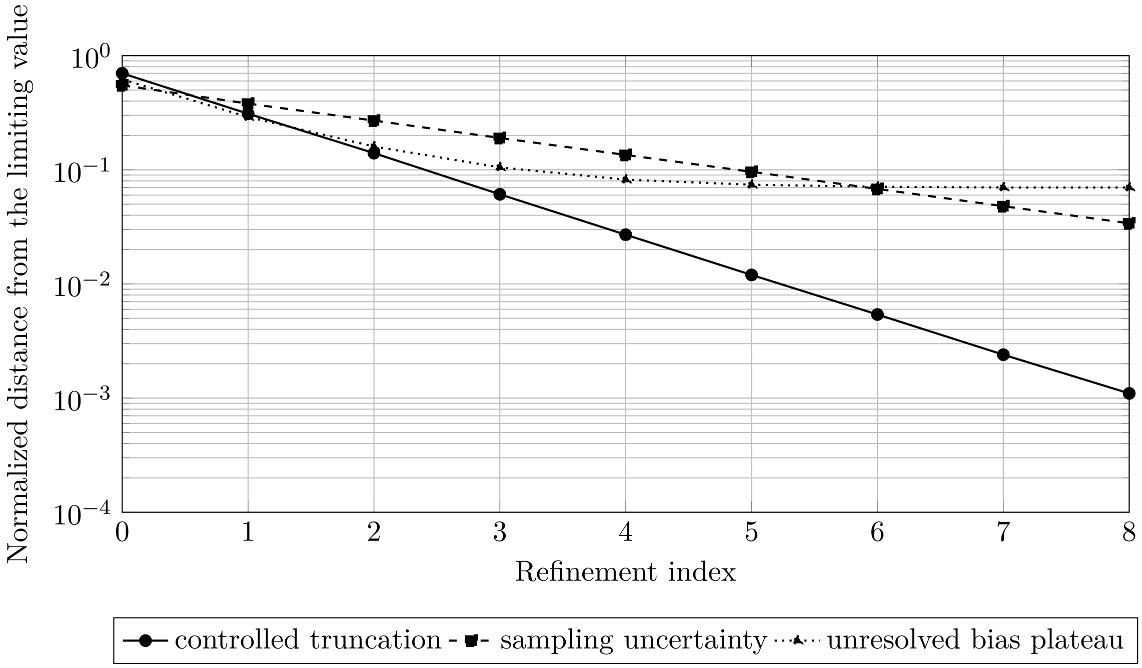

Figure 14.1: Schematic convergence patterns. A controlled truncation decreases toward the limiting value, stochastic uncertainty decreases more slowly, and an unresolved systematic contribution produces a plateau. The curves are pedagogical and carry no solver-specific fit.

Practical rule of thumb

Use at least one equation-based diagnostic, one refinement study, and one independent comparison. A stable sequence with a large residual, an unknown autocorrelation time, or an unchanged constraint can still carry a hidden error.

14.3 Exact Diagonalization and Full Configuration Interaction

14.3.1 Exact diagonalization

Physical setting and assumptions

Exact diagonalization (ED) applies to a Hamiltonian represented as a finite Hermitian matrix. The basis can describe spins, fermionic occupations, bosonic modes with cutoffs, lattice gauge variables, or a compiled quantum circuit mapped to an effective Hamiltonian. Symmetries can divide the full Hilbert space into smaller sectors with fixed particle number, momentum, parity, or total spin.

Let the selected sector have dimension D and basis states $\{|i\rangle\}_{i=1}^{D}$. The state is

$$|\psi_n\rangle = \sum_{i=1}^{D} c_i^{(n)} |i\rangle ,$$

and the eigenproblem is

$$H\boldsymbol{c}^{(n)} = E_n \boldsymbol{c}^{(n)}.$$

Dense diagonalization returns the full spectrum at cost that grows steeply with D. Sparse Krylov methods, including the Lanczos method, target extremal eigenpairs using repeated matrix-vector products [71].

Governing relation and convergence controls

For an eigenpair produced by an iterative solver, use

$$\eta_n = \left\| H\boldsymbol{c}^{(n)} - E_n\boldsymbol{c}^{(n)} \right\|_2 .$$

The iterative solve can reach machine-level residuals inside the finite matrix. Ground truth for the physical model still requires refinement of any model-space controls,

$$\boldsymbol{\lambda}_{\mathrm{ED}} = (D, N_{\mathrm{site}}, N_{\mathrm{boson}}, E_{\mathrm{cutoff}}, \text{boundary condition}, \text{precision})\,.$$

Finite-size scaling can use a relation such as

$$O(L) = O_\infty + aL^{-p} + \cdots$$

for system length L, provided the expected scaling form is justified for the phase and boundary conditions under study.

Interpretation and experimental use

ED supplies the strongest finite-space anchor in this chapter. It gives eigenenergies, gaps, eigenvectors, reduced density matrices, correlation functions, transition matrix elements, and exact time evolution for the selected finite model. Quantum hardware studies can use ED to check small instances, circuit prefixes, symmetry sectors, local observables, and mitigation training circuits. A discrepancy with ED identifies a problem in the quantum implementation, estimator, or model alignment after basis conventions and compilation have been matched.

The phrase *exact diagonalization* refers to the declared finite matrix. Bosonic cutoffs, orbital truncations, finite clusters, omitted interactions, and parameter uncertainty stay separate.

14.3.2 Full configuration interaction

Physical setting and assumptions

Full configuration interaction (FCI) is exact diagonalization specialized to the electronic Hamiltonian in a finite one-particle basis. Under the Born–Oppenheimer approximation and a chosen orbital basis,

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2}\sum_{pqrs} \langle pq\|rs\rangle a_p^\dagger a_q^\dagger a_s a_r .$$

The one-electron integrals h_{pq} contain kinetic and electron-nuclear terms. The antisymmetrized two-electron integrals $\langle pq\|rs\rangle$ contain Coulomb and exchange interactions.

The FCI state spans every determinant compatible with the chosen electron number and spin sector,

$$|\Psi_{\mathrm{FCI}}\rangle = \sum_I C_I\,|\Phi_I\rangle\,, \qquad \boldsymbol{H}\boldsymbol{C} = E_{\mathrm{FCI}}\boldsymbol{C}.$$

For M spatial orbitals with N_α spin-up and N_β spin-down electrons, the determinant count is

$$D_{\mathrm{FCI}} = \binom{M}{N_\alpha}\binom{M}{N_\beta}.$$

This combinatorial growth limits direct FCI to compact orbital spaces. Determinant-based algorithms provide efficient matrix-vector operations without storing the dense Hamiltonian [72], [73].

Reference status and remaining controls

FCI solves the electronic Schrödinger equation exactly inside the selected one-particle basis and Hamiltonian. Its residual can be driven to a small value,

$$\eta_{\mathrm{FCI}} = \|\boldsymbol{H}\boldsymbol{C} - E_{\mathrm{FCI}}\boldsymbol{C}\|_2.$$

The remaining controls include

- orbital basis family and cardinal number;
- frozen-core or effective-core treatment;
- relativistic and spin-orbit terms;
- molecular geometry and nuclear model;
- integral thresholds and numerical precision.

FCI in a small basis can be numerically tighter than a larger-basis approximate method and still have a larger basis-set error. The comparison record should separate these two facts.

What FCI permits experimentally

FCI supplies exact finite-basis energies, reduced density matrices, natural orbital occupations, transition amplitudes, and determinant weights. These quantities can benchmark variational quantum eigensolvers, quantum phase-estimation prototypes, active-space encodings, measurement reconstruction, and state-preparation protocols. FCI is especially useful for small molecules, embedded fragments, and reduced active spaces whose determinant counts are still tractable.

Key takeaway

ED and FCI answer the finite model directly. Their strongest role is to anchor small instances and expose implementation errors. Their principal limitation is the exponential or combinatorial growth of the model space.

14.4 Density Matrix Renormalization Group and Time-Evolving Block Decimation

14.4.1 Density matrix renormalization group

Physical setting and assumptions

Density matrix renormalization group (DMRG) targets ground states and selected excited states whose entanglement can be represented by a matrix product state. It is especially effective for one-dimensional local Hamiltonians, long cylinders of moderate circumference, quantum-chemistry active spaces with a favorable orbital ordering, and other problems whose Schmidt spectra decay sufficiently fast [74], [75], [76].

Write an MPS as

$$|\Psi(\{A\})\rangle = \sum_{i_1,\ldots,i_n} A^{[1]i_1} A^{[2]i_2} \cdots A^{[n]i_n} |i_1 \cdots i_n\rangle .$$

The internal indices have maximum dimension χ. DMRG optimizes one or two tensors at a time and sweeps across the chain.

Governing variational relation

At fixed bond dimension, the ideal target is

$$E_\chi = \min_{\Psi \in \mathrm{MPS}(\chi)} \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle}.$$

For a ground-state calculation in a fixed symmetry sector,

$$E_\chi \geq E_0,$$

and the accessible variational space grows as χ increases. A converged sequence often satisfies

$$E_{\chi_2} \leq E_{\chi_1} \qquad (\chi_2 > \chi_1),$$

provided each optimization reaches a comparable minimum.

The discarded weight at a truncation is

$$w_{\mathrm{disc}} = \sum_{a > \chi} \lambda_a^2,$$

where λ_a are Schmidt coefficients. This local quantity helps diagnose compression, and the final observable still requires direct convergence in χ.

Convergence controls and failure signatures

A DMRG ground-truth study varies

$$\boldsymbol{\lambda}_{\mathrm{DMRG}} = (\chi, N_{\mathrm{sweep}}, \epsilon_{\mathrm{SVD}}, \text{orbital or site ordering}, \text{initial state}, L)\,.$$

Useful checks include

- energy change between sweeps;
- energy variance σ_H^2;
- discarded weight and maximum bond dimension;
- stability across initial states and sweep schedules;
- finite-size and boundary-condition scaling;
- agreement with ED on smaller systems.

A low discarded weight can coexist with a biased observable after many truncations. A local minimum can produce smooth sweep convergence. An unfavorable site ordering can force large bond dimensions. The convergence record should therefore combine several diagnostics.

What DMRG permits experimentally

DMRG can provide local energies, spin and density correlations, entanglement measures, reduced density matrices, spectral functions, and active-space electronic energies. These outputs support quantum simulators of spin chains, Hubbard models, correlated orbitals, and neutron-scattering observables. DMRG also gives converged initial states and short-time references for hardware dynamics before entanglement growth raises the required χ.

14.4.2 Time-evolving block decimation

Physical setting and symbols

Time-evolving block decimation (TEBD) propagates an MPS under a local one-dimensional Hamiltonian. Split the Hamiltonian into commuting sets of local terms,

$$H = H_{\text{even}} + H_{\text{odd}}.$$

For a timestep Δt, the second-order symmetric product formula is

$$U_2(\Delta t) = e^{-iH_{\text{even}}\Delta t/2} e^{-iH_{\text{odd}}\Delta t} e^{-iH_{\text{even}}\Delta t/2} + \mathcal{O}(\Delta t^3)$$

for one step [77]. Over fixed total time t, the accumulated product-formula contribution scales as $\mathcal{O}(t\Delta t^2)$ under the usual bounded-commutator assumptions.

After each local gate, TEBD factorizes the updated tensor and truncates the Schmidt spectrum. The numerical result therefore carries two primary approximations,

$$\delta_O(t) = \delta_{\text{Trotter}}(\Delta t, t) + \delta_{\text{MPS}}(\chi, \epsilon_{\text{SVD}}, t).$$

Double convergence study

A reliable TEBD reference refines both axes. One practical sequence is

1. fix a large χ and reduce Δt until the observable stabilizes;
2. fix the selected Δt and increase χ or reduce the truncation threshold;
3. repeat the timestep study at the larger χ;
4. compare short times with ED and check conserved quantities throughout the evolution.

The norm, energy for time-independent H, particle number, and symmetry sectors provide inexpensive diagnostics. Long-time entanglement growth can cause a rapid rise in χ. A plateau in memory or a rising discarded weight then marks the loss of the declared tolerance.

What TEBD permits experimentally

TEBD can benchmark quench dynamics, local correlators, lightcone spreading, response functions, and short-to-intermediate-time spectra in one-dimensional systems. Quantum hardware experiments can compare time traces before Fourier transformation, which separates dynamical error from windowing and spectral reconstruction.

Key takeaway

DMRG controls a variational state space with bond dimension χ. TEBD adds timestep error to the same entanglement-controlled representation. Ground truth requires convergence in the observable across every active control.

14.5 Variational, Diffusion, and Path-Integral Monte Carlo

14.5.1 Variational Monte Carlo

Physical setting and assumptions

Variational Monte Carlo (VMC) evaluates and optimizes a parameterized many-body wavefunction in configuration space. Let R collect all particle coordinates and spin labels. A trial state $\Psi_T(R;\boldsymbol{\theta})$ defines the probability density

$$\pi(R) = \frac{|\Psi_T(R)|^2}{\int dR\, |\Psi_T(R)|^2}.$$

The local energy is

$$E_L(R) = \frac{H\Psi_T(R)}{\Psi_T(R)}.$$

Sampling $R \sim \pi$ gives

$$E_{\mathrm{VMC}} = \int dR\, \pi(R) E_L(R) = \frac{\langle \Psi_T|H|\Psi_T\rangle}{\langle \Psi_T|\Psi_T\rangle} \geq E_0.$$

Monte Carlo evaluation of variational wavefunctions has a long history in quantum fluids and many-body physics [78], [79].

Variance and optimization

The local-energy variance is

$$\sigma_{E_L}^2 = \left\langle (E_L - E_{\mathrm{VMC}})^2 \right\rangle_\pi .$$

It vanishes when Ψ_T is an exact eigenstate in every sampled region. Energy and variance therefore provide complementary optimization targets. Modern trial states can use Jastrow factors, Slater determinants, backflow, tensor networks, and neural networks.

The controls include

- ansatz class and parameter count;
- optimizer, regularization, and stopping rule;
- Markov-chain burn-in, length, and autocorrelation;
- number of independent replicas;
- finite cell, boundary conditions, basis, and pseudopotential.

An optimizer can settle in a local basin. A low sampling error then describes the chosen basin accurately and leaves ansatz bias unresolved. Energy variance, restarts, larger ansatz families, and comparison with projector methods address this issue.

What VMC permits experimentally

VMC supplies variational energies, pair correlations, densities, trial nodes, and flexible many-body wavefunctions. It can benchmark quantum-state ansatzes, initialize DMC, define trial states for AFQMC, and test whether a quantum-prepared state improves a classical variational family.

14.5.2 Diffusion Monte Carlo

Imaginary-time projection

Diffusion Monte Carlo (DMC) filters excited-state components by propagating in imaginary time. With a reference energy E_T,

$$-\frac{\partial \Psi(R,\tau)}{\partial \tau} = (H - E_T)\,\Psi(R,\tau),$$

which has the formal solution

$$|\Psi(\tau)\rangle = e^{-\tau(H-E_T)}\,|\Psi(0)\rangle\,.$$

If the initial state overlaps the ground state, higher-energy components decay relative to the ground component. Random walks, diffusion, drift from importance sampling, and branching implement the short-time propagator [79], [80].

Fermionic nodes and systematic controls

For fermions in real space, sign changes create an unstable cancellation problem. Fixed-node DMC holds the nodal surface to that of a trial state. The resulting fixed-node energy obeys a variational upper bound for the chosen nodes. Its difference from the exact fermionic energy is the fixed-node bias [79], [81].

A DMC convergence study varies

$$\boldsymbol{\lambda}_{\mathrm{DMC}} = (\Delta\tau, N_{\mathrm{walker}}, \tau_{\mathrm{proj}}, \Psi_T, L, \text{pseudopotential treatment})\,.$$

The expected short-time relation can be fitted as

$$E(\Delta\tau) = E(0) + a\Delta\tau^p + \cdots\,,$$

where p follows from the propagator and implementation. Population-control studies repeat the run with larger walker ensembles. Trial-node studies compare several optimized wavefunctions. Finite-size corrections and twist averaging address periodic cells.

For observables that fail to commute with H, the mixed estimator can carry trial-state dependence. Pure estimators, extrapolated estimators, or forward walking are then required, with their own convergence controls.

What DMC permits experimentally

DMC can provide accurate continuum ground-state energies, energy differences, structural trends, and selected observables for atoms, molecules, quantum fluids, and solids. It supplies a representation very different from orbital FCI or tensor networks, which makes it valuable for cross-checking quantum-chemistry and materials calculations.

14.5.3 Path-integral Monte Carlo

Finite-temperature setting

Path-integral Monte Carlo (PIMC) targets the thermal density operator and partition function,

$$\rho_\beta = \frac{e^{-\beta H}}{Z}, \qquad Z = \mathrm{Tr}\Big(e^{-\beta H}\Big)\,, \qquad \beta = \frac{1}{k_B T}.$$

Split the inverse temperature into M slices,

$$\Delta\tau = \frac{\beta}{M}.$$

For $H = T + V$, a symmetric factorization gives

$$e^{-\Delta\tau(T+V)} = e^{-\Delta\tau V/2} e^{-\Delta\tau T} e^{-\Delta\tau V/2} + \mathcal{O}(\Delta\tau^3).$$

The complete path contains M imaginary-time replicas, often called beads. The continuum limit is approached as M increases and $\Delta\tau$ decreases [82].

Convergence controls and the fermion sign problem

PIMC controls include

- imaginary-time step or bead count;
- system size and boundary conditions;
- Monte Carlo chain length and autocorrelation;
- permutation sampling for identical particles;
- pair-action or higher-order propagator approximations;
- temperature and finite-size extrapolation.

For fermions, positive and negative exchange contributions can cancel. The average sign can decrease rapidly with particle number or inverse temperature, causing an exponential growth in sampling cost. Restricted-path or fixed-node constructions control this growth and introduce a nodal constraint.

What PIMC permits experimentally

PIMC provides finite-temperature energies, heat capacities, superfluid fractions, pair correlations, density distributions, and equilibrium response functions. It is a natural reference for quantum simulators that target thermal equilibrium, quantum fluids, bosonic lattices, and distinguishable-particle models.

Key takeaway

VMC is variational and ansatz limited. DMC projects toward a ground state and carries timestep, population, and nodal controls. PIMC represents thermal equilibrium and carries imaginary-time and sign controls. Each Monte Carlo result needs an effective sample size and an explicit systematic-error study.

14.6 Auxiliary-Field Quantum Monte Carlo

14.6.1 Physical setting and determinant-space projection

Auxiliary-field quantum Monte Carlo (AFQMC) projects a many-fermion state in imaginary time and samples a stochastic ensemble of Slater determinants. It is used for lattice Hamiltonians, molecules, and periodic solids. The method begins from

$$|\Psi(\tau)\rangle = e^{-\tau(H-E_T)} |\Psi_I\rangle .$$

A Trotter factorization separates one-body and two-body pieces. After expressing the two-body interaction as a sum of squares, a Hubbard–Stratonovich transformation converts the propagator into an integral over auxiliary fields. Schematically,

$$e^{\frac{1}{2}\Delta\tau \hat{v}^2} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, e^{-x^2/2} e^{\sqrt{\Delta\tau}\, x \hat{v}}.$$

Each sampled field applies a one-body propagator, so a Slater determinant remains a Slater determinant after every step.

14.6.2 Sign, phase, and trial-state constraints

Free projection is exact in the finite basis after timestep, population, and projection convergence. Fermionic signs or complex phases produce rapidly growing variance. Constrained-path AFQMC controls real sign changes with a trial-state overlap condition [83]. Phaseless AFQMC extends the idea to complex overlaps and phase rotations [84]. The trial state stabilizes the calculation and introduces a trial-state-dependent constraint bias. Phaseless AFQMC lacks a general variational upper-bound guarantee.

The principal controls are

$$\boldsymbol{\lambda}_{\mathrm{AFQMC}} = (\Delta\tau, N_{\mathrm{walker}}, \tau_{\mathrm{proj}}, \Psi_T, \text{population-control interval}, \text{basis}, L)\,.$$

A complete study includes

- timestep extrapolation;
- walker-population and projection-length convergence;
- several trial states with increasing quality;
- independent replicas and autocorrelation analysis;
- release-constraint or free-projection checks on smaller instances;
- ED, FCI, DMRG, or coupled-cluster comparison where feasible.

Reviews of molecular AFQMC describe this hierarchy and its use across weakly and strongly correlated systems [85].

14.6.3 Interpretation and experimental use

AFQMC can provide ground-state energies, correlation functions, energy differences, and finite-temperature properties in determinant space. Its polynomial algebra per walker and broad Hamiltonian coverage make it useful between FCI-scale anchors and large quantum-hardware targets. Agreement between AFQMC and a method with a different trial-state dependence can sharply narrow the reference interval.

Practical rule of thumb

Treat the AFQMC trial state as a convergence control. Repeat the calculation with trial states from distinct constructions and include the observed spread in the constraint-bias assessment.

14.7 Multiconfiguration Self-Consistent Field and Coupled Cluster

14.7.1 Multiconfiguration self-consistent field

Physical setting and wavefunction

Multiconfiguration self-consistent field (MCSCF) methods optimize both the configuration coefficients and the molecular orbitals. They are designed for systems where several electronic configurations carry comparable weight, such as bond breaking, open shells, transition-metal complexes, conical intersections, and excited states.

Write

$$|\Psi_{\mathrm{MCSCF}}\rangle = \sum_I C_I \, |\Phi_I(\{\phi_p\})\rangle \, .$$

The coefficients C_I describe configuration mixing, and the orbitals $\{\phi_p\}$ are varied at the same time. An orbital rotation can be written

$$|\phi_p(\boldsymbol{\kappa})\rangle = e^{-\hat{\kappa}} \, |\phi_p\rangle \, , \qquad \hat{\kappa} = \sum_{p>q} \kappa_{pq} \left(E_{pq} - E_{qp}\right) .$$

The stationary conditions are

$$\frac{\partial E}{\partial C_I} = 0, \qquad \frac{\partial E}{\partial \kappa_{pq}} = 0.$$

Complete active space self-consistent field (CASSCF) divides orbitals into inactive, active, and external groups. It performs full CI for the chosen active electrons and active orbitals while optimizing all orbital subspaces [86].

Convergence and active-space controls

MCSCF is variational inside the declared orbital basis and configuration space. Its primary controls are

- active electron and orbital counts;
- orbital basis and frozen-core treatment;
- state-specific or state-averaged objective;
- root tracking across geometry or external parameters;
- orbital-gradient norm and energy convergence;
- initial orbitals and occupation constraints.

Active-space selection acts as a model choice in addition to a numerical control. Enlarging the active space tests static-correlation completeness. Dynamic correlation outside the active space often requires a later perturbative, configuration-interaction, coupled-cluster, DMRG, or AFQMC treatment.

What MCSCF permits experimentally

MCSCF provides balanced states across bond breaking, orbital occupations, transition densities, state characters, and compact active-space Hamiltonians. These outputs help define quantum-computing encodings, prepare multireference initial states, identify observables, and avoid comparing a quantum result with a qualitatively wrong single-determinant state.

14.7.2 CCSD(T)

Physical setting and exponential ansatz

Coupled-cluster theory starts from a reference determinant $|\Phi_0\rangle$ and uses

$$|\Psi_{\mathrm{CC}}\rangle = e^{T} \, |\Phi_0\rangle \, , \qquad T = T_1 + T_2 + T_3 + \cdots \, .$$

The similarity-transformed Hamiltonian is

$$\overline{H} = e^{-T} H e^{T}.$$

Projecting onto excited determinants gives the amplitude equations

$$\langle \Phi_\mu | \overline{H} | \Phi_0 \rangle = 0,$$

and the energy is

$$E_{\mathrm{CC}} = \langle \Phi_0 | \overline{H} | \Phi_0 \rangle.$$

CCSD retains T_1 and T_2. CCSD(T) adds a perturbative estimate of connected triple excitations. The original formulation and later review establish the method as a central high-accuracy approach for molecules with a strong single-reference description [87], [88].

Convergence hierarchy and limitations

The exponential ansatz gives size extensivity for separated fragments under the standard assumptions. CCSD(T) lacks a general upper-bound property. Its reliability weakens as the reference determinant loses dominance or near-degeneracy grows.

A ground-truth study varies

- orbital basis cardinal number and complete-basis extrapolation;
- frozen-core, core-correlation, relativistic, and spin-orbit treatments;
- integral and amplitude thresholds;
- reference orbitals and open-shell formulation;
- excitation hierarchy, such as CCSD, CCSD(T), CCSDT, and selected quadruple corrections;
- geometry, state character, and multireference diagnostics.

A smooth basis sequence and a small triples correction support the single-reference regime. Large amplitude diagnostics, unstable solutions, strong reference dependence, or disagreement with MCSCF, DMRG, FCI, DMC, or AFQMC signal the need for a multireference treatment.

What CCSD(T) permits experimentally

CCSD(T) supplies accurate molecular energies, reaction energies, equilibrium structures, vibrational properties, and response quantities for favorable single-reference systems. It can benchmark small quantum chemistry experiments, test active-space corrections, and provide a strong classical comparator near equilibrium geometries.

Key takeaway

MCSCF resolves static correlation by optimizing orbitals and multiple configurations. CCSD(T) resolves dynamic correlation efficiently from a dominant reference determinant. Their overlap is useful, and their disagreement can identify a change in electronic structure before the quantum comparison is interpreted.

14.8 The Numerov Algorithm for Spatial Wavefunctions

Prerequisite

Before reading this section, ensure that boundary-value eigenproblems, finite grids, and wavefunction normalization are familiar.

Numerov integration is useful for one-dimensional or radial Schrödinger equations in quantum wells, heterostructures, confinement potentials, and nanostructures. These calculations can supply energies and matrix elements for a reduced qubit Hamiltonian. Superconducting circuits usually begin from circuit quantization and basis truncation. Semiconductor and spin devices can require a spatial wavefunction first.

For a particle of mass m in a one-dimensional potential,

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x).$$

Define

$$k^2(x;E) = \frac{2m}{\hbar^2}[E - V(x)],$$

so that

$$\psi''(x) + k^2(x;E)\psi(x) = 0.$$

On a uniform grid with spacing h, the Numerov recurrence is

$$\psi_{n+1} = \frac{2\left(1 - \frac{5}{12}h^2k_n^2\right)\psi_n - \left(1 + \frac{1}{12}h^2k_{n-1}^2\right)\psi_{n-1}}{1 + \frac{1}{12}h^2k_{n+1}^2}.$$

The local defect is $\mathcal{O}(h^6)$ and the standard multistep scheme has global fourth-order convergence under smoothness and stability conditions [89].

The recurrence propagates a trial solution across the grid, and the boundary conditions determine the eigenvalue. A complete bound-state calculation proceeds as follows.

1. Choose the domain, boundary conditions, grid spacing, and an energy bracket.
2. Propagate inward from both boundaries for a trial energy.
3. Use node count to identify the eigenstate index.
4. Match logarithmic derivatives at a stable interior point.
5. Drive the matching residual to zero with bisection or a safeguarded root finder.
6. Normalize the wavefunction and repeat on a finer grid or larger domain.

Propagation deep into a classically forbidden region can amplify the unwanted exponential solution. Inward–outward matching or a renormalized Numerov formulation improves stability [90]. Discontinuous potentials, singular radial terms, variable mass, and nonuniform grids require modified interface or discretization rules.

The calculation permits experiment-facing quantities after convergence,

$$E_n, \qquad \langle n|x|m\rangle, \qquad \langle n|p|m\rangle, \qquad |\psi_n(x)|^2.$$

These values support transition-frequency estimates, dipole couplings, tunnel splittings, and reduced Hamiltonian parameters. The converged eigenenergies and matrix elements become inputs to the reduced Hamiltonian used by the circuit or open-system calculation in the next section.

14.9 Open Quantum Dynamics with QuTiP

Prerequisite

Before reading this section, ensure that state vectors, density operators, Hamiltonians, collapse operators, and basis truncation are familiar.

QuTiP provides numerical tools for closed and open quantum dynamics [91], [92]. The mathematical object determines the solver.

Table 14.1: Representative QuTiP solver roles.

Solver	Mathematical problem	Reference checks
`sesolve`	Schrödinger evolution of a state vector	Norm conservation, timestep or integrator comparison, analytical limits
`mesolve`	Unitary or Lindblad evolution of ρ	Trace, positivity diagnostics, tolerance study, basis convergence
`steadystate`	Stationary density operator	Residual $\|\mathcal{L}(\rho_{ss})\|$, trace, positivity
`mcsolve`	Quantum-jump trajectories	Trajectory-count convergence and comparison with master-equation averages
`smesolve`	Stochastic master equation	Ensemble convergence and measurement-record statistics
`ssesolve`	Stochastic Schrödinger equation	Norm behavior, ensemble convergence, unraveling consistency

QuTiP 5 introduced a flexible data layer and optional integrations with array ecosystems that include JAX and CuPy. Acceleration varies with solver, operator structure, sparsity, precision, and installation. A GPU label therefore supplies no timing result by itself.

A numerical reference should report

- QuTiP version and data backend;
- solver, integrator, and absolute and relative tolerances;
- Hilbert-space truncation and operator sparsity;
- timestep or output grid;
- residual, norm, trace, and positivity checks;
- comparison with an analytical limit, refined truncation, or second integrator.

For a truncated oscillator or multilevel qubit, increase the basis size N_{cut} and track the target observable,

$$\Delta_O(N_{\mathrm{cut}}) = \left|\langle O\rangle_{N_{\mathrm{cut}}+\Delta N} - \langle O\rangle_{N_{\mathrm{cut}}}\right|.$$

The result can serve as a reference after Δ_O falls below the numerical uncertainty allocated to the comparison.

14.10 A Branched Solver Map

Solver selection starts from the mathematical structure of the target, followed by the observable and tolerance. Width alone gives too little information. The same particle count can favor FCI, DMRG, AFQMC, or DMC under different bases, geometries, entanglement patterns, and accuracy goals.

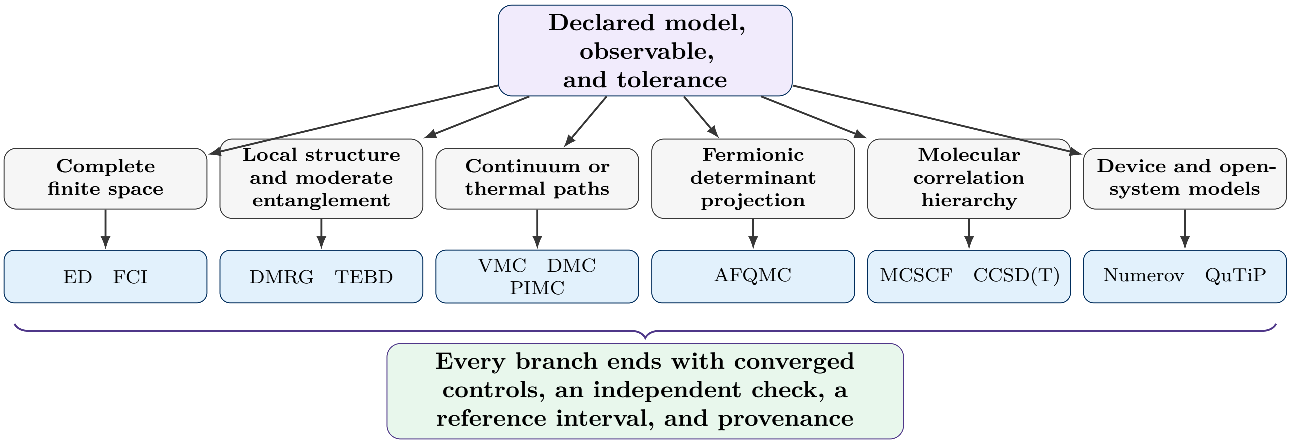

Figure 14.2: Branched solver map for constructing a numerical ground-truth record. The final reference interval still requires convergence and an independent check.

Table 14.2: Advanced numerical solvers for ground truth and convergence studies.

Method	Mathematical target	Strong regime	Main convergence controls	Dominant unresolved error	Use in a quantum comparison
ED	Eigenpairs and dynamics of a finite Hamiltonian matrix	Small lattice, circuit, spin, fermion, or truncated-boson models	Matrix residual, basis cutoff, finite size, symmetry sector, precision	Finite model space and omitted physics	Exact small-instance energies, states, correlators, and circuit prefixes
FCI	Electronic Schrödinger equation in a finite orbital basis	Small molecules and compact active spaces	Davidson or Lanczos residual, orbital basis, frozen core, relativistic model	One-particle basis incompleteness and determinant growth	Exact finite-basis energies, reduced density matrices, and state-preparation targets
DMRG	Variational MPS ground or excited state	One-dimensional and low-entanglement systems, favorable orbital orderings	Bond dimension, sweeps, discarded weight, variance, finite size, initial state	Entanglement truncation, ordering, local minima	Converged local observables, active-space energies, and many-body references
TEBD	Real or imaginary-time MPS evolution	Local one-dimensional dynamics before severe entanglement growth	Timestep, product-formula order, bond dimension, truncation threshold, conservation checks	Trotter and accumulated tensor truncation	Time traces, response functions, lightcones, and spectra
VMC	Variational expectation over a sampled trial state	Flexible continuum and lattice ansatzes	Ansatz family, optimizer, replicas, chain length, autocorrelation, finite size	Variational ansatz and optimization bias	Trial-state quality, variational energies, densities, and correlations
DMC	Imaginary-time ground-state projection in real space	Continuum ground states of atoms, molecules, fluids, and solids	Timestep, walkers, projection time, trial nodes, finite size, pseudopotentials	Fixed-node and population-control bias	Accurate ground-state energy cross-check with a distinct representation
PIMC	Thermal density operator and partition function	Bosons, distinguishable particles, and finite-temperature equilibrium	Imaginary-time slices, propagator order, permutations, sample length, finite size	Fermion sign or nodal constraint, thermal finite size	Thermal energies, correlations, superfluid response, and equilibrium observables
AFQMC	Imaginary-time projection using Slater-determinant walkers	Lattice fermions, molecules, and solids	Timestep, walkers, projection length, trial state, population control, basis, finite size	Constrained-path or phaseless bias	Large finite-basis energies and correlators beyond FCI scale
MCSCF	Joint orbital and multiconfiguration optimization	Bond breaking, open shells, excited states, and static correlation	Active space, state averaging, root tracking, orbital gradient, basis	Active-space choice and missing dynamic correlation	Balanced states, compact quantum active spaces, and state-character checks
CCSD(T)	Exponential single-reference correlation hierarchy	Weak to moderate correlation near a dominant determinant	Basis sequence, triples size, higher excitations, core and relativistic treatment, thresholds	Single-reference breakdown and excitation-rank truncation	High-accuracy molecular energies and reaction benchmarks in favorable regimes
Numerov	One-dimensional or radial Schrödinger boundary problem	Quantum wells, heterostructures, confinement, and radial states	Grid spacing, domain size, boundary residual, matching point	Discretization, domain, and interface model	Device eigenenergies, wavefunctions, dipoles, and tunnel couplings
QuTiP	Closed, Lindblad, steady-state, and stochastic finite-dimensional dynamics	Few-mode and few-level open quantum systems	Basis cutoff, integrator, tolerance, timestep, trajectories, trace and positivity	Hilbert-space truncation and environmental model	Pulse, decoherence, readout, and reduced-device dynamics

14.11 A Step-by-Step Ground-Truth Protocol for Quantum Computation

Prerequisite

Before reading this section, ensure that the target observable, quantum estimator, classical reference, and acceptance tolerance have been stated in compatible units and conventions.

A useful protocol builds the reference before the largest hardware run. The sequence below applies to energies, correlation functions, spectra, reduced density matrices, transport coefficients, thermal observables, and algorithmic success metrics.

1. **Register the model.** State the Hamiltonian, parameters, basis or lattice, boundary conditions, particle number, symmetry sector, geometry, and any frozen degrees of freedom.
2. **Register the observable.** Define the operator, estimator, normalization, Fourier convention, broadening, postprocessing, and requested tolerance. A total energy, energy difference, local correlation, and spectral image require different reference resources.
3. **Choose the smallest exact anchor.** Use ED or FCI for the largest instance that fits. Verify residuals and exact symmetries. This anchor tests code paths, conventions, and the first points in every scalable solver sequence.
4. **Choose a scalable primary solver.** Match the solver to the structure. DMRG or TEBD serves low-entanglement local systems. DMC serves continuum ground states. PIMC serves thermal equilibrium. AFQMC serves determinant-space fermionic projection. MCSCF and CCSD(T) serve complementary electronic-correlation regimes.
5. **List every active convergence control.** Write the vector $\boldsymbol{\lambda}$ before inspecting the final result. Include basis, finite size, timestep, bond dimension, walker population, sample length, trial state, active space, excitation rank, and solver tolerances as applicable.
6. **Perform one-dimensional sweeps first.** Refine each control separately far enough to identify its direction and scale. Plot the target observable against the control, together with residuals, variances, discarded weights, effective sample sizes, or constraint diagnostics.
7. **Perform coupled refinement.** Repeat the dominant sweep after refining the second-largest source. Examples include χ at two timesteps, AFQMC timestep at two trial states, DMC timestep at two walker populations, and basis size at two correlation levels.
8. **Add an independent method.** Select a method whose principal bias differs from the primary solver. Compare ED with DMRG, FCI with AFQMC, CCSD(T) with MCSCF, DMC with orbital methods, or TEBD with Krylov time evolution on smaller instances.
9. **Construct the reference interval.** Combine statistical uncertainty with bounded numerical contributions. Keep one-sided variational bounds, extrapolation spreads, fit sensitivity, and constraint spreads visible.
10. **Compare with the quantum estimator.** Align the model, ordering, units, symmetry sector, initial state, observable, and postprocessing. Preserve raw and corrected quantum estimates, together with the reference interval.
11. **Archive the record.** Store input files, code and library versions, seeds, convergence tables, failed runs, resource use, and the script that generates the final comparison.

Let the quantum estimate be $\widehat{O}_Q$ with standard uncertainty u_Q and bounded systematic contribution b_Q. Let the reference have $\widehat{O}_{\mathrm{ref}}$, u_{ref}, and b_{ref}. For independent estimates,

$$u_D = \sqrt{u_Q^2 + u_{\mathrm{ref}}^2}.$$

A conservative two-sided comparison quantity is

$$\Delta_{\mathrm{comp}} = \left|\widehat{O}_Q - \widehat{O}_{\mathrm{ref}}\right| + z_{1-\alpha/2} u_D + b_Q + b_{\mathrm{ref}}.$$

The registered criterion is

$$\Delta_{\mathrm{comp}} \le \epsilon_{\mathrm{req}}.$$

Shared samples, common fitted parameters, or correlated calibration data require the covariance form for u_D. A failed criterion narrows the supported conclusion and identifies which budget term needs more work.

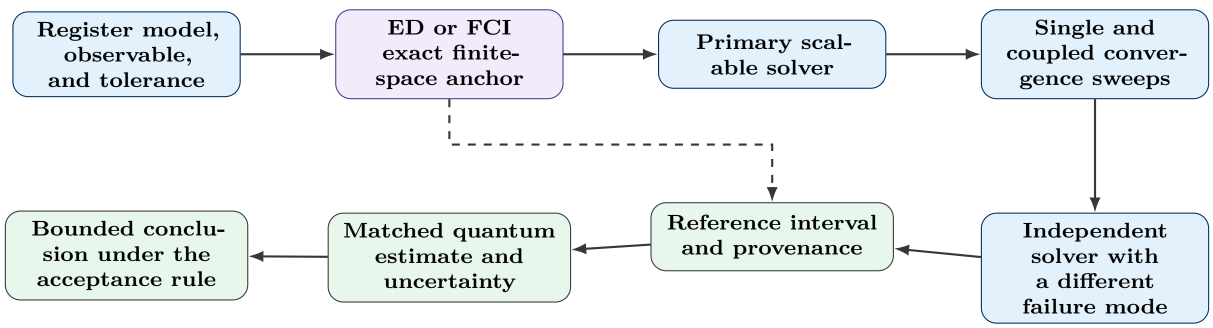

Figure 14.3: A ground-truth workflow designed from the model and observable toward the final quantum comparison.

14.12 Worked Example: A Molecular Energy Difference Across Weak and Strong Correlation

14.12.1 Physical setting

Consider an electronic energy difference

$$\Delta E(R) = E_B(R) - E_A(R)$$

for two states or structures along a geometry coordinate R. Near equilibrium, one determinant may dominate. At stretched bonds or near a state crossing, several configurations can carry comparable weight. The goal is a reference curve with uncertainty below ϵ_{req} over the full interval $R \in [R_{\min}, R_{\max}]$.

14.12.2 Scaffolded solver sequence

1. Build the same second-quantized Hamiltonian and integral convention for every solver.
2. Run FCI in a small orbital basis at selected geometries. Record eigenvalue residuals, state symmetries, natural occupations, and overlaps between neighboring geometries.
3. Run MCSCF or CASSCF across the full curve. Increase the active space until state characters, occupations, and $\Delta E(R)$ stabilize. Use state averaging near crossings to preserve a balanced orbital description.

4. Run DMRG inside the larger active spaces. Increase χ, reduce the discarded weight, and track the energy variance. Compare with FCI wherever both fit.
5. Run CCSD(T) in geometries with a dominant reference. Compare CCSD, CCSD(T), and a higher-excitation correction on smaller basis sets. Flag geometries where the triples correction or reference sensitivity grows sharply.
6. Run AFQMC with trial states from a single determinant and from the converged active-space calculation. Extrapolate the timestep and walker population. Use the trial-state spread as one contribution to the constraint assessment.
7. Add DMC for selected geometries if a real-space cross-check is valuable. Optimize several nodal forms and extrapolate the timestep and population.
8. Extrapolate the orbital basis or apply a documented basis correction. Keep the correlation-method spread and basis correction as separate entries.

The resulting reference can be written

$$\Delta E_{\mathrm{ref}}(R) = \widehat{\Delta E}(R) \pm \Delta_{\mathrm{num}}(R),$$

with

$$\Delta_{\mathrm{num}}(R) = \Delta_{\mathrm{basis}}(R) + \Delta_{\mathrm{solver}}(R) + \Delta_{\mathrm{stat}}(R) + \Delta_{\mathrm{constr}}(R).$$

Near equilibrium, CCSD(T), AFQMC, and active-space methods may agree tightly. Near strong correlation, MCSCF and DMRG define the state structure, and AFQMC or DMC adds a distinct stochastic check. The change in solver agreement across R carries physical information about the electronic state.

14.12.3 What the result permits experimentally

A quantum processor can prepare the same active-space states and measure energies, reduced density matrices, or transition properties. The comparison can then ask whether the quantum estimate lies inside the reference interval at every geometry, whether errors grow with multireference character, and whether mitigation preserves the ordering of the two states. This question is more informative than comparing one equilibrium energy alone.

Check your understanding

1. An ED calculation has an eigenvalue residual of 10^{-12} and uses a bosonic cutoff $N_{\mathrm{boson}} = 8$. State which error has been tightly controlled and which convergence study is still necessary.
2. A DMRG energy changes by 10^{-7} between sweeps, and the energy variance remains 10^{-3}. Explain why the sweep difference alone supplies an incomplete convergence record.
3. A DMC study has small statistical uncertainty and uses one fixed nodal surface. Identify the principal remaining control and the calculation that can probe it.
4. A phaseless AFQMC energy agrees with FCI on a small basis and shifts after the trial state is changed in a larger basis. Explain how both observations enter the reference interval.
5. A CCSD(T) curve agrees with MCSCF near equilibrium and diverges near bond breaking. State the physical interpretation and the solver hierarchy that should carry the stretched-bond reference.
6. Design a two-axis TEBD convergence study for a correlation function $C(t)$. Specify the timestep sequence, bond-dimension sequence, and one conserved quantity.

Key takeaway

Numerical ground truth is built from a declared model, a solver suited to its structure, a visible refinement path, and an independent method with a different failure mode. ED and FCI provide finite-space anchors. DMRG and TEBD control entanglement. VMC, DMC, PIMC, and AFQMC expose statistical and constraint contributions. MCSCF and CCSD(T) separate multireference and single-reference correlation regimes. Numerov and QuTiP connect the reference chain to device parameters and open-system dynamics.

Practical rule of thumb

Use ED or FCI whenever the complete finite model fits. For larger systems, select a solver with a measurable convergence axis and pair it with an independent method whose dominant error differs. Accept a reference after the target observable, residuals, stochastic uncertainty, and active constraints all satisfy the allocated numerical budget.

Chapter 15

Quantum Processors as Scientific Computing Accelerators

How the argument develops

A QPU contributes one kernel inside a larger scientific workflow. The chapter develops validation steps from hardware execution to deployment, defines matched accuracy and time metrics, and examines concrete workflows in chemistry, band theory, magnetic neutron spectroscopy, correlated materials, many-body dynamics, and quantum lattice Boltzmann CFD.

15.1 The Physical Setting of a Hybrid Workflow

Prerequisite

Before reading this chapter, ensure that you can distinguish a quantum circuit from a complete scientific workflow and that numerical error, statistical uncertainty, and wall-clock time are familiar.

Scientific computing is a natural setting for a quantum accelerator. Classical processors prepare models, reduce active spaces, compile circuits, schedule experiments, optimize parameters, process samples, solve projected equations, estimate uncertainty, and verify the result. The QPU supplies a selected kernel whose state space, interference, entanglement, or sampling structure can become costly to reproduce at the required scale with a classical method.

Within a heterogeneous architecture, this role is termed a **quantum co-processor**. In practice, the workflow resembles a standard HPC calculation utilizing an accelerator call; each QPU invocation returns finite experimental data and incurs associated overheads, including calibration, queueing, error control, and readout.

15.1.1 Partitioning the workload

Let W denote the full workload. A useful decomposition is

$$W = W_{\text{verify}} \circ W_{\text{post}} \circ W_{\text{QPU}} \circ W_{\text{pre}}.$$

The stages have the following roles.

- W_{pre} prepares the physical model, basis, active space, initial data, and circuit inputs.
- W_{QPU} executes the selected quantum kernel.

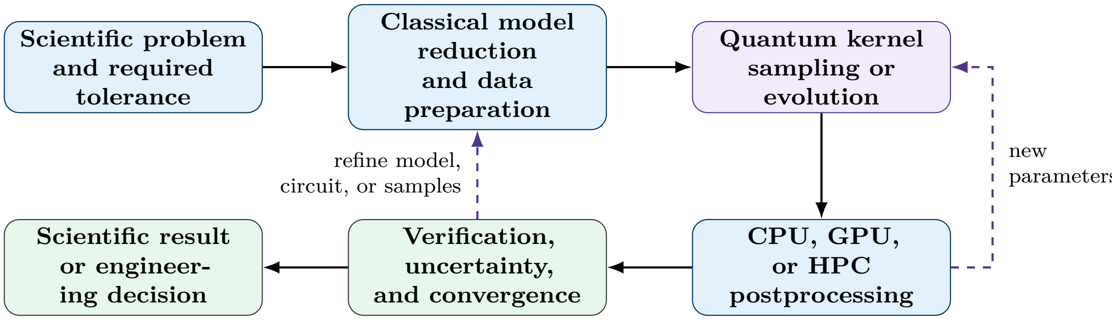

Figure 15.1: A quantum processor embedded in a scientific computing loop. The QPU supplies a selected kernel and classical computation carries model preparation, postprocessing, verification, and control.

- W_{post} performs reconstruction, optimization, embedding, or projected solution steps.
- W_{verify} tests the output against numerical and domain-specific acceptance conditions.

The composition symbol emphasizes data dependence. A change in the quantum sample set can alter a classical subspace, which can trigger another QPU call.

Suppose the quantum kernel returns N_s samples of b bits and N_o observable estimates with metadata volume V_{meta}. The stored data volume is approximately

$$V_{\text{QPU}} \approx N_s b + N_o V_o + V_{\text{meta}},$$

where V_o is the storage per observable record. Sample-based eigensolvers can return a sparse basis extracted from bitstrings. Variational workflows can exchange parameter vectors and expectation values. The accelerator boundary should sit where the quantum kernel adds useful information faster, more accurately, or at a larger accessible scale after transfer and orchestration costs are included.

15.2 Application Progress as an Evidence Hierarchy

A quantum circuit can execute successfully and still fall short of a useful scientific calculation. Application assessment therefore separates several questions.

- **Hardware executability** asks whether the complete compiled workflow runs on the processor with a documented data and control path.
- **Scientific validity** asks whether the result satisfies a domain accuracy target, uncertainty budget, conservation law, residual test, or reference comparison.
- **Scientific utility** asks whether the result supports a prediction, explanation, design choice, or scientifically relevant regime.
- **Classical competitiveness** compares the end-to-end workflow with a strong classical alternative at matched accuracy and confidence.
- **Economic deployment** adds cost, throughput, uptime, staffing, energy, data governance, and service reliability.

These questions form the application coordinate A introduced in table 11.1.

Table 15.1: Application qualification stages for quantum scientific computing.

Grade	Qualification	Evidence required
A0	Problem relevance	A scientific observable, decision, or design target is stated with the accuracy and time scale required for use.
A1	Quantum transformability	Encoding, state preparation, data loading, measurement, output extraction, and classical reduction costs are specified.
A2	Hardware executability	The full compiled workflow runs on physical hardware with a complete record of circuits, shots, calibration, and failures.
A3	Scientific validity	Domain-specific error, uncertainty, residual, symmetry, conservation, and convergence tests satisfy registered tolerances.
A4	Scientific utility	Repeated runs support a useful prediction, inference, design decision, or access to a scientifically relevant regime.
A5	Classical competitiveness	Comparison uses strong classical algorithms, matched inputs, matched accuracy, declared hardware, and a fixed comparison date.
A6E	Quantum enhancement	A quantum resource improves a declared quality, accuracy, sampling, or accessible-regime metric relative to the matched baseline.
A6A	Quantum acceleration	End-to-end time-to-solution or asymptotic resource scaling improves relative to the matched classical baseline.
A7	Economic deployment	Cost per valid result, throughput, uptime, reproducibility, energy, and labor satisfy the deployment requirements.

A6E and A6A are parallel grades. An experiment can improve accuracy at equal time, improve time at equal accuracy, or expand the accessible regime. The report should state the demonstrated condition.

The heat map explains the change in evidence as a workflow matures. Hardware execution emphasizes successful data return. Scientific validity gives the greatest weight to accuracy and reproducibility. Classical competitiveness elevates time and scaling. Deployment treats all five criteria as joint acceptance conditions.

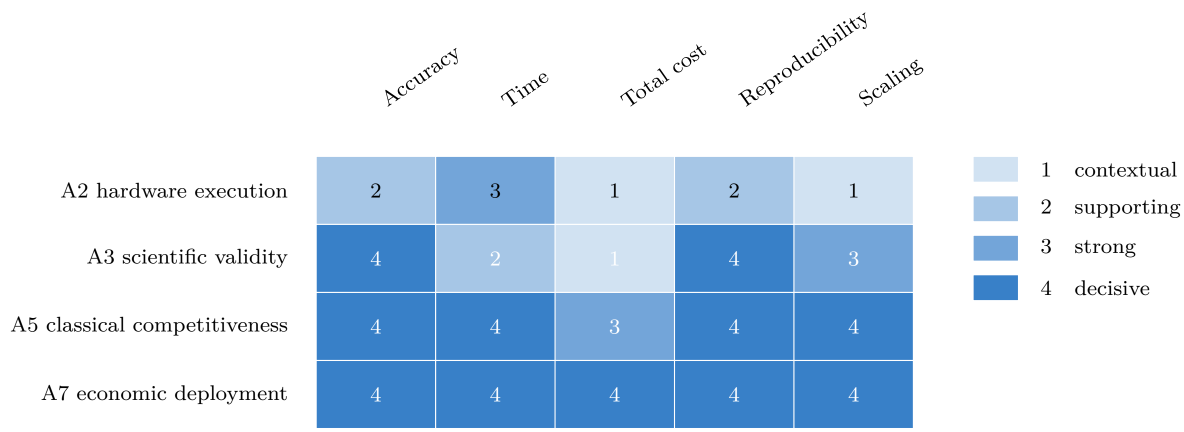

Figure 15.2: Pedagogical evidence heat map. Cell intensity indicates how decisive each criterion is for the corresponding qualification stage. The cell values are qualitative teaching weights and carry no empirical calibration.

15.3 Quantum Enhancement, Acceleration, and Error Mitigation

Let Q denote the quantum workflow and C the matched classical baseline. Let

$$L(\widehat{y}, y_\star)$$

be a domain loss relative to a trusted target $y_\star$, with

$$\epsilon_Q = L(\widehat{y}_Q, y_\star), \qquad \epsilon_C = L(\widehat{y}_C, y_\star).$$

At matched resource and confidence, an accuracy-enhancement factor is

$$A_\epsilon = \frac{\epsilon_C}{\epsilon_Q}.$$

Values $A_\epsilon > 1$ indicate a lower quantum-workflow error under the registered comparison.

At matched accuracy and confidence, the end-to-end acceleration factor is

$$A_T = \frac{T_C}{T_Q}.$$

Values $A_T > 1$ indicate a shorter quantum-assisted time-to-solution. An asymptotic comparison also needs the input model, oracle assumptions, precision dependence, data-access cost, and output requirement.

QEM changes the estimator. For an observable μ,

$$\mathrm{MSE}(\widehat{\mu}) = \mathrm{Bias}(\widehat{\mu})^2 + \mathrm{Var}(\widehat{\mu}).$$

A mitigation method can reduce bias and increase variance. It can also add circuit variants, calibration data, fitting, and model error. The strongest comparison fixes a target tolerance ϵ and confidence $1-\alpha$, then asks for the total resource required to reach it.

Define

$$T_{\mathrm{sol}}(\epsilon, \alpha) = \inf\left\{T : \Pr[L(\widehat{y}(T), y_\star) \le \epsilon] \ge 1-\alpha\right\},$$

and

$$C_{\mathrm{sol}}(\epsilon, \alpha) = \inf\left\{C : \Pr[L(\widehat{y}(C), y_\star) \le \epsilon] \ge 1-\alpha\right\}.$$

A mitigation protocol creates practical value if it lowers T_{sol}, lowers C_{sol}, or lowers the attainable error under the available budget.

15.4 A Hierarchy of Success Probabilities

The shot-level probability from chapter 11 is the first member of a larger hierarchy,

$$p_{\text{shot}} \longrightarrow p_{\text{observable}} \longrightarrow p_{\text{workflow}} \longrightarrow p_{\text{scientific}} \longrightarrow p_{\text{deploy}}.$$

Each level defines a different repeated experiment.

- p_{shot} is the probability that one measurement shot satisfies its scoring rule.
- $p_{\text{observable}}$ is the probability that a finite-shot estimator meets its uncertainty and bias target.
- p_{workflow} is the probability that one complete quantum–classical execution converges and returns a numerically acceptable output.
- $p_{\text{scientific}}$ is the probability that the workflow satisfies the domain validation protocol.
- p_{deploy} is the probability that the result is scientifically valid, timely, affordable, and reproducible under service conditions.

For workload W, define

$$\mathcal{S}_{\text{sci}}(W) = \{L(\widehat{y}, y_{\text{ref}}) \le \epsilon_{\text{req}}, \quad V_{\text{checks}} = 1\},$$

where V_{checks} represents the required residuals, symmetries, conservation laws, convergence tests, or reference comparisons. Then

$$p_{\text{scientific}}(W) = \Pr[\mathcal{S}_{\text{sci}}(W)].$$

Deployment adds resource and reliability limits,

$$\mathcal{S}_{\text{deploy}}(W) = \mathcal{S}_{\text{sci}}(W) \cap \{T_Q \le T_{\max}, \quad C_Q \le C_{\max}, \quad R_{\text{service}} \ge R_{\min}\},$$

so that

$$p_{\text{deploy}}(W) = \Pr[\mathcal{S}_{\text{deploy}}(W)].$$

The independent unit at this level is a complete job or session. Shot-level uncertainty should be propagated inside each job.

15.5 Representative Scientific Computing Workflows

Current experiments and algorithm studies illustrate several accelerator boundaries. Their strongest contribution is often the validation of encodings, quantum kernels, measurement methods, mitigation, and quantum–HPC integration. Classical competitiveness and deployment require separate matched comparisons.

Table 15.2: Representative quantum scientific computing workflows in electronic structure and materials, together with the evidence they contribute.

Scientific setting	Quantum kernel	Classical remainder	Qualification supplied
Molecules and metal clusters	Sample correlated electronic configurations from a prepared state	Active-space construction, distributed sample processing, projected diagonalization, and energy verification	Large hybrid execution, classically checkable upper bounds, and quantum–HPC integration [93]
Graphene and hexagonal boron nitride band structures	Sample an RBM distribution or solve momentum-dependent eigensystems with variational circuits	Tight-binding or Wannier model construction, optimization, k-path assembly, and comparison with classical bands	Hardware benchmark for reduced valence-band models and later multi-band simulator studies [94], [95]
Magnetic spectra for $KCuF_3$	Prepare a spin-chain state, evolve correlators, and reconstruct the dynamical structure factor	Model fitting, Fourier processing, MPS references, image and physics metrics, and comparison with inelastic neutron scattering	Direct comparison between a 50-qubit simulation and laboratory neutron-scattering spectra in a 2026 preprint [96]
DFT plus DMFT for a correlated material	Solve a reduced Anderson impurity model and estimate Green-function information	DFT, Wannier construction, DMFT self-consistency, bath fitting, spectroscopy comparison, and exact references	Real-material workflow, hardware execution up to 14 qubits, and comparison with exact and experimental data [97]

Table 15.3: Representative quantum scientific computing workflows in dynamics and computational fluid dynamics, together with the evidence they contribute.

Scientific setting	Quantum kernel	Classical remainder	Qualification supplied
Quantum lattice Boltzmann CFD	Encode lattice populations and apply collision and streaming operators	Initial-state construction, boundary conditions, moment extraction, field reconstruction, and classical validation	Nonlinear-flow algorithm validation, selected hardware demonstrations, and explicit end-to-end bottleneck analysis [98], [99], [100], [101]
Two-dimensional Ising dynamics	Execute Trotterized dynamics and estimate selected observables	Compilation, mitigation, tensor-network references, and observable analysis	Wide and deep hardware execution with application-specific verification [65]
Classical reassessment of the Ising workflow	Replace the QPU kernel with sparse-Pauli and tensor-network methods	CPU and GPU convergence studies	Demonstrates that the best classical baseline can change after the quantum experiment [66]

15.5.1 Electronic structure and chemical modeling

The active-space Hilbert space grows combinatorially with electron and orbital count. This growth makes electronic structure a natural accelerator target after core orbitals and weakly relevant degrees of freedom have been reduced classically.

In sample-based quantum diagonalization, the QPU generates bitstrings associated with electronic configurations. Classical computing repairs symmetries, selects a compact subspace, constructs the projected Hamiltonian,

$$H_{\mathrm{proj}} = V^{\dagger} H V,$$

and solves

$$H_{\mathrm{proj}} \boldsymbol{c} = E \boldsymbol{c}.$$

Here the columns of V span the sampled subspace. The resulting energy is checked against variational bounds, active-space convergence, and independent classical references.

Experiments using an IBM Heron processor with the Fugaku supercomputer reported circuits up to 77 qubits and 10,570 gates for N_2 dissociation and iron–sulfur clusters [93]. The workflow demonstrates scale and a useful division of labor. Classical competitiveness and deployment are part of later qualification stages.

The scientific target should state the chemical observable, energy or force tolerance, conformational coverage, environment model, and downstream decision. The QPU receives credit for the part of the workflow that changes the final scientific capability.

15.5.2 Electronic band structures of graphene and hexagonal boron nitride

A band structure records the allowed single-particle energies $E_n(\boldsymbol{k})$ across crystal momentum $\boldsymbol{k}$,

$$H(\boldsymbol{k}) \left| u_{n\boldsymbol{k}} \right\rangle = E_n(\boldsymbol{k}) \left| u_{n\boldsymbol{k}} \right\rangle .$$

For a two-sublattice honeycomb model, a compact tight-binding Hamiltonian is

$$H(\boldsymbol{k}) = \begin{pmatrix} \Delta & -t f(\boldsymbol{k}) \\ -t f^*(\boldsymbol{k}) & -\Delta \end{pmatrix}, \qquad f(\boldsymbol{k}) = \sum_{j=1}^{3} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}_j}.$$

The band energies are

$$E_{\pm}(\boldsymbol{k}) = \pm\sqrt{\Delta^2 + t^2 |f(\boldsymbol{k})|^2}.$$

Here t is the nearest-neighbor hopping energy, Δ is half the on-site energy difference between the two sublattices, and $\boldsymbol{\delta}_j$ are the three nearest-neighbor bond vectors. The structure factor $f(\boldsymbol{k})$ adds the phase accumulated along those bonds. A band diagram is constructed by solving $E_{\pm}(\boldsymbol{k})$ along a chosen path through the Brillouin zone.

Graphene has equivalent carbon sublattices in the minimal model, so $\Delta = 0$. Its two branches meet at momenta where $f(\boldsymbol{k}) = 0$, producing a Dirac crossing. Hexagonal boron nitride has inequivalent boron and nitrogen sublattices, so $\Delta \neq 0$. The same crossing is then separated by an energy gap of $2|\Delta|$ in this minimal description.

A 2021 hardware study used a restricted Boltzmann machine and a quantum circuit that sampled a Gibbs distribution on IBM-Q devices. It evaluated reduced tight-binding and Hubbard models for monolayer hBN and graphene, with warm starts and measurement-error mitigation [94]. The reported valence-band energies agreed with the corresponding conventional calculations inside the reduced models.

A later variational study used VQE and variational quantum deflation to reconstruct several bands for prototype solids that included graphene and boron nitride [95]. That work used state-vector and noisy

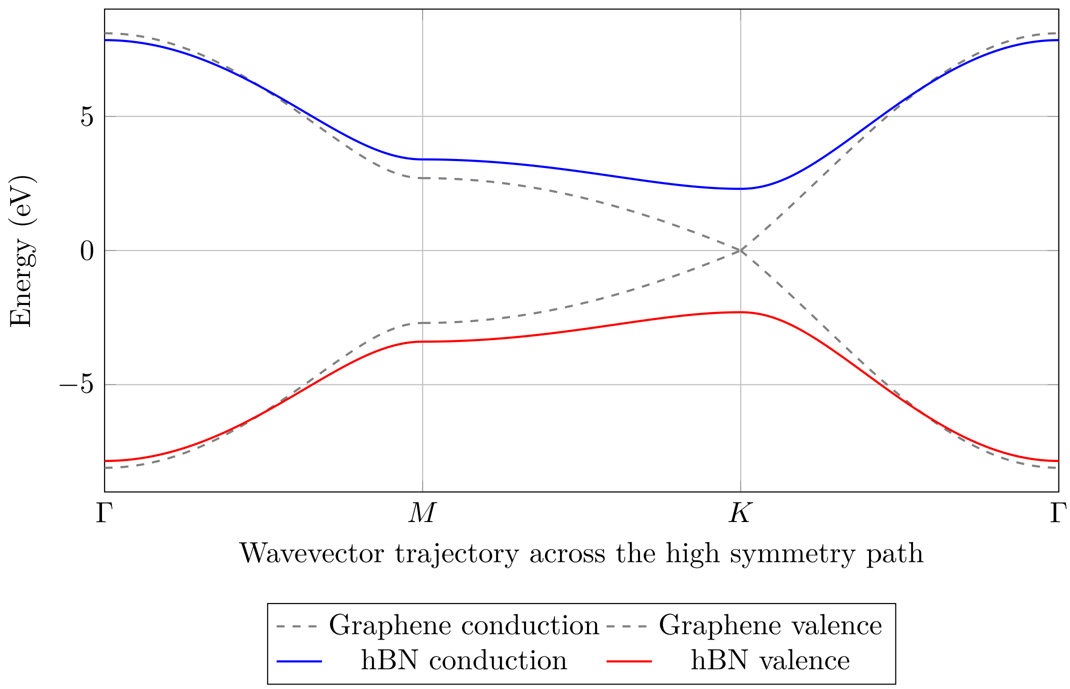

Figure 15.3: Comprehensive one dimensional tight binding dispersion encompassing the macroscopic high symmetry path. Equivalent sublattices yield the continuous graphene crossing precisely at the reciprocal K boundary, whereas the significant sublattice energy offset permanently opens the wide hBN bandgap.

simulation evidence. Together, the studies show the progression from a reduced hardware benchmark to a multi-band algorithm study.

The experimental value lies in validating

- periodic-system encodings and k-point continuation;
- warm-start and optimizer strategies;
- excited-state orthogonality constraints;
- measurement mitigation and noise sensitivity;
- band-gap and dispersion reconstruction.

Classical tight-binding and density-functional calculations are still strong baselines for these prototype systems. The examples therefore qualify the workflow and leave broad advantage as an open comparison.

15.5.3 Magnetic spectra benchmarked against neutron scattering

Inelastic neutron scattering measures how incident neutrons exchange momentum and energy with a magnetic material. The measured intensity is related to a dynamical structure factor. For spin components α and β,

$$S^{\alpha\beta}(\boldsymbol{q},\omega) = \frac{1}{2\pi N}\sum_{j,\ell} e^{-i\boldsymbol{q}\cdot(\boldsymbol{r}_j-\boldsymbol{r}_\ell)} \int_{-\infty}^{\infty} dt\, e^{i\omega t} \left\langle S_j^{\alpha}(t) S_\ell^{\beta}(0) \right\rangle .$$

The magnetic scattering cross section has the schematic dependence

$$\frac{d^2\sigma}{d\Omega\, dE'} \propto |F(\boldsymbol{q})|^2 S(\boldsymbol{q},\omega),$$

where $F(\boldsymbol{q})$ is a magnetic form factor.

Here N is the number of spin sites, $\boldsymbol{r}_j$ is the position of site j, and $S_j^\alpha(t)$ is the Heisenberg-picture spin component at time t. The spatial Fourier factor converts site separation into momentum transfer $\boldsymbol{q}$, and the temporal Fourier factor converts the time correlation into angular frequency ω. The form factor $F(\boldsymbol{q})$ accounts for the spatial distribution of magnetic density around an ion. Experimental geometry, polarization factors, instrumental resolution, and thermal population also enter a quantitative neutron-scattering comparison.

A common model for a one-dimensional quantum magnet is the XXZ chain,

$$H = J \sum_j \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right).$$

In this Hamiltonian, J sets the exchange-energy scale and Δ sets the anisotropy between the longitudinal and transverse spin couplings. The quantum processor prepares a ground-state approximation, applies a local perturbation, evolves the state, and measures a time-domain correlation. A classical Fourier transform converts the correlation into $S(\boldsymbol{q}, \omega)$.

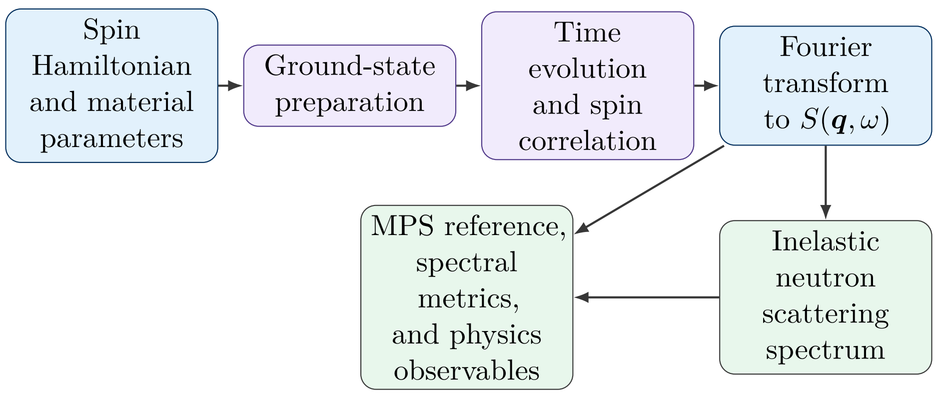

Figure 15.4: A quantum–classical workflow for comparing a simulated magnetic spectrum with an inelastic neutron-scattering experiment.

A 2026 IBM-led preprint used up to 50 qubits on Heron processors to reconstruct the spectrum of $KCuF_3$, a quasi-one-dimensional antiferromagnet, and compared the result with inelastic neutron-scattering data [96]. The study used MPS references, spectral-image metrics, line scans, and entanglement-related quantities. The quantum result reproduced key features of the spinon continuum under the stated finite-size, finite-time, compilation, and noise conditions.

This example strengthens the evidence chain in two ways. The target is a laboratory observable instead of an internal quantum-computing score. The reference set includes experiment, tensor-network calculations, and physics-based spectral diagnostics.

Finite lattice size limits momentum resolution. Finite evolution time limits frequency resolution. Gate noise suppresses long-time correlations and broadens the Fourier spectrum. A strong report therefore records qubit count, time steps, Trotter error, state-preparation fidelity, circuit compression, mitigation, windowing, spectral resolution, and comparison metrics.

The workflow permits direct testing of spin Hamiltonians against measured spectra. Future value will come from materials and parameter regimes where the relevant long-time correlations exceed classical tensor-network reach.

15.5.4 Embedding a quantum impurity solver in a materials workflow

Strongly correlated materials often use embedding methods that isolate a difficult local many-body problem inside a larger classical calculation. Density-functional theory plus dynamical mean-field theory supplies a clear accelerator boundary.

A local Green function can be written

$$G_{\mathrm{loc}}(i\omega_n) = \frac{1}{N_{\boldsymbol{k}}} \sum_{\boldsymbol{k}} \left[(i\omega_n + \mu)I - H_{\boldsymbol{k}} - \Sigma(i\omega_n)\right]^{-1}.$$

Here $H_{\boldsymbol{k}}$ is the one-electron Hamiltonian, Σ is the local self-energy, and μ is the chemical potential. DMFT maps the lattice problem to an Anderson impurity model whose bath is updated until the impurity Green function agrees with the local lattice Green function.

The QPU can act as the impurity solver. Classical computing carries DFT, Wannier construction, bath fitting, self-consistency, analytic continuation, and comparison with spectroscopy.

A hybrid implementation used quantum hardware with up to 14 qubits to solve impurity models associated with $Ca_2CuO_2Cl_2$, then compared the result with exact references and photoemission spectroscopy [97]. The final test concerns spectral functions, quasiparticle renormalization, bath discretization, DMFT convergence, and agreement with experiment. Circuit accuracy supplies one layer of that validation.

15.5.5 Quantum lattice Boltzmann methods for CFD

The lattice Boltzmann method represents a fluid with discrete populations $f_i(\boldsymbol{x}, t)$ moving along lattice velocities $\boldsymbol{c}_i$. A single-relaxation-time update is

$$f_i(\boldsymbol{x} + \boldsymbol{c}_i \Delta t, t + \Delta t) = f_i(\boldsymbol{x}, t) - \frac{\Delta t}{\tau}\left[f_i(\boldsymbol{x}, t) - f_i^{\mathrm{eq}}(\rho, \boldsymbol{u})\right].$$

Here $f_i(\boldsymbol{x}, t)$ is the population associated with discrete velocity $\boldsymbol{c}_i$, Δt is the timestep, τ is the relaxation time, and $f_i^{\mathrm{eq}}(\rho, \boldsymbol{u})$ is the local equilibrium distribution determined by density ρ and velocity $\boldsymbol{u}$. The update first relaxes each population toward equilibrium and then streams it to the neighboring lattice node selected by $\boldsymbol{c}_i$.

The macroscopic fields are recovered from moments,

$$\rho = \sum_i f_i, \qquad \rho \boldsymbol{u} = \sum_i \boldsymbol{c}_i f_i.$$

For the single-relaxation-time model in a common lattice normalization,

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2}\right),$$

where c_s is the lattice sound speed. Positive viscosity in this model requires $\tau > \Delta t/2$. Classical LBM theory and implementation are developed in standard references such as [102].

The structure appears attractive for quantum computing. Streaming is a permutation-like operation. Collision is local. The difficult parts are equally important:

- the equilibrium distribution depends nonlinearly on ρ and $\boldsymbol{u}$;
- relaxation is generally nonunitary;
- arbitrary field loading can require expensive state preparation;
- full-field readout can erase an amplitude-encoding advantage;
- realistic walls, inlets, outlets, forcing, and turbulence enlarge the circuit and data path.

Wang and coauthors proposed a node-level ensemble description that maps nonlinear fluid dynamics to linear transport and collision operations in an enlarged representation [98]. Their numerical validation included vortex-pair merging and decaying turbulence on grids up to 16.8 million points. This result

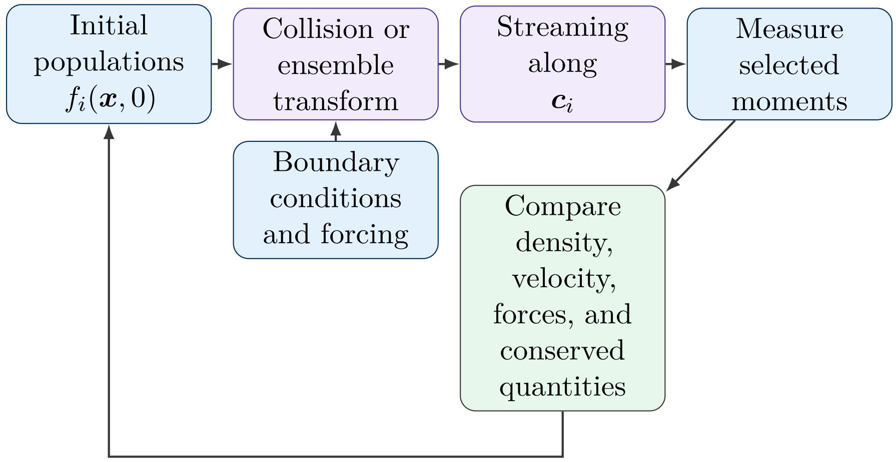

Figure 15.5: A quantum lattice Boltzmann workflow. Practical performance is governed by state preparation, collision and streaming depth, boundary treatment, selected-observable readout, and reinitialization.

validates an algorithmic construction on classical simulation resources; it is distinct from an end-to-end quantum-hardware CFD calculation.

A separate 2025 preprint reported a trapped-ion hardware demonstration for a two-dimensional advection–diffusion field. It used compressed state preparation, reduced-depth collision and streaming operators, selected-observable readout, and error mitigation [99]. The physical problem was linear transport-diffusion, which serves as a hardware step toward fluid solvers.

Schalkers' 2026 dissertation develops quantum lattice Boltzmann methods from streaming, collision, encoding, boundary treatment, and complexity viewpoints [100]. An end-to-end analysis by Jennings and coauthors examined nonlinear incompressible LBM, identified bottlenecks in earlier proposals, and found that a modest advantage can survive for selected observables in a high-error-tolerance regime [101].

These results support a precise experimental interpretation. Near-term work can test transport, collision, low-dimensional field parametrizations, and boundary primitives. Fault-tolerant studies must include state loading, timestep count, condition numbers, readout, and the cost of the requested observable. A quantum CFD advantage is most plausible for a small set of global or local functionals, such as force, flux, or correlation measures, instead of a complete flow field at every grid point.

15.5.6 Many-body dynamics and a moving classical baseline

A 127-qubit experiment studied Trotterized Ising dynamics with circuits containing up to 2,880 CNOT gates and used error mitigation to estimate selected observables [65]. Subsequent classical work developed sparse-Pauli and tensor-network calculations that converged the same observables more accurately and with lower runtime for the studied instances [66].

The sequence teaches an important lesson. Classical competitiveness is time-indexed. A fair comparison record should state

- classical algorithms and software versions;
- CPU, GPU, memory, and interconnect resources;
- numerical tolerances and convergence parameters;
- matched scientific output and confidence target;
- comparison date.

A later classical advance can change an A5 or A6A assessment without changing the quality of the original hardware experiment.

15.6 End-to-End Time and Cost

The quantum-assisted wall-clock time should include every stage on the critical path,

$$\begin{aligned} T_Q = {} & T_{\text{data}} + T_{\text{mapping}} + T_{\text{compile}} + T_{\text{queue}} + T_{\text{calibration}} + T_{\text{QPU}} \\ & + T_{\text{mitigation}} + T_{\text{classical}} + T_{\text{verification}}. \end{aligned}$$

Parallel stages can be represented with a critical-path graph. A microsecond circuit can coexist with an hour-scale result if queueing, repeated sampling, calibration, or classical reconstruction dominates.

The financial ledger has the form

$$\begin{aligned} C_Q = {} & C_{\text{QPU}} + C_{\text{CPU/GPU}} + C_{\text{storage}} + C_{\text{calibration}} \\ & + C_{\text{verification}} + C_{\text{labor}} + C_{\text{infrastructure}}. \end{aligned}$$

A cost-advantage factor at matched scientific acceptance is

$$A_C = \frac{C_C}{C_Q}.$$

Deployment can require $A_C > 1$ or another business criterion established by the user. Throughput and service reliability can dominate single-job cost, so valid results per day and failure-recovery time also belong in the record.

The cost of statistical evidence is

$$C_{\text{evidence}} = C_{\text{setup}} + c_{\text{shot}} n_{\text{required}} + C_{\text{classical}} + C_{\text{calibration}}.$$

For logical experiments, one nominal shot can contain many syndrome cycles, mid-circuit measurements, resets, feedforward operations, and decoder calls.

15.7 Platform-Independent Assessment with Platform-Specific Contracts

The same qualification logic applies across hardware families, and the benchmark contract should reflect native physics. Superconducting and spin processors expose calibrated pulse sequences and gate layers. Trapped-ion and neutral-atom systems offer different connectivity, interaction range, measurement, and reconfiguration. Photonic systems emphasize source probability, loss, interference visibility, feedforward, and detector behavior. Analog simulators expose Hamiltonian preparation, control schedules, local observables, and model-parameter calibration. Annealers require an embedding, schedule, sampling, and postprocessing contract.

15.8 A Decision Record for Scientific Deployment

A complete application report can center on one decision sentence. The sentence states the workload, output, tolerance, confidence, classical baseline, total time, total cost, and reproducibility window.

The supporting record includes

- workload decomposition and accelerator boundary;

- benchmark contract for every QPU kernel;
- domain validation, convergence, and uncertainty tests;
- classical baseline and comparison date;
- enhancement, acceleration, or mitigation metrics under matched conditions;
- success probabilities from shots to complete jobs;
- end-to-end quantum and classical resource ledger.

This structure permits careful near-term progress and stronger future conclusions. Hardware demonstrations can qualify at A2 or A3. Scientific utility can qualify at A4. A5, A6, and A7 are still available as evidence accumulates.

Key takeaway

A quantum processor creates application value as one component of a verified scientific workflow. Hardware execution, scientific validity, utility, classical competitiveness, acceleration, and economic deployment are separate qualifications.

Practical rule of thumb

Compare quantum and classical workflows at the same scientific tolerance and confidence. Include preparation, queueing, calibration, mitigation, classical processing, verification, and failed jobs in the time and cost ledger.

Chapter 16

A Scaffolded Verification Workflow

How the argument develops

Verification starts from a precise scientific conclusion and works backward toward the tests required to support it. This chapter connects analytical checks, classical references, hardware diagnostics, holdout circuits, drift-aware sampling, uncertainty propagation, and a registered acceptance rule.

16.1 The Physical Setting and the Object Being Tested

Prerequisite

Before reading this chapter, ensure that statistical uncertainty, numerical error, experimental systematic error, and model discrepancy are familiar. Review chapters 2 and 7 if these categories need reinforcement.

A large quantum experiment rarely has one complete classical reference. Credibility therefore comes from several tests that operate at different scales. Exact simulation can cover small instances and circuit prefixes. Tensor networks or Pauli propagation can cover selected observables. Hardware diagnostics can expose leakage, crosstalk, assignment error, and drift. Domain invariants can test whether the final result respects known physics.

The workflow begins with a **conclusion record**

$$\mathcal{C} = (M,\, I,\, U,\, O,\, D,\, \widehat{\mu},\, \epsilon_{\mathrm{req}},\, 1-\alpha,\, \mathcal{A})\,.$$

The symbols carry the following meanings.

- M specifies the mathematical model and its parameters.
- I specifies the initial state or input ensemble.
- U specifies the logical and compiled evolution.
- O specifies the measured observable or scoring rule.
- D specifies the processor region, calibration window, and data-processing path.
- $\widehat{\mu}$ is the reported estimator.
- ϵ_{req} is the required accuracy.
- $1-\alpha$ is the required confidence level.

- $\mathcal{A}$ is the registered acceptance rule.

This record prevents a result from drifting away from the experiment that generated it. A change in compilation, qubit map, mitigation model, postselection policy, or calibration window creates a new record.

16.2 Building the Uncertainty Budget

Four uncertainty categories recur across the handbook. They arise from different mechanisms and require different diagnostics.

Table 16.1: Uncertainty categories and the tests that constrain them.

Contribution	Physical source	Primary diagnostic	Typical response
Statistical	Finite samples, accepted-shot loss, serial correlation	Confidence interval, effective sample size, block bootstrap	Add independent sessions or improve the estimator
Numerical	Basis truncation, tensor truncation, solver tolerance, finite precision	Grid, bond-dimension, timestep, or tolerance refinement	Refine the classical calculation until the target observable stabilizes
Experimental systematic	Drift, leakage, crosstalk, pulse error, assignment error	Sentinel circuits, repeated calibration, randomized order, device diagnostics	Recalibrate, redesign the acquisition, or enlarge the interval
Model discrepancy	Missing interactions, imperfect noise model, uncertain boundary conditions	Competing models, residual patterns, held-out observables, parameter sensitivity	Narrow the conclusion or enlarge the physical model

Let the estimator error be decomposed as

$$\widehat{\mu} - \mu_\star = \delta_{\text{stat}} + \delta_{\text{num}} + \delta_{\text{sys}} + \delta_{\text{model}},$$

where $\mu_\star$ is the target quantity defined by the scientific protocol. This equation is an accounting identity. It carries no assumption of independence.

If independent zero-mean approximations are defensible for a preliminary budget, the variance terms combine as

$$\sigma_{\text{tot}}^2 = \sigma_{\text{stat}}^2 + \sigma_{\text{num}}^2 + \sigma_{\text{sys}}^2 + \sigma_{\text{model}}^2.$$

Known bias bounds should remain visible,

$$|\widehat{\mu} - \mu_\star| \leq z_{1-\alpha/2}\sigma_{\text{tot}} + b_{\text{num}} + b_{\text{sys}} + b_{\text{model}}.$$

Correlated contributions require covariance terms or a joint resampling model. A root-sum-square budget can otherwise look precise and still understate the uncertainty.

16.3 A Nine-Step Verification Procedure

16.3.1 Design the evidence before the target run

The first four steps define the conclusion and build references before the final data are inspected.

1. **Register the conclusion.** Record $\mathcal{C}$, including the observable, compiled circuit, processor region, precision target, confidence level, and acceptance rule.
2. **Derive analytical anchors.** Identify normalization, symmetry, conservation, positivity, monotonicity, limiting cases, and dimensional relations that the target result must satisfy.
3. **Construct classical references.** Use dense simulation for small widths, structured simulation for favorable instances, and independent numerical implementations for solver validation.
4. **Characterize the processor.** Measure readout assignment, leakage, drift, gate and layer behavior, simultaneous-operation effects, and sentinel circuits under the target compilation conditions.

16.3.2 Separate method development from evaluation

Mitigation and model selection can overfit tractable circuits. The next two steps keep tuning data separate from evaluation data.

5. **Tune on training circuits.** Select noise scales, truncation thresholds, regression models, calibration windows, and shot allocation using circuits reserved for method development.
6. **Reserve holdout circuits.** Choose circuits that resemble the target in width, depth, observable support, gate composition, and noise exposure. Their exact or trusted values reside outside the tuning loop.

16.3.3 Acquire, analyze, and decide

The final steps preserve temporal information and apply the registered rule.

7. **Acquire interleaved data.** Randomize circuit order and interleave target circuits, holdouts, sentinels, calibrations, and noise-scaled variants.
8. **Report every estimator stage.** Preserve raw counts and report unmitigated, corrected, and mitigated values with shots, acceptance rates, fit residuals, effective sample size, and uncertainty propagation.
9. **Apply the acceptance rule.** Compare the final interval and validation checks with $\mathcal{A}$. Record failed checks and the resulting restriction on the conclusion.

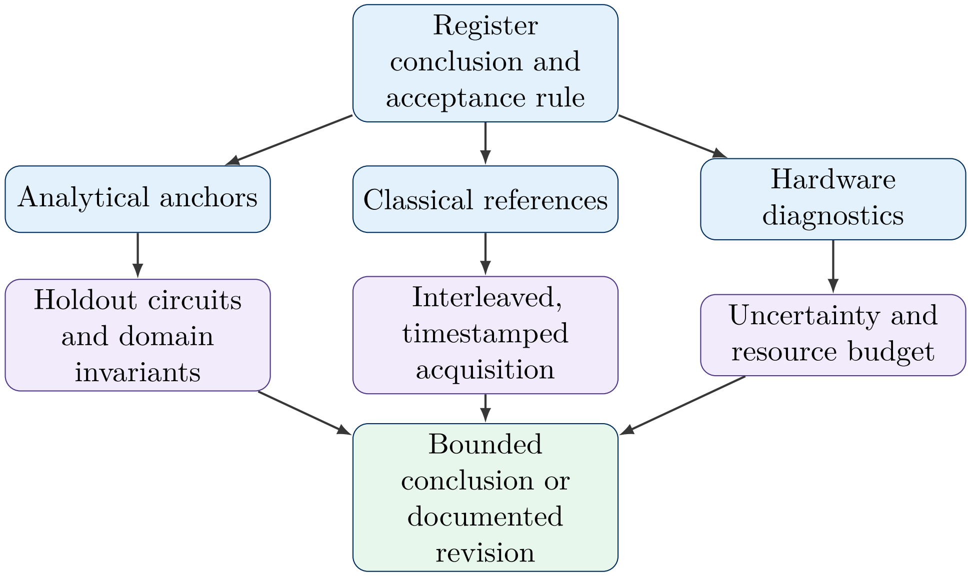

Figure 16.1: Verification designed from the conclusion backward. Independent tests constrain distinct parts of the uncertainty budget before the acceptance rule is applied.

16.4 Worked Example for a Local Observable

Consider a 100-qubit Trotter circuit that estimates a local magnetization

$$\mu_\star(t) = \langle Z_j(t) \rangle.$$

The acceptance target is an absolute error below $\epsilon_{\text{req}} = 0.03$ with 95% confidence over a declared time window. The evidence can be distributed as follows.

- Exact state-vector calculations cover smaller widths and short prefixes.
- Tensor-network calculations cover selected times where bond-dimension convergence stays favorable.
- A symmetry check tests the conserved parity sector.
- Mirror or randomized circuits probe the compiled layers used by the target.
- Sentinel observables are repeated across the acquisition to estimate drift.
- Raw and mitigated estimates are evaluated on holdout circuits before the target result is accepted.

Treat the statistical term, 0.012, and the session-to-session drift term, 0.009, as approximately zero-mean standard uncertainties. Treat the numerical term, 0.006, and the model-discrepancy term, 0.008, as magnitude bounds. Under these assumptions, a conservative 95% error bound is

$$\Delta_{\text{tot}} = 1.96\sqrt{0.012^2 + 0.009^2} + 0.006 + 0.008 \approx 0.043.$$

The combined bound exceeds the registered tolerance of 0.03. The current evidence therefore supports a narrower conclusion. A revised acquisition design, a stronger reference calculation, a reduced model discrepancy, or additional independent sessions must lower the bound before the original acceptance target is supported. Extra digits in the central estimate leave this conclusion unchanged.

Key takeaway

A large experiment gains credibility from several tests with different failure sensitivities. Exact references, structured calculations, hardware diagnostics, holdouts, drift analysis, and domain invariants contribute separate pieces of evidence.

Practical rule of thumb

Write the conclusion record, holdout set, and acceptance rule before inspecting the target result. A later methodological change creates a new analysis version and should be documented as such.

Chapter 17

Resource Accounting and Final Reporting

How the argument develops

Scientific comparison requires more than qubit count or circuit duration. This chapter builds a resource ledger, separates sum-of-times accounting from critical-path timing, defines equal-accuracy comparisons, and ends with a reusable reporting template.

17.1 From Asymptotic Scaling to a Complete Ledger

Prerequisite

Before reading this chapter, ensure that asymptotic scaling, fixed overhead, wall-clock time, peak memory, and confidence intervals are distinct concepts in your resource model.

Asymptotic complexity describes how a selected resource changes with problem size under stated assumptions. A practical experiment also incurs compilation, calibration, queueing, data movement, sampling, mitigation, classical reconstruction, and verification. A useful resource ledger is

$$\mathcal{R} = (n_{\mathrm{q}}, D, G, N_{\mathrm{circ}}, N_{\mathrm{shot}}, t_{\mathrm{QPU}}, t_{\mathrm{cal}}, t_{\mathrm{classical}}, M_{\mathrm{peak}}, V_{\mathrm{data}}, E_{\mathrm{total}}, C_{\mathrm{total}})\,.$$

Here n_{q} is qubit count, D is compiled depth, G is a declared noisy-gate count, N_{circ} is the number of circuit variants, and N_{shot} is the total recorded shot count. The remaining entries record execution time, calibration time, classical time, peak memory, retained data, energy, and financial cost.

The ledger should also identify the accounting boundary. A laboratory study may exclude fabrication and staff time. A service-level study may include queue latency, failed jobs, recovery time, verification labor, and infrastructure amortization. Comparisons are still meaningful after both workflows use the same boundary.

17.2 Wall-Clock Time and the Critical Path

A serial checklist writes the end-to-end time as

$$\begin{aligned} T_Q = {} & T_{\mathrm{data}} + T_{\mathrm{mapping}} + T_{\mathrm{compile}} + T_{\mathrm{queue}} + T_{\mathrm{calibration}} + T_{\mathrm{QPU}} \\ & + T_{\mathrm{mitigation}} + T_{\mathrm{classical}} + T_{\mathrm{verification}}. \end{aligned}$$

This expression is useful for identifying omitted stages. Parallel work requires a directed acyclic graph. If task v has duration t_v and predecessor set $\mathrm{pred}(v)$, its earliest finish time satisfies

$$F(v) = t_v + \max_{u \in \mathrm{pred}(v)} F(u).$$

The workflow wall-clock time is

$$T_{\mathrm{wall}} = \max_{v \in V} F(v).$$

The longest predecessor chain is the critical path. Accelerating a task outside that path can leave the final time unchanged.

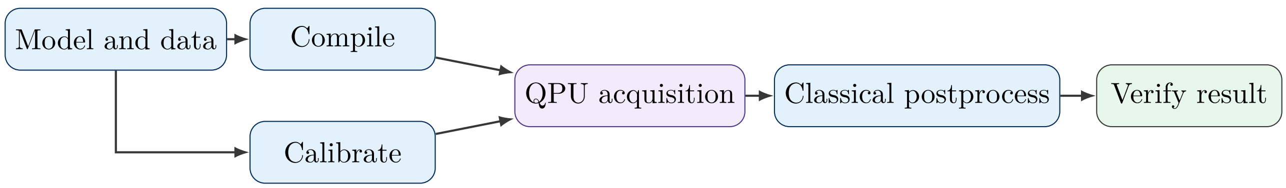

Figure 17.1: A simple critical path graph. Compilation and calibration can proceed in parallel after the model is prepared, and QPU acquisition waits for both branches.

17.3 Equal-Accuracy and Equal-Confidence Comparisons

Let Q denote the quantum-assisted workflow and C the classical baseline. At required error ϵ and confidence $1-\alpha$, define

$$T_{\mathrm{sol}}^{(X)}(\epsilon, \alpha) = \inf \left\{ T : \Pr[L(\widehat{y}_X(T), y_\star) \le \epsilon] \ge 1 - \alpha \right\},$$

for $X \in \{Q, C\}$. The matched time ratio is

$$A_T(\epsilon, \alpha) = \frac{T_{\mathrm{sol}}^{(C)}(\epsilon, \alpha)}{T_{\mathrm{sol}}^{(Q)}(\epsilon, \alpha)}.$$

A value above one indicates a shorter quantum-assisted time under the registered comparison.

The corresponding cost ratio is

$$A_C(\epsilon, \alpha) = \frac{C_{\mathrm{sol}}^{(C)}(\epsilon, \alpha)}{C_{\mathrm{sol}}^{(Q)}(\epsilon, \alpha)}.$$

The comparison record should state the algorithms, software versions, hardware, numerical tolerances, data-access assumptions, output quantity, failure policy, and comparison date. A faster kernel can lose its system advantage if preparation, sampling, or verification dominates the critical path.

17.4 How Mitigation Changes the Ledger

QEM changes several resources at once.

- ZNE adds noise-scaled circuit variants, regression, and repeated calibration across the scaling schedule.
- PEC adds weighted samples according to the quasiprobability norm and requires a learned noise representation.
- Readout mitigation adds calibration circuits and can increase estimator variance after inversion.

- Virtual distillation adds copies or permutation measurements and introduces a ratio estimator.
- OBP and PNA reduce quantum depth by increasing Pauli propagation, measurement groups, classical memory, and truncation analysis.

The fair comparison fixes ϵ and $1 - \alpha$, then reports the total resources required to reach them. A central value closer to the reference carries limited value if its interval is wider or its cost is much larger.

17.5 Refining Common Assertions

Table 17.1: More precise forms of common quantum-computing statements.

Informal statement	Scientific form
Classical simulation gives an exact answer	A solver evaluates a stated model with numerical error, approximation error, and possible model discrepancy
Quantum hardware returns the answer	A calibrated processor produces finite data, followed by estimation, correction, uncertainty analysis, and validation
Fifty qubits define a universal classical boundary	Dense complex128 storage becomes severe near this scale; tractability can stay favorable for shallow, local, weakly entangled, or observable-focused problems
Error mitigation removes decoherence	QEM redesigns an estimator and can trade lower bias for larger variance, more circuits, calibration burden, and model sensitivity
A benchmark certifies an application	A benchmark constrains specified failure modes under its circuit ensemble; application accuracy requires workload-specific validation
Lightcone methods remove QEM scaling barriers	Locality can reduce the mitigated region; the retained cone can still carry rapid sampling or Pauli-growth costs
A fast circuit implies a fast result	End-to-end time includes preparation, queueing, calibration, sampling, postprocessing, and verification along the critical path

17.6 A Reusable Final Report

A concise report can follow the sequence below.

1. State the scientific workload and the required output.
2. Identify the model, processor, qubit map, compiler, calibration window, and benchmark contract.
3. Report the estimator, interval, effective sample size, sessions, drift diagnostics, and mitigation settings.
4. Report analytical checks, classical references, convergence tests, holdout circuits, and domain invariants.
5. Report the scaling range and the evidence supporting any logical or application qualification.
6. Publish the resource ledger and the matched classical baseline.
7. State the narrowest conclusion supported by the registered acceptance rule.

Decision sentence template

For workload W, processor D, compiled circuit family U, and observable O, the registered estimator achieved the stated tolerance and confidence across the declared sessions, passed the listed reference and holdout checks over scaling range S, and used the reported quantum and classical resource ledger.

Chapter 18

Economic Value from Near-Term Prototypes to Fault-Tolerant Computing

How the argument develops

Chapter 15 placed the QPU inside a complete scientific workflow. This chapter asks how that workflow can create value before utility-scale fault-tolerant computing is available by connecting direct scientific benefit, information gained from a prototype, cost per scientifically accepted result, expected net value, useful-result throughput, and expansion of the feasible scientific domain. The final sections relate these quantities to scientific workloads and to the broad range of quantum hardware families active in 2026.

18.1 A Primer on Technical and Economic Value

Prerequisite

Before reading this chapter, ensure that you can distinguish a quantum kernel from a complete workflow. Review probability, expected value, confidence level, time to solution, and the resource ledger in chapters 15 and 17. An economics course is unnecessary.

A quantum processor can create useful work long before it runs a large fault-tolerant application. The first useful work is usually narrow. A device may estimate one observable, sample one structured distribution, evolve one reduced model, or solve one embedded subproblem. The surrounding classical computation prepares the model, chooses the representation, schedules the processor, reconstructs the result, and checks whether the output satisfies the scientific requirement. This full-stack view has guided scientific quantum-computing roadmaps and application co-design studies [103], [104].

The maturity labels used in this chapter describe a complete capability record. These labels carry no universal qubit-count scale. An analog simulator may control hundreds or thousands of quantum degrees of freedom and still have a narrow observable set. A gate processor may control fewer qubits and support deeper programmability, mid-circuit measurement, or early logical operations. The useful label follows from the task, error-management stage, accepted output, and resource cost.

- A **near-term processor** uses physical qubits, modes, atoms, ions, spins, or other quantum degrees of freedom to execute bounded experiments under finite coherence, finite sampling, and limited error control. The category includes gate-based, analog, photonic, and annealing approaches.

- An **advanced pre-fault-tolerant processor** supports a stronger full stack, such as improved native operations, dynamic circuits, mid-circuit measurement, error detection, early logical elements, more stable calibration, or wider integration with classical computing. It has yet to sustain the logical depth and logical width required by the target utility-scale application.
- A **fault-tolerant processor** executes logical operations under an error-correction construction whose logical error is controlled by increasing protected resources. The application still needs enough logical qubits, logical depth, throughput, and output extraction to meet the scientific contract [105].
- A **quantum kernel** is the selected part of a workflow assigned to the quantum processor. It may prepare a state, evolve a Hamiltonian, estimate a Green function, sample a distribution, or solve an embedded eigensystem.
- **Scientific value** is created when the accepted result changes a prediction, explanation, design choice, parameter estimate, or experimental plan.
- **Economic value** is the expected benefit of the decision or scientific capability after total cost, delay, unsuccessful runs, uncertainty, and downstream loss are included.

The distinction between scientific and economic value prevents two common errors. A device can produce an informative scientific result whose execution cost stays above the deployment target. A prototype can also produce little direct scientific output and still create positive economic value by resolving a costly design uncertainty. Both cases deserve a precise record.

Table 18.1: Symbols used in the economic-value analysis.

Symbol	Mathematical role	Workflow interpretation
W	Workload or workload family	Scientific problem, model, inputs, and requested output
$X \in \{Q, C\}$	Competing workflow	Quantum-assisted workflow Q or matched classical workflow C
ϵ_{req}	Accuracy tolerance	Largest accepted domain loss or observable error
$1 - \alpha$	Confidence requirement	Required probability coverage for the reported result
p_X	Acceptance probability	Probability that one complete attempt satisfies the registered scientific and service conditions
C_X	Cost per attempt	QPU, CPU or GPU, calibration, checks, labor, storage, and infrastructure within the chosen accounting boundary
T_X	Time per attempt	Critical-path wall-clock time, including queueing and recovery
B_X	Benefit of an accepted result	Scientific, engineering, or financial value assigned by the decision context
L_X	Loss from an unsuccessful result	Delay, discarded work, missed opportunity, or incorrect downstream decision
$\mathcal{D}_X$	Feasible scientific domain	Workloads that meet the registered accuracy, reliability, time, and cost limits

18.2 Why Small Hardware Experiments Can Carry Real Value

A prototype experiment has a physical setting, a bounded quantum system coupled to a classical control and analysis stack. Its scientific output may be a small energy calculation, a correlation function, a sampled subspace, a transport coefficient, or a benchmarked primitive. The result gains meaning from the same records developed earlier in this handbook. They include the encoded model, compiled operations, calibration interval, estimator, uncertainty, reference calculations, and resource ledger.

Small experiments create value through several channels.

- **Direct scientific output.** A reduced model can answer a real question inside its stated scope. Examples include active-space energies, small impurity models, spin correlations, reduced band calculations, and laboratory-linked spectra. The model size can be small and the scientific question can still be real.
- **Information about scale.** A prototype measures state-preparation cost, circuit depth, measurement growth, drift, mitigation overhead, decoder demand, and classical orchestration. These quantities replace uncertain estimates with measured distributions.
- **Reusable technical assets.** Encodings, compiler passes, calibration routines, pulse libraries, measurement groupings, decoders, data schemas, and acceptance tests can transfer to later processors. Their value persists after one hardware generation is retired.
- **Integration capability.** A working QPU-to-HPC path tests authentication, scheduling, data movement, provenance, failure recovery, and job-level reproducibility. These systems tasks can dominate deployment time after the quantum kernel improves.
- **Decision information.** A prototype can determine whether a larger experiment, fabrication run, algorithm program, or procurement path deserves further investment.
- **Strategic option value.** A team that has already built the model, software, controls, and acceptance record can exploit a hardware improvement earlier than a team beginning after the improvement arrives.

The channels interact. A small materials calculation can expose a weak active-space choice, which improves the classical model. A noisy dynamics experiment can reveal which observables keep stable, which changes the measurement strategy. A logical-memory demonstration can establish decoder latency and syndrome-data volume, which changes the cryogenic and HPC design. The scientific result and the engineering lesson arise from the same experiment.

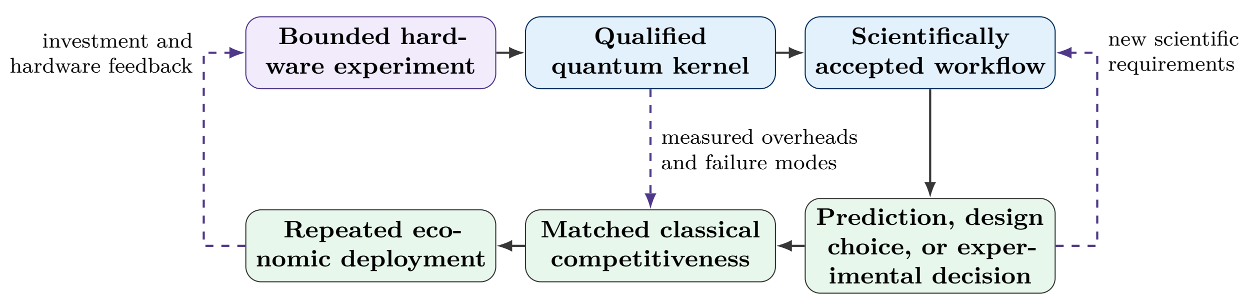

Figure 18.1: A prototype can create value at several points in the development loop. Direct scientific output, measured overheads, reusable infrastructure, and investment information can appear before broad deployment.

Experiments already span several points in this loop. Hybrid chemistry studies have coupled quantum processors with large classical systems, magnetic simulations have been compared with neutron-scattering data, and protected-operation studies have tested repeated correction, logical gates, or stronger processor-scale control on superconducting, trapped-ion, neutral-atom, silicon-spin, bosonic, and diamond-spin hardware [54], [93], [96], [106], [107], [108], [109], [110], [111], [112]. Each experiment answers a bounded question and supplies measured quantities for the next design cycle.

18.3 The Value of Information from a Prototype Experiment

18.3.1 Physical setting and assumptions

Consider a laboratory or company choosing an action before a costly uncertainty has been resolved. The action could select a qubit modality, a fabrication stack, an algorithm family, a cryogenic controller, a target molecule, or a larger experimental campaign. Let θ denote the unknown state of the world. Let $a \in \mathcal{A}$ denote an available action. The utility $U(a, \theta)$ records the consequence of choosing a when the true state is θ.

The analysis assumes that the decision maker has a prior distribution $p(\theta)$, a model for possible prototype data D, and a consistent utility scale. Money is one possible unit. Scientific opportunity, schedule risk, energy use, or a weighted multi-objective score can also be used.

Before collecting new data, the best expected utility is

$$V_0 = \max_{a \in \mathcal{A}} \mathbb{E}_\theta[U(a, \theta)].$$

After a prototype returns data D, Bayes' rule updates the distribution to $p(\theta \mid D)$. The optimal action can then depend on the observed data. The expected utility with the prototype information is

$$V_1 = \mathbb{E}_D \left[\max_{a \in \mathcal{A}} \mathbb{E}_{\theta|D}[U(a, \theta)] \right].$$

The expected value of sample information is

$$\text{EVSI} = V_1 - V_0.$$

This construction is a standard result in decision analysis [113]. The decision maker may ignore the data, so access to D leaves the optimum unchanged or raises it under the idealized model. Thus

$$\text{EVSI} \geq 0.$$

The prototype itself consumes resources. It can also produce direct scientific output and reusable technical assets. A fuller value equation is

$$V_{\text{proto}} = \text{EVSI} + V_{\text{direct}} + V_{\text{transfer}} - C_{\text{proto}}.$$

Here V_{direct} is the value of the accepted scientific result, V_{transfer} is the value of reusable software, data, process knowledge, and trained personnel, and C_{proto} is the complete cost of the experiment.

18.3.2 Step-by-step numerical example

Suppose a research program must choose between two hardware paths. Before a prototype run, the best prior decision has an expected value of \$3.20 million. The experiment can return two data classes.

- With probability 0.40, the data favor path A and the optimized posterior decision has expected value \$4.20 million.

- With probability 0.60, the data favor path B and the optimized posterior decision has expected value \$3.80 million.

The expected value after observing the prototype is

$$V_1 = 0.40(4.20) + 0.60(3.80) = 3.96 \text{ million dollars.}$$

The expected value of sample information is

$$\text{EVSI} = 3.96 - 3.20 = 0.76 \text{ million dollars.}$$

If the prototype costs \$0.25 million and its direct result and transferable assets are omitted from this illustrative calculation, the net decision value is

$$V_{\text{proto}} = 0.76 - 0.25 = 0.51 \text{ million dollars.}$$

The numbers are pedagogical and carry no hardware calibration. The calculation shows why a small experiment can be economically positive without delivering a speedup. Its value comes from changing a larger decision.

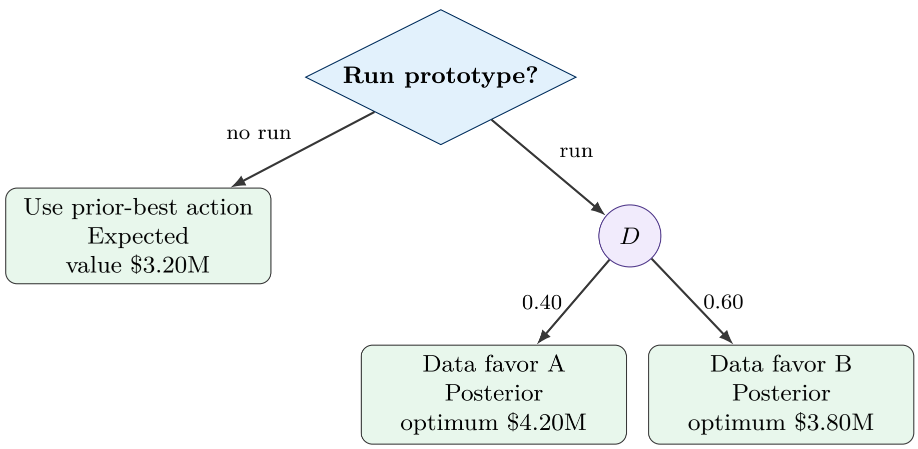

Figure 18.2: Illustrative value-of-information decision tree. The prototype is useful if the expected gain from improved decisions and transferable assets exceeds the complete experiment cost.

18.3.3 What the result permits experimentally

The value-of-information view changes prototype design. The experiment should target the uncertainty that can alter a decision. A larger circuit is useful only if it improves that discrimination. Useful design questions include

- which measurement separates the competing models most strongly;
- which hardware metric changes the architecture decision;
- which uncertainty has the largest downstream cost;
- which result can be reused across several applications;
- which calibration or scaling experiment narrows the resource forecast.

A prototype optimized for decision value can be smaller than a demonstration optimized for qubit count. Its conclusion is also easier to audit.

18.4 Cost per Scientifically Accepted Result

18.4.1 The acceptance event

A low cost per circuit execution can coexist with a large cost per useful result. The unit of comparison should therefore be a complete accepted job. For workflow X, define the acceptance event

$$\mathcal{A}_X = \{L(\widehat{y}_X, y_\star) \le \epsilon_{\mathrm{req}}, \quad V_{\mathrm{checks}} = 1, \quad T_X \le T_{\max}\}.$$

The first condition enforces the scientific tolerance. The second requires the registered residuals, conservation laws, reference checks, or reproducibility tests. The third enforces the service deadline. The acceptance probability is

$$p_X = \Pr(\mathcal{A}_X).$$

Assume independent attempts with constant acceptance probability p_X. Let K be the number of attempts required to obtain the first accepted result. Then K follows a geometric distribution,

$$\Pr(K = k) = (1 - p_X)^{k-1} p_X, \qquad k = 1, 2, \ldots$$

and

$$\mathbb{E}[K] = \frac{1}{p_X}.$$

If every attempt has cost C_X and duration T_X, the expected cost and time per accepted result are

$$\overline{C}_X^{\mathrm{acc}} = \frac{C_X}{p_X}, \qquad \overline{T}_X^{\mathrm{acc}} = \frac{T_X}{p_X}.$$

The useful-result throughput is

$$\lambda_X = \frac{p_X}{T_X}.$$

Availability can be included with a factor $A_X \in [0, 1]$,

$$\lambda_X^{\mathrm{service}} = A_X \frac{p_X}{T_X}.$$

Unsuccessful attempts can have a different cost. Let $C_{X,\mathrm{s}}$ be the cost of the successful attempt and $C_{X,\mathrm{f}}$ the cost of one failed attempt. The expected number of failures before the first success is $(1 - p_X)/p_X$, which gives

$$\overline{C}_X^{\mathrm{acc}} = C_{X,\mathrm{s}} + \frac{1 - p_X}{p_X} C_{X,\mathrm{f}}.$$

This form handles early-aborted jobs, rejected calibration windows, decoder failures, and experiments that fail before all postprocessing has been incurred.

18.4.2 Matched quantum and classical comparison

Let

$$c = \frac{C_Q}{C_C}, \qquad r_p = \frac{p_Q}{p_C}.$$

The accepted-cost ratio is

$$\Gamma_C = \frac{C_Q/p_Q}{C_C/p_C} = \frac{c}{r_p}.$$

The quantum-assisted workflow has a lower expected cost per accepted result when

$$\Gamma_C < 1, \qquad \text{equivalently} \qquad r_p > c.$$

A quantum attempt may therefore cost more and still win if its accepted-result probability rises by a larger factor. The reverse is also possible. A low-cost attempt can lose after retries, calibration rejection, and scientific acceptance are included.

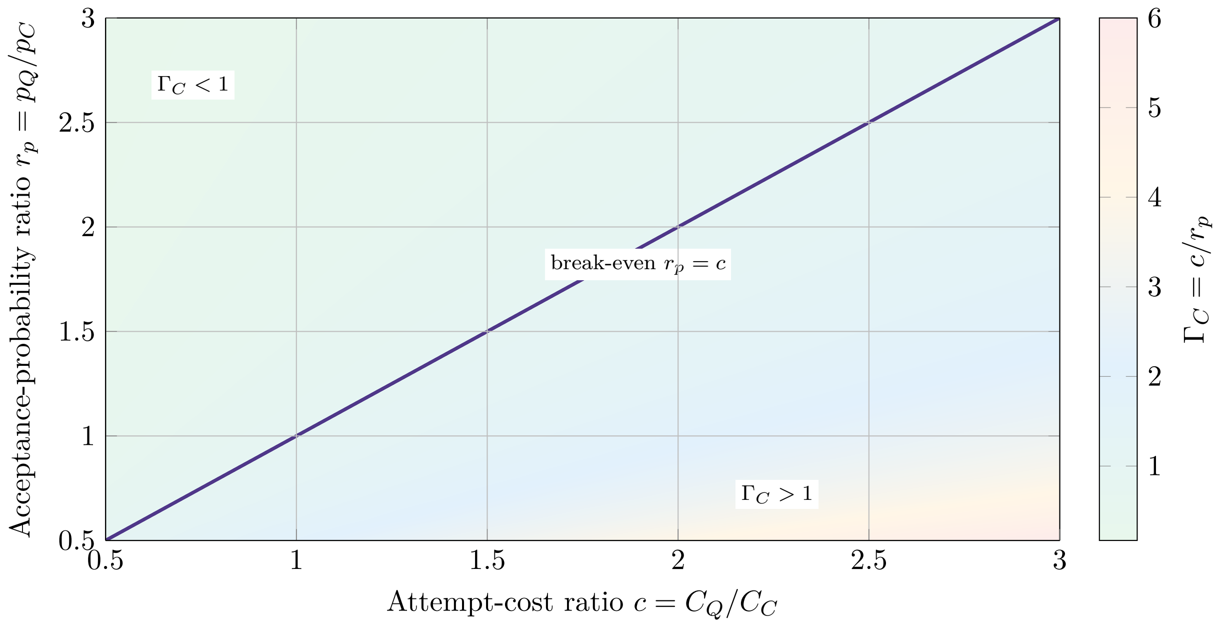

Figure 18.3: Accepted-cost heat map. A quantum-assisted workflow lies in the lower-cost region when its acceptance-probability gain exceeds its attempt-cost ratio. The surface is analytical and uses no empirical hardware fit.

18.4.3 Building the attempt cost

A complete attempt cost should include the resources consumed on the critical path and the resources required to make the result credible.

- model construction, active-space selection, embedding, and input preparation;
- circuit synthesis, compilation, pulse or schedule generation, and queueing;
- calibration, sentinel circuits, error detection, mitigation, and accepted-shot loss;
- QPU execution, resets, mid-circuit measurements, syndrome cycles, and decoder calls;
- CPU, GPU, or HPC reconstruction, optimization, Fourier transforms, and uncertainty analysis;
- independent references, convergence studies, storage, provenance, and personnel time;
- failed jobs, repeated sessions, service interruption, and recovery.

This accounting permits a direct experimental target. Improve the physical processor, estimator, or orchestration until C_X/p_X and T_X/p_X meet the accepted-result requirement.

18.5 Expected Net Value Under Accuracy, Reliability, and Delay

18.5.1 Governing relation

The accepted-cost metric treats every accepted result as equally valuable. Many decisions assign different benefits and losses to different workflows. Let B_X be the benefit of an accepted result, C_X the cost of one attempt, and L_X the loss associated with an unsuccessful attempt. The expected net value is

$$\mathcal{V}_X = p_X B_X - C_X - (1 - p_X) L_X.$$

Each term has a direct interpretation. The first is the expected benefit. The second is the attempt cost. The third is the expected loss from rejection, delay, or an unusable output.

For matched benefit B and matched loss L, the quantum-minus-classical value difference is

$$\Delta\mathcal{V} = \mathcal{V}_Q - \mathcal{V}_C = (p_Q - p_C)(B + L) - (C_Q - C_C).$$

The quantum-assisted workflow has larger expected value when

$$(p_Q - p_C)(B + L) > C_Q - C_C.$$

This relation shows how reliability and consequence interact. A modest increase in acceptance probability can be valuable in a high-consequence design problem. The same increase may carry little value for a cheap exploratory calculation.

18.5.2 Delay and time-sensitive benefit

Scientific and industrial benefits can decay with delay. A simple model is

$$B_X(T_X) = B_0 e^{-rT_X},$$

where r is a discount or obsolescence rate. Substitution gives

$$\mathcal{V}_X = p_X B_0 e^{-rT_X} - C_X - (1 - p_X)L_X.$$

A workflow with an excellent kernel can lose value if queueing, sampling, calibration, or reconstruction delays the decision. This is why the critical path matters more than pulse duration alone.

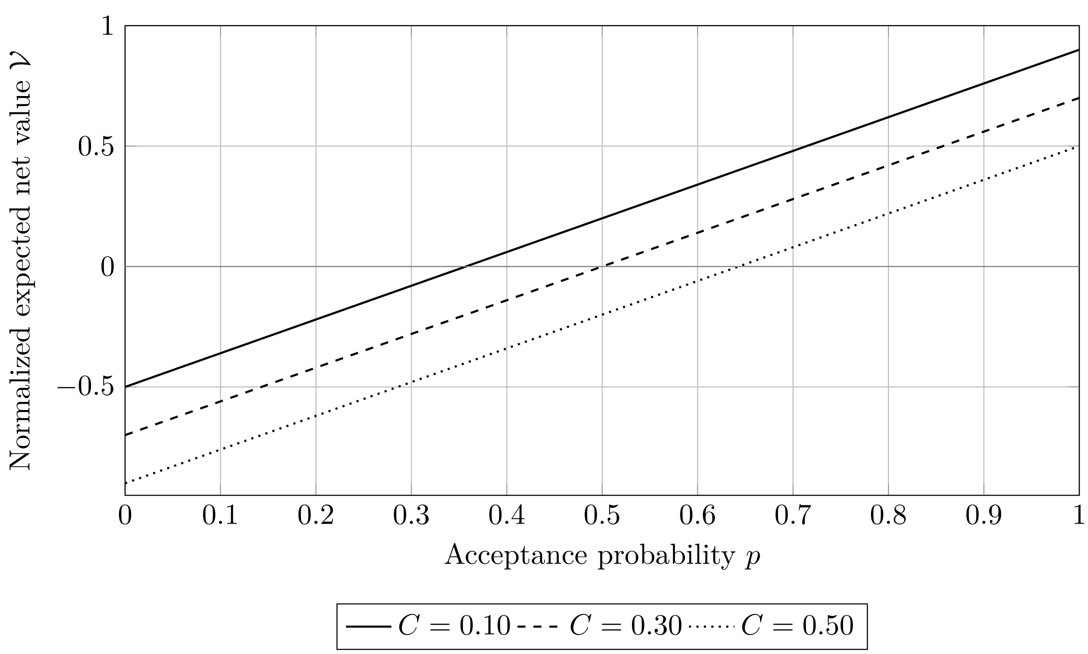

Figure 18.4: Illustrative expected net value for normalized benefit $B = 1$ and failure loss $L = 0.4$. Higher acceptance probability can compensate for a larger attempt cost. The curves are pedagogical and carry no hardware fit.

18.5.3 What the relation permits experimentally

The equation translates system improvement into decision value. Gate fidelity, logical error, accepted-shot fraction, queue time, decoder latency, and calibration stability matter through their effect on p_X, C_X, and T_X. An experiment can therefore report a value sensitivity,

$$\frac{\partial \mathcal{V}_X}{\partial p_X} = B_X + L_X, \qquad \frac{\partial \mathcal{V}_X}{\partial C_X} = -1.$$

The first derivative identifies workloads where reliability deserves the greatest engineering effort. The second shows that every unit of reduced cost adds one unit of value inside the accounting boundary.

18.6 The Development Continuum

18.6.1 Stage definitions from demonstrated capability

The path from prototype computing to utility-scale fault tolerance is continuous. The stages below organize the dominant questions without assigning a universal width threshold.

Table 18.2: Development stages and their dominant value mechanisms.

Stage	Technical state	Primary value channel	Decisive measurement
Near-term prototype	Physical qubits or quantum modes execute bounded circuits, analog evolutions, sampling tasks, or embedded kernels under finite coherence and finite sampling	Direct small-scale science, encoding tests, control learning, estimator design, and information value	Accepted observables, uncertainty, drift, overhead, reproducibility, and transfer of the learned method
Advanced pre-fault-tolerant	Stronger native control, dynamic circuits, mid-circuit measurement, error detection, early logical elements, or improved analog programmability support larger full-stack experiments	Reusable workflow assets, hardware-algorithm co-design, early domain utility, and more accurate resource forecasts	Cost per accepted result, depth or time reach, logical or detected-error behavior, and job-level repeatability
Early fault-tolerant	Protected logical qubits execute repeated correction and a useful logical operation set over small workloads	Reliable logical primitives, reduced retry cost, trustworthy longer circuits, and transfer from mitigation to correction	Logical error scaling, decoder latency, logical throughput, state injection cost, and complete logical resource ledger

Stage	Technical state	Primary value channel	Decisive measurement
Utility-scale fault-tolerant	Logical width, depth, throughput, and output extraction meet the scientific workload contract	Classical competitiveness, expansion of the feasible scientific domain, and repeated deployment	Equal-accuracy time and cost, accepted-result throughput, uptime, and scientific or engineering impact

Near-term and advanced pre-fault-tolerant processors already support real experiments. Preskill's NISQ description emphasized useful scientific exploration under limited error correction [19]. Later experiments and system studies have added dynamic control, deeper circuits, hybrid supercomputing, early logical processing, and broad platform diversity [54], [93], [109], [110], [114]. The long-term goal is fault-tolerant quantum computation at a scale that meets the application contract.

18.6.2 How the value mix changes

The dominant value mechanism shifts with maturity. Early experiments create information and transferable assets. Advanced prototypes add direct scientific utility and integration experience. Fault-tolerant systems increase reliability and circuit reach. Utility-scale systems must satisfy all resource and service conditions together.

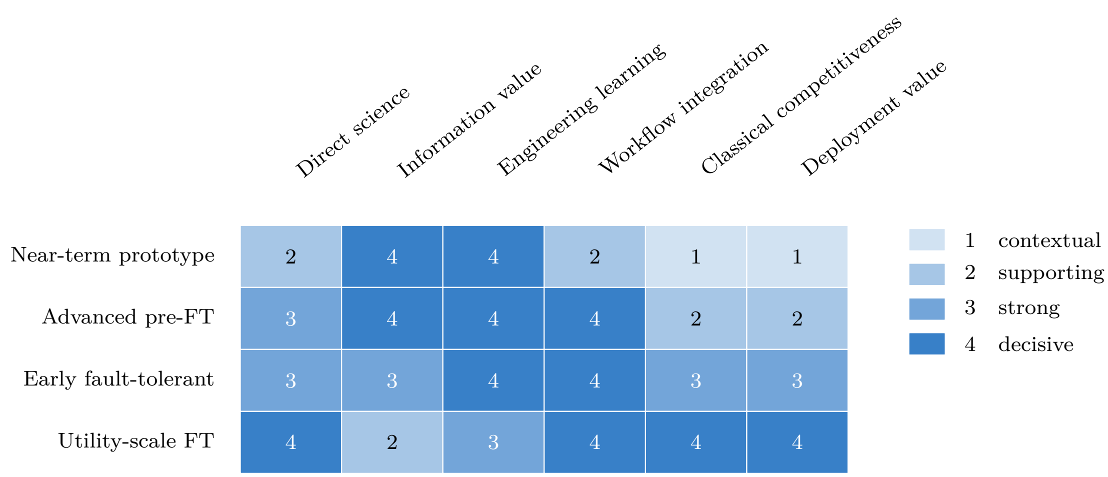

Figure 18.5: Pedagogical heat map of the value mix across development stages. The weights organize the discussion and carry no empirical calibration.

18.7 Expansion of the Feasible Scientific Domain

18.7.1 A workload-space definition

Speedup is one route to value. Access to a previously inaccessible scientific regime is another. Let $\mathcal{W}$ be a set of workloads. For workflow X, define the feasible scientific domain

$$\mathcal{D}_X = \{W \in \mathcal{W} : \epsilon_X(W) \leq \epsilon_{\text{req}}, \quad p_X(W) \geq p_{\min}, \quad T_X(W) \leq T_{\max}, \quad C_X(W) \leq C_{\max}\}.$$

The domain contains the workloads that meet every registered condition. A quantum processor creates domain-expansion value if it adds useful workloads outside the classical feasible domain,

$$\mathcal{D}_{Q\setminus C} = \mathcal{D}_Q \setminus \mathcal{D}_C.$$

This set is prospective until an experiment demonstrates the required accuracy, reliability, time, and cost.

A portfolio can weight workloads by their probability or importance. Let π_j be the weight of workload W_j. The best available portfolio value is

$$\mathcal{V}_{\text{portfolio}} = \sum_j \pi_j \max_{X\in\{Q,C\}} \mathcal{V}_X(W_j).$$

A heterogeneous computing center can therefore route each workload to the workflow with the larger accepted value. The QPU earns a place in the system through the workloads for which it changes the maximum.

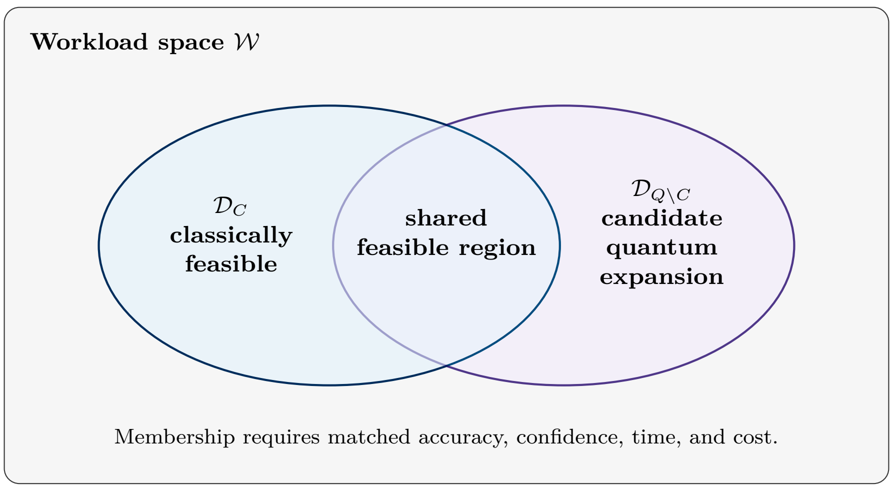

Figure 18.6: Conceptual feasible domain diagram. The candidate quantum expanded region gains economic meaning after the full scientific and resource contract has been met.

18.7.2 Experimental interpretation

A domain-expansion experiment should state the coordinate that moved the boundary. The improvement may come from accuracy, system size, evolution time, temperature, interaction strength, sign-problem severity, sampling quality, or a selected observable. A larger feasible domain can create value on workloads that both methods solve, including cases where the quantum workflow uses more time.

The comparison also has a date. Classical algorithms, accelerators, tensor methods, stochastic solvers, and learned surrogates continue to improve. Chapter 15 illustrated this movement with the classical reassessment of a many-body dynamics experiment. The boundary $\mathcal{D}_C$ should therefore record the algorithms, hardware, tolerances, and comparison date.

18.8 Scientific Domains with Plausible Economic Leverage

The strongest candidates share a common structure. A small part of the scientific calculation is limited by a quantum many-body representation, a difficult sampling distribution, a long-time correlation, or an optimization landscape. Classical computation is responsible for model construction, embedding, parameter fitting, postprocessing, and scientific acceptance. Quantum chemistry, materials science, and high-energy physics occupy a large fraction of the NERSC production workload and appear prominently in a recent capability assessment [104].

Table 18.3: Scientific domains, candidate quantum kernels, and economic triggers.

Scientific setting	Physical target	Candidate quantum kernel	Near-term and advanced-pre-FT role	Economic trigger
High-accuracy electronic structure	Ground and excited energies, forces, response properties, and correlated configurations	State preparation, sampled configuration subspaces, variational energy estimation, or later phase estimation	Test active spaces, mappings, symmetry repair, measurement allocation, projected diagonalization, and quantum-HPC orchestration [93], [115]	Accepted energies or forces enter a chemical decision outside the best classical accuracy-cost envelope
Quantum-enhanced DFT and embedding	Density functionals, Kohn–Sham potentials, impurity Green functions, and self-energies	Learn functionals from quantum states or solve a reduced impurity problem	Test density extraction, bath discretization, self-consistency, Green-function estimators, and spectral reconstruction [97], [116]	The quantum kernel improves a reusable functional, embedding solver, or material prediction across many classical calculations
Molecular and reaction modeling	Reaction energies, transition states, nonadiabatic structure, and conformer ordering	Correlated active-space solution and selected excited-state calculations	Qualify chemically relevant reduced models, ansatz transfer, excited-state constraints, and uncertainty propagation [115], [117]	Better ranking of candidates reduces synthesis, assay, or process-development cycles
Pharmaceutical discovery and molecular design	Binding-relevant conformers, protonation states, reaction pathways, excited states, and property predictions	Correlated active-space solution, selected response calculations, or quantum-generated features within a classical discovery pipeline	Test chemically faithful encodings, active-space transfer, measurement allocation, uncertainty propagation, and links to docking, molecular dynamics, or assay models [115], [117]	Accepted predictions reduce the number of synthesized candidates, assay rounds, or lead-optimization cycles
Catalysis and transition-metal chemistry	Strongly correlated catalytic intermediates and reaction barriers	Accurate energy estimation for large active spaces	Use prototype calculations to reduce resource uncertainty and test low-rank Hamiltonian constructions [118]	A more accurate catalyst decision changes yield, selectivity, energy demand, or feedstock use
Quantum transport and nanodevice design	Current, transmission, spectral density, nonequilibrium response, and open-system dynamics	Green-function, scattering, or time-evolution kernel for the active device region	Test small molecular junctions, boundary embeddings, current estimators, and noise sensitivity [119]	Device geometry or materials selection improves before costly fabrication and characterization

Scientific setting	Physical target	Candidate quantum kernel	Near-term and advanced-pre-FT role	Economic trigger
Strongly correlated materials	Spectral functions, magnetic phases, superconducting tendencies, and impurity physics	Impurity solver, spin-model dynamics, or analog Hamiltonian evolution	Compare reduced models with exact references, spectroscopy, and scattering data [96], [97]	The workflow identifies a useful regime that classical methods cannot reach at the required fidelity and turnaround time
Quantum dynamics and spectroscopy	Long-time correlation functions, response spectra, and nonequilibrium evolution	Hamiltonian simulation followed by selected-observable measurement	Test Trotterization, circuit compression, operator propagation, Fourier reconstruction, and laboratory-linked observables [96], [114]	A measured spectrum or dynamical response changes a material model, experiment plan, or device design
Sampling and constrained optimization	Structured distributions, rare events, spin-glass states, and constrained design variables	Photonic sampling, annealing, variational sampling, or specialized analog evolution	Establish sample quality, constraint satisfaction, embedding cost, and application-linked utility under strong classical baselines [67], [120]	Better samples or decisions reduce search cost after data loading, checking, retries, and classical postprocessing are counted

18.8.1 Electronic structure, DFT, and reaction science

Electronic structure has a direct numerical reason for attracting quantum-computing research. The many-electron state can require a combinatorial number of determinants in a chosen orbital basis. Classical approximations are still powerful, so a useful quantum kernel must target the portion of the problem where correlation error changes the scientific decision. Reviews of quantum computational chemistry describe the spectrum from small variational experiments to fault-tolerant phase-estimation targets [115], [117].

The economic trigger is rarely an isolated energy value. A useful result changes a ranking, force, barrier, spectrum, or uncertainty interval used downstream. Quantum-enhanced DFT offers a second route. A quantum calculation can supply information used to construct a reusable functional or embedding component, after which many classical calculations exploit the learned object [116]. This reuse can amplify the value of one expensive quantum calculation.

18.8.2 Transport, devices, and correlated materials

Nanodevice design connects microscopic quantum mechanics with fabrication cost. A quantum transport kernel may estimate transmission, current, or a Green function for an active region embedded between classically described contacts. Early algorithms for single-molecule transport show how this boundary can be formulated [119]. The economic value arises if the result changes geometry, contact material, layer sequence, or bias conditions before fabrication.

Strongly correlated materials provide a similar accelerator boundary. DFT can construct a one-electron environment, DMFT can define an impurity problem, and a QPU can target the correlated local solver. The output still passes through bath fitting, self-consistency, analytic continuation, and comparison with spectroscopy. A hardware result gains value after the complete loop converges and the material prediction changes.

18.8.3 Dynamics, sampling, and optimization

Long-time dynamics can challenge tensor methods as entanglement grows. Quantum processors naturally represent the evolving state, yet finite coherence, state preparation, measurement demand, and Fourier resolution can move the bottleneck. Laboratory-linked observables such as a dynamical structure factor provide a direct acceptance target.

Sampling and optimization require extra care. A hard-to-simulate distribution can establish a computational separation and still have limited application value. The sampled object must enter a decision, estimator, or scientific observable. Photonic sampling and programmable spin-glass experiments illustrate platform-specific reach [67], [120]. Economic qualification then adds sample checking, embedding, classical search, failure recovery, and output usefulness.

18.9 Platform Diversity and Workload Fit

18.9.1 Why the platform matters

Quantum hardware families expose different state spaces, native interactions, connectivity, measurement models, cycle times, and error mechanisms. These differences shape the best accelerator boundary. A platform-independent economic metric can compare accepted results. The benchmark contract and resource ledger still need platform-specific terms.

The table below spans major and emerging computational families active in the 2026 research landscape. Categories overlap. Bosonic encodings can live inside superconducting or photonic hardware, defect spins can act as processors or network nodes, and hybrid systems can couple several carriers. The table organizes workload fit instead of assigning a universal rank. Broad quantum-simulator roadmaps provide additional architectural context [103], [121].

Table 18.4: Major and emerging quantum-computing hardware families and the value they can create before utility-scale fault tolerance.

Hardware family	Physical carrier and native interaction	Computational form	Near-term or transferable value	Central bottleneck for economic deployment
Superconducting Josephson circuits	Anharmonic microwave modes coupled through capacitive, inductive, resonator, or tunable interactions	Fast digital gates, bosonic modes, analog lattices, dynamic circuits, and surface-code style correction	Large compiled experiments, quantum-HPC integration, mitigation studies, repeated syndrome extraction, and below-threshold scaling [54], [65]	Cryogenic control density, coherence, leakage, crosstalk, calibration load, wiring, decoder integration, and logical-resource overhead
Trapped atomic ions	Internal atomic states coupled by collective motional modes or photonic links	High-fidelity digital gates, flexible connectivity, qudits, modular nodes, and analog spin models	Precise algorithm tests, fault-tolerant primitives, logical gates, metrology, and long coherent circuits [106], [107], [122]	Gate duration, optical control scale, motional-mode management, shuttling or networking, and system throughput
Neutral-atom and Rydberg arrays	Optical-tweezer atoms with Rydberg blockade, transport, and local measurement	Reconfigurable digital processors, analog spin Hamiltonians, erasure-aware logic, and logical arrays	Large programmable geometries, materials and many-body simulation, moving-qubit architectures, and logical processing [108], [109]	Atom loss, gate and measurement fidelity, optical complexity, cycle time, reloading, and sustained logical throughput
Photonic qubits and continuous-variable photonics	Single photons, squeezed modes, interferometers, nonlinearities, detectors, and feedforward	Boson sampling, Gaussian and non-Gaussian processing, measurement-based computing, and networked computation	Programmable sampling, graph and molecular demonstrations, room-temperature transport, and natural networking [67]	Source and detector efficiency, loss, indistinguishability, deterministic interactions, feedforward latency, and error-corrected resource generation
Gate-defined semiconductor spin qubits	Electron or hole spins in silicon, germanium, or related quantum dots with exchange and microwave control	Dense digital processors, shuttling architectures, encoded spin qubits, and cryogenic-CMOS integration	Foundry-compatible process learning, compact control, multi-qubit logic, early correction, and logical operations [110], [123]	Device variability, charge noise, calibration scale, readout, exchange control, interconnect, and cryogenic electronics
Donor electron and nuclear spins in silicon	Donor-bound electron and nuclear spins coupled by exchange, hyperfine control, or mediated links	Long-lived memory, multilevel registers, digital gates, and local processor nodes	High-coherence memory, atomically engineered devices, donor-array control, and hybrid electron-nuclear protocols	Deterministic placement, coupling range, readout, yield, wiring, and scaling from local registers to connected arrays
Diamond and other defect-center spins	Electronic and nuclear spins around optically active defects with microwave, optical, and hyperfine control	Small registers, network nodes, sensors, logical qubits, and distributed processing	Fault-tolerant primitives, quantum memory, remote entanglement, nanoscale sensing, and processor-network co-design [112]	Photon collection, spectral stability, deterministic fabrication, nuclear-register control, network rate, and device uniformity
Bosonic cavity, cat, binomial, GKP, and qudit encodings	Oscillator states in microwave or optical modes with nonlinear control and ancilla-assisted measurement	Hardware-efficient logical qubits, qudits, autonomous correction, and analog-digital hybrids	Break-even error correction, reduced ancilla count, compact logical storage, and new gate constructions [111]	State preparation, loss and dephasing channels, ancilla faults, nonlinear control, leakage across levels, and universal fault-tolerant gate overhead
Ultracold atoms in optical lattices	Neutral atoms with tunneling, contact interactions, synthetic gauge fields, and quantum-gas microscopy	Analog Hubbard, spin, gauge, and many-body simulators with selected digital control	Direct exploration of phase diagrams, transport, thermalization, and correlated matter across large lattices [121]	Hamiltonian calibration, temperature, state preparation, model mapping, observable access, and quantitative comparison with target materials
Polar molecules and molecular tweezer arrays	Rotational states with long-range dipolar interactions and optical or microwave control	Analog spin models, quantum simulation, qudits, and prospective digital gates	Long-range interaction studies, chemistry-informed simulation, and new connectivity patterns	Cooling, trapping, coherence, state-resolved detection, gate fidelity, molecule loss, and array scale

Hardware family	Physical carrier and native interaction	Computational form	Near-term or transferable value	Central bottleneck for economic deployment
NMR and ensemble molecular spins	Nuclear spins controlled by radio-frequency pulses in molecular ensembles	Small digital algorithms, quantum control, spectroscopy, and ensemble information processing	Foundational algorithm demonstrations, pulse design, control theory, and quantum sensing [124]	Pseudopure-state scaling, ensemble readout, polarization, addressability, and limited scalable entanglement
Quantum annealers and adiabatic processors	Superconducting flux variables or other Ising degrees of freedom with programmable couplings and schedules	Annealing, sampling, optimization, spin-glass studies, and quantum simulation	Large embedded optimization experiments, sampling studies, control-schedule learning, and application screening [120]	Embedding overhead, analog control error, thermal effects, fair classical baselines, readout, and mapping from samples to useful decisions
Topological and Majorana-based devices	Candidate non-Abelian excitations, parity states, superconducting-semiconductor structures, or related topological modes	Protected qubits, braiding or measurement-based gates, and topological memory proposals	Materials qualification, parity measurement, poisoning studies, device-network design, and tests of protection mechanisms [125]	Unambiguous mode identification, quasiparticle poisoning, controllable gates, scalable readout, materials reproducibility, and complete logical demonstrations
Electrons on helium and trapped-electron routes	Electron charge or spin states above cryogenic surfaces or in electromagnetic traps	Digital qubits, analog electron arrays, and hybrid charge-spin processing	Clean material environment, unusual coupling mechanisms, and alternative scaling studies	Trapping stability, motion and heating, fabrication, control integration, readout, and reproducible multi-electron operation
Rare-earth-ion and solid-state optical ensembles	Optical and spin transitions in doped crystals or color-center ensembles	Quantum memory, optical processing, network repeaters, and spectral-multiplexed registers	Long-lived storage, transduction, multiplexing, and distributed-computing infrastructure	Efficient interfaces, spectral control, deterministic coupling, local gates, collection efficiency, and processor-level programmability
Hybrid acoustic, magnonic, optomechanical, and transducer systems	Phonons, magnons, microwave photons, optical photons, and matter qubits coupled across domains	Quantum buses, memories, converters, analog simulators, and modular links	Interconnect experiments, frequency conversion, distributed sensing, memory, and heterogeneous architecture development	End-to-end efficiency, added noise, bandwidth, thermal occupation, impedance matching, and scalable fabrication

18.9.2 Platform-specific contracts

The economic metric stays common and the acceptance event changes with the hardware. A photonic job records source probability, transmission loss, interference visibility, detector efficiency, feedforward, and accepted-event rate. An annealing job records embedding, schedule, gauge choices, chain breaks, sample quality, and classical repair. An analog simulator records Hamiltonian calibration, preparation fidelity, temperature, local observables, and mapping to the target model. A logical gate processor records code distance, syndrome cycles, decoder, logical error, magic-state or non-Clifford cost, and logical throughput.

This approach permits fair comparison without compressing every platform into gate fidelity. The common output is a scientifically accepted result. The path to that output is still tied to the native physics.

18.10 What Near-Term Progress Must Transfer Forward

Fault-tolerant hardware will inherit more than algorithms. It will inherit the representations, software, measurements, controls, and acceptance rules developed during the near-term phase. The most valuable transfer assets are those that stay useful after the physical processor changes.

- **Problem representations.** Active spaces, lattice encodings, impurity mappings, symmetry sectors, boundary conditions, and output observables determine the logical workload.
- **Compiler and scheduling knowledge.** Gate synthesis, qubit placement, pulse scheduling, dynamic control, measurement grouping, and circuit compression influence logical depth and throughput.
- **Estimator design.** Shot allocation, observable grouping, classical shadows, amplitude estimation choices, ratio estimators, and covariance-aware analysis is central after physical noise diminishes.
- **Error-management interfaces.** QEM experiments reveal sensitivity to calibration and estimator variance. QEC experiments add syndrome extraction, decoding, logical acceptance, and leakage handling. The classical control path must carry both.
- **Control and readout electronics.** Timing, waveform generation, cryogenic delivery, amplification, digitization, classification, and feedback scale with logical hardware.
- **Quantum-HPC orchestration.** Model preparation, job graphs, queueing, data movement, provenance, checkpointing, and recovery are still part of time to solution.
- **Resource estimation.** Prototype measurements replace assumed gate rates, decoder costs, and measurement counts with observed distributions. Resource forecasts improve as these distributions mature.
- **Domain acceptance rules.** Chemical accuracy, conservation residuals, spectral resolution, material trends, transport tolerances, and design margins define whether a logical calculation is useful.

A later system truly surpasses the near-term phase after it absorbs these lessons. The same software or method may require revision, yet the questions established by the earlier work remain. What is the accepted output? Which error source controls the decision? How many repeated jobs are needed? Which classical stages dominate? Which hardware property changes the feasible scientific domain?

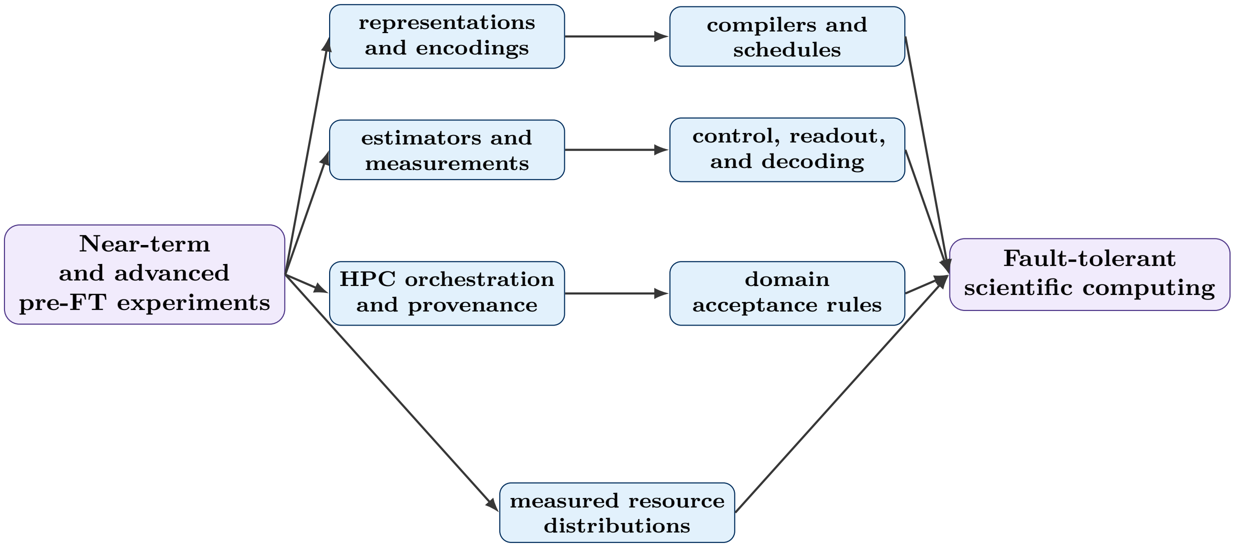

Figure 18.7: Transfer assets connect prototype computing to fault-tolerant scientific workflows. Their value lies in the reduction of future integration, estimation, and decision uncertainty.

18.11 A Reusable Economic Decision Record

A technical result can be converted into an economic decision record without hiding its physics. For workload W, define

$$\mathcal{E}_Q = (W, K, P, \epsilon_{\mathrm{req}}, 1-\alpha, C_Q, T_Q, p_Q, B_Q, L_Q, \mathcal{V}_Q, \mathcal{D}_Q, \tau)\,.$$

The entries are

- K, the quantum kernel and its classical boundary;
- P, the hardware platform, protection modality, processor region, and software stack;
- ϵ_{req} and $1-\alpha$, the scientific accuracy and confidence contract;
- C_Q and T_Q, the complete attempt cost and critical-path time;
- p_Q, the probability that a complete job meets the scientific and service conditions;
- B_Q and L_Q, the benefit of acceptance and the loss associated with rejection or delay;
- $\mathcal{V}_Q$, the resulting expected net value;
- $\mathcal{D}_Q$, the feasible scientific domain under the registered limits;
- τ, the comparison date and reproducibility window.

The classical record $\mathcal{E}_C$ uses the same accounting boundary and acceptance contract. The comparison can then report

$$\Delta\mathcal{V} = \mathcal{V}_Q - \mathcal{V}_C, \qquad \Gamma_C = \frac{C_Q/p_Q}{C_C/p_C}, \qquad \Gamma_T = \frac{T_Q/p_Q}{T_C/p_C}.$$

A positive $\Delta\mathcal{V}$ identifies the larger expected value. Ratios below one identify a lower expected cost or time per accepted result.

Decision sentence template

For workload W, the quantum-assisted kernel K on platform P met the registered accuracy and confidence conditions with acceptance probability p_Q, expected cost C_Q/p_Q per accepted result, expected time T_Q/p_Q, and expected net value $\mathcal{V}_Q$. The matched classical workflow produced the corresponding quantities under the same accounting boundary and comparison date. The decision follows from the value difference, accepted-result ratios, and any scientifically important workloads added to the feasible domain.

Check your understanding

1. A prototype produces no speedup and changes a hardware investment decision. Write the terms in V_{proto} that can still make the experiment economically positive.
2. A quantum attempt costs twice as much as a classical attempt. Its probability of producing an accepted result is three times larger. Evaluate Γ_C and interpret the result.
3. Two workflows have the same cost and acceptance probability. One returns a result in one day and the other in ten days. Explain how a time-sensitive benefit B_0e^{-rT} can change the preferred workflow.
4. Choose one platform from table 18.4. State a platform-specific acceptance contract and one transferable asset that are still useful for a later fault-tolerant system.

Key takeaway

Near-term quantum value has several measurable forms. A prototype can answer a bounded scientific question, reduce uncertainty in a larger decision, produce reusable technical assets, lower the cost of future integration, or expand the scientific regimes that can be studied. Fault-tolerant computing raises the accessible depth and reliability, yet its economic value still comes from accepted results inside complete quantum-classical workflows.

18.12 Concluding Perspective

To accurately validate quantum performance, we must employ a comprehensive evidence system. This begins with exact classical simulations that serve as decisive benchmarks for small-scale operations, which are then extended via structured algorithms tailored to specific circuit geometries and entanglement patterns. While hardware diagnostics identify physical failure modes, statistical qualification transforms this raw data into meaningful intervals, effective sample sizes, and reproducibility metrics. Ultimately, logical scaling allows us to distinguish a genuine success fraction from behavior that falls below the required threshold.

However, the QPU is only one component of the broader scientific computing workflow. The actual value of a result is determined by the entire pipeline: from model preparation and classical reconstruction to convergence, matched baselines, total execution time, and cost. Because the QPU acts as a specific kernel within this larger system, quantum enhancement, acceleration, and mitigation must be treated as distinct conclusions, each requiring its own specific acceptance criteria

Chapter 19

Glossary

Table 19.1: Compact glossary of quantum simulation, verification, and scientific-computing terms.

Term	Meaning
Amplitude	Complex coefficient of a basis state in a pure quantum state
Application qualification	Evidence stage spanning hardware execution, scientific validity, utility, classical competitiveness, and deployment
Assignment matrix	Conditional-probability matrix mapping prepared basis labels to reported measurement labels
Band structure	Energy eigenvalues $E_n(\mathbf{k})$ of a periodic system as functions of crystal momentum
Benchmark contract	Specification of inputs, operations, width, depth, duration, measurement, decoding, and postselection for a reported score
Bitstring	Binary data sample obtained from computational-basis measurement
Bloch state	Eigenstate of a periodic Hamiltonian written as a plane-wave phase multiplied by a lattice-periodic function
Bond dimension	Internal tensor-network index dimension controlling the representable entanglement across a cut
Breakeven logical qubit	Encoded memory or operation that outperforms a matched physical reference under a declared metric
Brillouin zone	Primitive reciprocal-space cell used to label crystal momentum in a periodic solid
Calibration	Procedure that tunes control pulses, frequencies, readout thresholds, and hardware parameters
Causal lightcone	Earlier circuit region that can influence a chosen terminal observable
Classical shadow	Randomized-measurement estimator that supports many compatible observables from a shared data set
Clifford circuit	Circuit whose gates map Pauli operators to Pauli operators under conjugation
Cost to solution	Minimum total cost required to meet a scientific tolerance at a declared confidence

Term	Meaning
Cross-entropy benchmarking	Random-circuit benchmark that compares sampled bitstrings with ideal probabilities
Density matrix	Positive semidefinite, unit-trace matrix representing a pure or mixed quantum state
Dynamical mean-field theory	Embedding framework that maps a correlated lattice problem to a self-consistent quantum impurity problem
Dynamical structure factor	Frequency- and momentum-resolved correlation spectrum $S(\mathbf{q}, \omega)$ measured in scattering experiments
Effective sample size	Independent-sample count that gives the same estimator variance as a correlated data set
Economic deployment	Repeated scientific computation meeting requirements for cost, throughput, uptime, reliability, and reproducibility
Error mitigation	Estimator-design strategy that uses extra circuits, calibration, modified measurements, or postprocessing to reduce selected noise bias
Estimator	Function of finite data used to infer a model or experimental quantity
Fault-tolerant logical operation	Encoded operation whose construction preserves logical protection under the applicable correction protocol
Green function	Correlation or response function that encodes propagation, spectral weight, and excitation information
Inelastic neutron scattering	Experimental probe that measures momentum and energy transfer between neutrons and material excitations
Layer fidelity	Benchmark that estimates simultaneous layer quality over connected qubits
Lattice Boltzmann method	Mesoscopic CFD method that evolves discrete particle populations by collision and streaming
Matrix product state	Tensor-network representation suited to many one-dimensional or weakly entangled states
Mirror circuit	Circuit containing a randomized sequence and an inverse construction with an accessible ideal output
Model discrepancy	Difference between a mathematical model and the physical system it represents
Numerov algorithm	Multistep method for second-order differential equations lacking a first-derivative term
OBP	Operator backpropagation, which conjugates a measured observable by a removed circuit suffix
PEC	Probabilistic error cancellation using quasiprobability sampling of an inverse-noise representation
PNA	Propagated noise absorption, which incorporates a learned inverse-noise model into a modified observable
POVM	Positive-operator-valued measure describing probabilities of general quantum outcomes
QEM	Quantum error mitigation

Term	Meaning
QLBM	Quantum lattice Boltzmann method, a quantum algorithmic family for collision, streaming, encoding, and readout of lattice populations
Quantum acceleration	Reduction in end-to-end time to solution or asymptotic resource scaling relative to a matched classical baseline
Quantum co-processor	QPU used as a selected numerical kernel inside a larger classical workflow
Quantum enhancement	Improvement in a declared accuracy, quality, sampling, or accessible-regime metric associated with a quantum resource
Quantum volume	System benchmark based on square randomized circuits and heavy-output generation
QuTiP	Open-source Quantum Toolbox in Python for closed and open quantum dynamics
Readout mitigation	Calibration and postprocessing method that corrects selected measurement-assignment errors
Reference calculation	High-confidence calculation inside a specified mathematical model
Scientific utility	Ability of a validated workflow to support a useful prediction, inference, explanation, or engineering decision
Shaded lightcone	Error-channel influence bounds used to target mitigation resources within a causal region
Shot	One execution and measurement of a quantum circuit
Tensor network	Factorized representation of a high-dimensional quantum object
Time to solution	Minimum wall-clock time required to meet a scientific tolerance at a declared confidence
Variational quantum eigensolver	Hybrid algorithm that optimizes a parameterized quantum state to minimize an energy expectation
Variational quantum deflation	Excited-state extension that adds overlap penalties against lower-energy states
Wilson interval	Binomial confidence interval obtained by inverting a score test
Zero-failure bound	One-sided binomial lower bound inferred after a run with zero observed failures
ZNE	Zero-noise extrapolation using estimates measured at scaled noise levels

Appendix A

Python Snippets

A.1 QuTiP Master-Equation Solver

```python
"""Simulate a relaxing driven qubit with QuTiP."""

from dataclasses import dataclass

import matplotlib.pyplot as plt
import numpy as np
import qutip as qt


@dataclass(frozen=True)
class SimulationConfig:
    """Control knobs for the open-system simulation."""

    omega: float = 1.0
    relaxation_rate: float = 0.10
    start_time: float = 0.0
    stop_time: float = 10.0
    num_time_points: int = 250
    figure_dpi: int = 250


def run_simulation(config: SimulationConfig) -> None:
    """Solve the master equation and render sigma-z values."""
    plt.rcParams["figure.dpi"] = config.figure_dpi

    hamiltonian = 0.5 * config.omega * qt.sigmax()
    collapse_operators = [
        np.sqrt(config.relaxation_rate) * qt.sigmam()
    ]
    initial_state = qt.basis(2, 1)
    times = np.linspace(
        config.start_time,
        config.stop_time,
        config.num_time_points,
    )

    result = qt.mesolve(
        hamiltonian,
```

```
        initial_state,
        times,
        collapse_operators,
        e_ops=[qt.sigmaz()],
    )

    figure, axis = plt.subplots()
    axis.plot(
        times,
        result.expect[0],
        label=r"$\langle\sigma_z\rangle$",
    )
    axis.set_xlabel("Time")
    axis.set_ylabel("Expectation value")
    axis.legend()
    figure.tight_layout()
    plt.show()


run_simulation(SimulationConfig())
```

A.2 Wilson Interval Helper

```
"""Compute a two-sided Wilson interval for a Bernoulli proportion."""

from math import sqrt
from statistics import NormalDist


def wilson_interval(
    successes: int,
    trials: int,
    confidence: float = 0.95,
) -> tuple[float, float]:
    """Return the lower and upper Wilson score limits."""
    if trials <= 0:
        raise ValueError("trials must be positive")
    if successes < 0 or successes > trials:
        raise ValueError(
            "successes must lie between zero and trials"
        )
    if confidence <= 0.0 or confidence >= 1.0:
        raise ValueError(
            "confidence must lie between zero and one"
        )

    proportion = successes / trials
    z_score = NormalDist().inv_cdf(
        0.5 + confidence / 2.0
    )
    denominator = 1.0 + z_score**2 / trials
    center = (
        proportion + z_score**2 / (2.0 * trials)
    ) / denominator
```

```
    half_width = (
        z_score
        * sqrt(
            proportion * (1.0 - proportion) / trials
            + z_score**2 / (4.0 * trials**2)
        )
        / denominator
    )
    return center - half_width, center + half_width
```

A.3 Extended Primer Exercises

Check your understanding

1. For
$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$
write the density operator $\rho_+ = |+\rangle\langle+|$. Identify the entries that determine computational-basis probabilities and the entries that carry coherence.

2. Compare ρ_+ with the incoherent mixture
$$\rho_{\mathrm{mix}} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|.$$
Both states give equal probabilities for 0 and 1 in the computational basis. Explain how an X-basis measurement distinguishes them.

3. Distinguish the model expectation $\mu = \mathrm{Tr}(\rho O)$ from the finite-data estimator $\widehat{\mu}$. Name one source of statistical uncertainty, one source of systematic error, and one source of model discrepancy.

4. For confidence level 95% and additive precision $\delta = 0.01$, evaluate the Hoeffding shot bound. Compare it with the variance-based estimate near $\mu = 0$. For the two-point ZNE estimator with weights 2 and -1, also calculate the variance multiplier and explain its experimental consequence.

5. Consider a system of $n = 12$ qubits. Calculate the raw memory required to store a `complex128` state vector versus a dense density matrix. If you increase the system to $n = 13$ qubits, by what factors do these two memory requirements increase? Why does this discrepancy create a "memory wall" for open-system simulations much sooner than for pure-state simulations?

6. A state is given by $|\psi\rangle = |0\rangle^{\otimes n}$. What is the minimum bond dimension χ required for an exact Matrix Product State (MPS) representation? Contrast this with a maximally entangled state where the Schmidt rank at the central cut is $2^{n/2}$. Explain why the "representation question" is more important than the qubit count when determining if a simulation is classically tractable.

7. Suppose a readout classifier has a confusion matrix with $P(b_1|1) = 0.92$ and $P(b_0|0) = 0.95$. If the hardware returns an observed probability $p_{\mathrm{obs}}(1) = 0.80$, calculate the corrected probability $p_{\mathrm{true}}(1)$. Describe how the condition number of the assignment matrix affects the statistical uncertainty of the mitigated result.

8. An experiment reports zero failures over $N = 300$ independent shots. Using the one-sided zero-failure bound at $\alpha = 0.05$, determine the lower confidence bound p_L for the success

probability. How many additional consecutive successes are required to support a claim that $p \geq 0.999$ at the same confidence level?

References

[1] Z. Cai, R. Babbush, S. C. Benjamin, S. Endo, W. J. Huggins, Y. Li, J. R. McClean, and T. E. O'Brien, "Quantum error mitigation," *Reviews of Modern Physics*, vol. 95, p. 045 005, 2023. DOI: 10.1103/RevModPhys.95.045005.

[2] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, "Completely positive dynamical semigroups of N-level systems," *Journal of Mathematical Physics*, vol. 17, no. 5, pp. 821–825, 1976. DOI: 10.1063/1.522979.

[3] G. Lindblad, "On the generators of quantum dynamical semigroups," *Communications in Mathematical Physics*, vol. 48, no. 2, pp. 119–130, 1976. DOI: 10.1007/BF01608499.

[4] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*. Oxford, U.K.: Oxford University Press, 2002. DOI: 10.1093/acprof:oso/9780199213900.001.0001.

[5] K. Temme, S. Bravyi, and J. M. Gambetta, "Error mitigation for short-depth quantum circuits," *Physical Review Letters*, vol. 119, p. 180 509, 2017. DOI: 10.1103/PhysRevLett.119.180509.

[6] Y. Li and S. C. Benjamin, "Efficient variational quantum simulator incorporating active error minimization," *Physical Review X*, vol. 7, p. 021 050, 2017. DOI: 10.1103/PhysRevX.7.021050.

[7] P. Moin and K. Mahesh, "Direct numerical simulation: A tool in turbulence research," *Annual Review of Fluid Mechanics*, vol. 30, pp. 539–578, 1998. DOI: 10.1146/annurev.fluid.30.1.539.

[8] M. M. Rai, "Utilizing direct numerical simulations of transition and turbulence in design optimization," NASA Ames Research Center, Tech. Rep. TM-2015-218932, Dec. 2015. [Online]. Available: https://ntrs.nasa.gov/citations/20150023011.

[9] T. Häner and D. S. Steiger, "0.5 petabyte simulation of a 45-qubit quantum circuit," in *Proc. Int. Conf. High Performance Computing, Networking, Storage and Analysis*, 2017, pp. 1–10. DOI: 10.1145/3126908.3126947.

[10] Google Quantum AI, *qsim and qsimh*, Accessed: Jul. 10, 2023, 2023. [Online]. Available: https://quantumai.google/qsim/overview.

[11] Qiskit Aer contributors, *AerSimulator*, Accessed: Jul. 10, 2023, 2023. [Online]. Available: https://qiskit.github.io/qiskit-aer/stubs/qiskit_aer.AerSimulator.html.

[12] Qiskit Aer contributors, *Matrix product state simulation method*, Accessed: Jul. 10, 2023, 2023. [Online]. Available: https://qiskit.github.io/qiskit-aer/tutorials/7_matrix_product_state_method.html.

[13] G. Vidal, "Efficient classical simulation of slightly entangled quantum computations," *Physical Review Letters*, vol. 91, p. 147 902, 2003. DOI: 10.1103/PhysRevLett.91.147902.

[14] I. L. Markov and Y. Shi, "Simulating quantum computation by contracting tensor networks," *SIAM Journal on Computing*, vol. 38, no. 3, pp. 963–981, 2008. DOI: 10.1137/050644756.

[15] J. Gray and S. Kourtis, "Hyper-optimized tensor network contraction," *Quantum*, vol. 5, p. 410, 2021. DOI: 10.22331/q-2021-03-15-410.

[16] T. Peng, A. W. Harrow, M. Ozols, and X. Wu, "Simulating large quantum circuits on a small quantum computer," *Physical Review Letters*, vol. 125, p. 150 504, 2020. DOI: 10.1103/PhysRevLett.125.150504.

[17] H.-Y. Huang, R. Kueng, and J. Preskill, "Predicting many properties of a quantum system from very few measurements," *Nature Physics*, vol. 16, pp. 1050–1057, 2020. DOI: 10.1038/s41567-020-0932-7.

[18] S. Endo, S. C. Benjamin, and Y. Li, "Practical quantum error mitigation for near-future applications," *Physical Review X*, vol. 8, p. 031 027, 2018. DOI: 10.1103/PhysRevX.8.031027.

[19] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018. DOI: 10.22331/q-2018-08-06-79.

[20] A. Kandala, K. Temme, A. D. Córcoles, A. Mezzacapo, J. M. Chow, and J. M. Gambetta, "Error mitigation extends the computational reach of a noisy quantum processor," *Nature*, vol. 567, pp. 491–495, 2019. DOI: 10.1038/s41586-019-1040-7.

[21] E. van den Berg, Z. K. Minev, A. Kandala, and K. Temme, "Probabilistic error cancellation with sparse Pauli–Lindblad models on noisy quantum processors," *Nature Physics*, vol. 19, pp. 1116–1121, 2023. DOI: 10.1038/s41567-023-02042-2.

[22] F. B. Maciejewski, Z. Zimborás, and M. Oszmaniec, "Mitigation of readout noise in near-term quantum devices by classical post-processing based on detector tomography," *Quantum*, vol. 4, p. 308, 2020. DOI: 10.22331/q-2020-04-24-308.

[23] S. Bravyi, S. Sheldon, A. Kandala, D. C. McKay, and J. M. Gambetta, "Mitigating measurement errors in multiqubit experiments," *Physical Review A*, vol. 103, p. 042 605, 2021. DOI: 10.1103/PhysRevA.103.042605.

[24] B. Koczor, "Exponential error suppression for near-term quantum devices," *Physical Review X*, vol. 11, p. 031 057, 2021. DOI: 10.1103/PhysRevX.11.031057.

[25] W. J. Huggins et al., "Virtual distillation for quantum error mitigation," *Physical Review X*, vol. 11, p. 041 036, 2021. DOI: 10.1103/PhysRevX.11.041036.

[26] X. Bonet-Monroig, R. Sagastizabal, M. Singh, and T. E. O'Brien, "Low-cost error mitigation by symmetry verification," *Physical Review A*, vol. 98, p. 062 339, 2018. DOI: 10.1103/PhysRevA.98.062339.

[27] R. Sagastizabal et al., "Error mitigation by symmetry verification on a variational quantum eigensolver," *Physical Review A*, vol. 100, 010302(R), 2019. DOI: 10.1103/PhysRevA.100.010302.

[28] P. Czarnik, A. Arrasmith, P. J. Coles, and L. Cincio, "Error mitigation with Clifford quantum-circuit data," *Quantum*, vol. 5, p. 592, 2021. DOI: 10.22331/q-2021-11-26-592.

[29] A. Lowe, M. H. Gordon, P. Czarnik, A. Arrasmith, P. J. Coles, and L. Cincio, "Unified approach to data-driven quantum error mitigation," *Physical Review Research*, vol. 3, p. 033 098, 2021. DOI: 10.1103/PhysRevResearch.3.033098.

[30] R. LaRose et al., "Mitiq: A software package for error mitigation on noisy quantum computers," *Quantum*, vol. 6, p. 774, 2022. DOI: 10.22331/q-2022-08-11-774.

[31] Mitiq contributors, *About error mitigation*, Accessed: Jul. 10, 2023, 2023. [Online]. Available: https://mitiq.readthedocs.io/en/stable/guide/error-mitigation.html.

[32] M. C. Tran, K. Sharma, and K. Temme, *Locality and error mitigation of quantum circuits*, arXiv:2303.06496 [quant-ph], 2023. DOI: 10.48550/arXiv.2303.06496. [Online]. Available: https://arxiv.org/abs/2303.06496.

[33] R. Takagi, S. Endo, S. Minagawa, and M. Gu, "Fundamental limits of quantum error mitigation," *npj Quantum Information*, vol. 8, p. 114, 2022. DOI: 10.1038/s41534-022-00618-z.

[34] K. Tsubouchi, T. Sagawa, and N. Yoshioka, "Universal cost bound of quantum error mitigation based on quantum estimation theory," *Physical Review Letters*, vol. 131, p. 210 601, 2023. DOI: 10.1103/PhysRevLett.131.210601.

[35] R. Takagi, H. Tajima, and M. Gu, "Universal sampling lower bounds for quantum error mitigation," *Physical Review Letters*, vol. 131, p. 210 602, 2023. DOI: 10.1103/PhysRevLett.131.210602.

[36] K. Wang et al., "Can error mitigation improve quantumness?" *PRX Quantum*, vol. 2, p. 030 350, 2021. DOI: 10.1103/PRXQuantum.2.030350.

[37] A. Eddins et al., "Stroboscopic qubit measurement with squeezing and parametric modulation," *Physical Review Letters*, vol. 120, p. 040 505, 2018. DOI: 10.1103/PhysRevLett.120.040505.

[38] Qiskit add-on contributors, *Operator backpropagation*, Accessed: Jul. 10, 2023, 2023. [Online]. Available: https://qiskit.github.io/qiskit-addon-obp/.

[39] M. Cerezo et al., "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, pp. 625–644, 2021. DOI: 10.1038/s42254-021-00348-9.

[40] P. D. Nation et al., "Scalable mitigation of measurement errors on quantum computers," *PRX Quantum*, vol. 2, p. 040 326, 2021. DOI: 10.1103/PRXQuantum.2.040326.

[41] Qiskit add-on contributors, *Propagated noise absorption*, Accessed: Jul. 10, 2023, 2023. [Online]. Available: https://qiskit.github.io/qiskit-addon-pna/.

[42] E. Magesan, J. M. Gambetta, and J. Emerson, "Scalable and robust randomized benchmarking of quantum processes," *Physical Review Letters*, vol. 106, p. 180 504, 2011. DOI: 10.1103/PhysRevLett.106.180504.

[43] E. Magesan et al., "Efficient measurement of quantum gate error by interleaved randomized benchmarking," *Physical Review Letters*, vol. 109, p. 080 505, 2012. DOI: 10.1103/PhysRevLett.109.080505.

[44] D. C. McKay, I. Hincks, E. J. Pritchett, M. Carroll, L. C. G. Govia, and S. T. Merkel, *Benchmarking quantum processor performance at scale*, arXiv:2311.05933 [quant-ph], 2023. DOI: 10.48550/arXiv.2311.05933. [Online]. Available: https://arxiv.org/abs/2311.05933.

[45] A. W. Cross, L. S. Bishop, S. Sheldon, P. D. Nation, and J. M. Gambetta, "Validating quantum computers using randomized model circuits," *Physical Review A*, vol. 100, p. 032 328, 2019. DOI: 10.1103/PhysRevA.100.032328.

[46] T. Proctor, S. Seritan, K. Rudinger, E. Nielsen, R. Blume-Kohout, and K. Young, "Scalable randomized benchmarking of quantum computers using mirror circuits," *Physical Review Letters*, vol. 129, p. 150 502, 2022. DOI: 10.1103/PhysRevLett.129.150502.

[47] S. Boixo et al., "Characterizing quantum supremacy in near-term devices," *Nature Physics*, vol. 14, pp. 595–600, 2018. DOI: 10.1038/s41567-018-0124-x.

[48] F. Arute et al., "Quantum supremacy using a programmable superconducting processor," *Nature*, vol. 574, pp. 505–510, 2019. DOI: 10.1038/s41586-019-1666-5.

[49] R. Blume-Kohout and K. C. Young, "A volumetric framework for quantum computer benchmarks," *Quantum*, vol. 4, p. 362, 2020. DOI: 10.22331/q-2020-11-15-362.

[50] T. Proctor et al., "Measuring the capabilities of quantum computers," *Nature Physics*, vol. 18, pp. 75–79, 2022. DOI: 10.1038/s41567-021-01409-7.

[51] D. P. DiVincenzo, “The physical implementation of quantum computation,” *Fortschritte der Physik*, vol. 48, no. 9–11, pp. 771–783, 2000. DOI: 10.1002/1521-3978(200009)48:9/11<771::AID-PROP771>3.0.CO;2-E.

[52] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. DOI: 10.1080/01621459.1927.10502953.

[53] C. J. Clopper and E. S. Pearson, “The use of confidence or fiducial limits illustrated in the case of the binomial,” *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934. DOI: 10.1093/biomet/26.4.404.

[54] Google Quantum AI, “Suppressing quantum errors by scaling a surface code logical qubit,” *Nature*, vol. 614, pp. 676–681, 2023. DOI: 10.1038/s41586-022-05434-1.

[55] A. Streltsov, G. Adesso, and M. B. Plenio, “Colloquium: Quantum coherence as a resource,” *Reviews of Modern Physics*, vol. 89, p. 041 003, 2017. DOI: 10.1103/RevModPhys.89.041003.

[56] A. Mohamed, E. Zohari, J. J. Pla, P. E. Barclay, and S. Barzanjeh, “Selective single- and double-mode quantum-limited amplifier,” *Physical Review Applied*, vol. 21, p. 064 052, 2024. DOI: 10.1103/PhysRevApplied.21.064052.

[57] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Physical Review Letters*, vol. 72, no. 22, pp. 3439–3443, 1994. DOI: 10.1103/PhysRevLett.72.3439.

[58] C. M. Caves, “Quantum limits on noise in linear amplifiers,” *Physical Review D*, vol. 26, no. 8, pp. 1817–1839, 1982. DOI: 10.1103/PhysRevD.26.1817.

[59] C. M. Caves, J. Combes, Z. Jiang, and S. Pandey, “Quantum limits on phase-preserving linear amplifiers,” *Physical Review A*, vol. 86, no. 6, p. 063 802, 2012. DOI: 10.1103/PhysRevA.86.063802.

[60] J. M. Arrazola, A. Delgado, B. R. Bardhan, and S. Lloyd, “Quantum-inspired algorithms in practice,” *Quantum*, vol. 4, p. 307, 2020. DOI: 10.22331/q-2020-08-13-307.

[61] V. Giovannetti, S. Lloyd, and L. Maccone, “Quantum random access memory,” *Physical Review Letters*, vol. 100, p. 160 501, 2008. DOI: 10.1103/PhysRevLett.100.160501.

[62] J. Biamonte et al., “Quantum machine learning,” *Nature*, vol. 549, pp. 195–202, 2017. DOI: 10.1038/nature23474.

[63] V. Giovannetti, S. Lloyd, and L. Maccone, “Advances in quantum metrology,” *Nature Photonics*, vol. 5, pp. 222–229, 2011. DOI: 10.1038/nphoton.2011.35.

[64] T. F. Rønnow et al., “Defining and detecting quantum speedup,” *Science*, vol. 345, no. 6195, pp. 420–424, 2014. DOI: 10.1126/science.1252319.

[65] Y. Kim et al., “Evidence for the utility of quantum computing before fault tolerance,” *Nature*, vol. 618, pp. 500–505, 2023. DOI: 10.1038/s41586-023-06096-3.

[66] T. Begušić, J. Gray, and G. K.-L. Chan, “Fast and converged classical simulations of evidence for the utility of quantum computing before fault tolerance,” *Science Advances*, vol. 10, no. 3, eadk4321, 2024. DOI: 10.1126/sciadv.adk4321.

[67] L. S. Madsen et al., “Quantum computational advantage with a programmable photonic processor,” *Nature*, vol. 606, pp. 75–81, 2022. DOI: 10.1038/s41586-022-04725-x.

[68] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, “A quantum engineer’s guide to superconducting qubits,” *Applied Physics Reviews*, vol. 6, p. 021 318, 2019. DOI: 10.1063/1.5089550.

[69] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, “Circuit quantum electrodynamics,” *Reviews of Modern Physics*, vol. 93, p. 025 005, 2021. DOI: 10.1103/RevModPhys.93.025005.

[70] Y. Y. Gao, M. A. Rol, S. Touzard, and C. Wang, "Practical guide for building superconducting quantum devices," *PRX Quantum*, vol. 2, p. 040 202, 2021. DOI: 10.1103/PRXQuantum.2.040202.

[71] C. Lanczos, "An iteration method for the solution of the eigenvalue problem of linear differential and integral operators," *Journal of Research of the National Bureau of Standards*, vol. 45, no. 4, pp. 255–282, 1950. DOI: 10.6028/jres.045.026.

[72] P. J. Knowles and N. C. Handy, "A new determinant-based full configuration interaction method," *Chemical Physics Letters*, vol. 111, no. 4–5, pp. 315–321, 1984. DOI: 10.1016/0009-2614(84)85513-X.

[73] P. J. Knowles and N. C. Handy, "A determinant based full configuration interaction program," *Computer Physics Communications*, vol. 54, no. 1, pp. 75–83, 1989. DOI: 10.1016/0010-4655(89)90033-7.

[74] S. R. White, "Density matrix formulation for quantum renormalization groups," *Physical Review Letters*, vol. 69, no. 19, pp. 2863–2866, 1992. DOI: 10.1103/PhysRevLett.69.2863.

[75] S. R. White, "Density-matrix algorithms for quantum renormalization groups," *Physical Review B*, vol. 48, no. 14, pp. 10 345–10 356, 1993. DOI: 10.1103/PhysRevB.48.10345.

[76] U. Schollwöck, "The density-matrix renormalization group in the age of matrix product states," *Annals of Physics*, vol. 326, no. 1, pp. 96–192, 2011. DOI: 10.1016/j.aop.2010.09.012.

[77] G. Vidal, "Efficient simulation of one-dimensional quantum many-body systems," *Physical Review Letters*, vol. 93, p. 040 502, 2004. DOI: 10.1103/PhysRevLett.93.040502.

[78] W. L. McMillan, "Ground state of liquid ^{4}He," *Physical Review*, vol. 138, no. 2A, A442–A451, 1965. DOI: 10.1103/PhysRev.138.A442.

[79] W. M. C. Foulkes, L. Mitas, R. J. Needs, and G. Rajagopal, "Quantum monte carlo simulations of solids," *Reviews of Modern Physics*, vol. 73, no. 1, pp. 33–83, 2001. DOI: 10.1103/RevModPhys.73.33.

[80] J. B. Anderson, "A random-walk simulation of the schrödinger equation: H_3^+," *The Journal of Chemical Physics*, vol. 63, no. 4, pp. 1499–1503, 1975. DOI: 10.1063/1.431514.

[81] P. J. Reynolds, D. M. Ceperley, B. J. Alder, and J. William A. Lester, "Fixed-node quantum monte carlo for molecules," *The Journal of Chemical Physics*, vol. 77, no. 11, pp. 5593–5603, 1982. DOI: 10.1063/1.443766.

[82] D. M. Ceperley, "Path integrals in the theory of condensed helium," *Reviews of Modern Physics*, vol. 67, no. 2, pp. 279–355, 1995. DOI: 10.1103/RevModPhys.67.279.

[83] S. Zhang, J. Carlson, and J. E. Gubernatis, "Constrained path monte carlo method for fermion ground states," *Physical Review B*, vol. 55, no. 12, pp. 7464–7477, 1997. DOI: 10.1103/PhysRevB.55.7464.

[84] S. Zhang and H. Krakauer, "Quantum monte carlo method using phase-free random walks with slater determinants," *Physical Review Letters*, vol. 90, p. 136 401, 2003. DOI: 10.1103/PhysRevLett.90.136401.

[85] M. Motta and S. Zhang, "Ab initio computations of molecular systems by the auxiliary-field quantum monte carlo method," *WIREs Computational Molecular Science*, vol. 8, no. 5, e1364, 2018. DOI: 10.1002/wcms.1364.

[86] B. O. Roos, P. R. Taylor, and P. E. M. Siegbahn, "A complete active space SCF method (CASSCF) using a density matrix formulated super-CI approach," *Chemical Physics*, vol. 48, no. 2, pp. 157–173, 1980. DOI: 10.1016/0301-0104(80)80045-0.

[87] K. Raghavachari, G. W. Trucks, J. A. Pople, and M. Head-Gordon, "A fifth-order perturbation comparison of electron correlation theories," *Chemical Physics Letters*, vol. 157, no. 6, pp. 479–483, 1989. DOI: 10.1016/S0009-2614(89)87395-6.

[88] R. J. Bartlett and M. Musiał, “Coupled-cluster theory in quantum chemistry,” *Reviews of Modern Physics*, vol. 79, no. 1, pp. 291–352, 2007. DOI: 10.1103/RevModPhys.79.291.

[89] B. V. Noumerov, “A method of extrapolation of perturbations,” *Monthly Notices of the Royal Astronomical Society*, vol. 84, no. 8, pp. 592–602, 1924. DOI: 10.1093/mnras/84.8.592.

[90] B. R. Johnson, “New numerical methods applied to solving the one-dimensional eigenvalue problem,” *Journal of Chemical Physics*, vol. 67, no. 9, pp. 4086–4093, 1977. DOI: 10.1063/1.435384.

[91] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP: An open-source Python framework for the dynamics of open quantum systems,” *Computer Physics Communications*, vol. 183, no. 8, pp. 1760–1772, 2012. DOI: 10.1016/j.cpc.2012.02.021.

[92] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP 2: A Python framework for the dynamics of open quantum systems,” *Computer Physics Communications*, vol. 184, pp. 1234–1240, 2013. DOI: 10.1016/j.cpc.2012.11.014.

[93] Google Quantum AI and Collaborators, “Hartree-Fock on a superconducting qubit quantum computer,” *Science*, vol. 369, no. 6507, pp. 1084–1089, 2020. DOI: 10.1126/science.abb9811.

[94] S. H. Sureshbabu, M. Sajjan, S. Oh, and S. Kais, “Implementation of quantum machine learning for electronic structure calculations of periodic systems on quantum computing devices,” *Journal of Chemical Information and Modeling*, vol. 61, no. 6, pp. 2667–2674, 2021. DOI: 10.1021/acs.jcim.1c00294.

[95] R. Babbush et al., “Low-depth quantum simulation of materials,” *Physical Review X*, vol. 8, p. 011 044, 2018. DOI: 10.1103/PhysRevX.8.011044.

[96] A. Chiesa et al., “Quantum hardware simulating four-dimensional magnetic phenomena,” *Nature Physics*, vol. 15, pp. 455–459, 2019. DOI: 10.1038/s41567-019-0437-4.

[97] B. Bauer et al., “Hybrid quantum-classical approach to correlated materials,” *Physical Review X*, vol. 6, p. 031 045, 2016. DOI: 10.1103/PhysRevX.6.031045.

[98] J. Yepez, “Quantum lattice-gas model for the Burgers equation,” *Journal of Statistical Physics*, vol. 107, pp. 203–224, 2002. DOI: 10.1023/A:1014588526554.

[99] S. Succi, “Lattice Boltzmann 2038,” *EPL (Europhysics Letters)*, vol. 109, p. 50 001, 2015. DOI: 10.1209/0295-5075/109/50001.

[100] J. Yepez, “Quantum lattice gas dynamics,” *Physical Review E*, vol. 63, p. 046 702, 2001. DOI: 10.1103/PhysRevE.63.046702.

[101] F. Gaitan, “Quantum algorithm for the Navier-Stokes equations,” *Physical Review A*, vol. 85, p. 032 311, 2012. DOI: 10.1103/PhysRevA.85.032311.

[102] T. Krüger, H. Kusumaatmaja, A. Kuzmin, O. Shardt, G. Silva, and E. M. Viggen, *The Lattice Boltzmann Method: Principles and Practice* (Graduate Texts in Physics). Cham, Switzerland: Springer, 2017. DOI: 10.1007/978-3-319-44649-3.

[103] Y. Alexeev et al., “Quantum computer systems for scientific discovery,” *PRX Quantum*, vol. 2, p. 017 001, 2021. DOI: 10.1103/PRXQuantum.2.017001.

[104] N. M. Linke et al., “Experimental comparison of two quantum computing architectures,” *Proceedings of the National Academy of Sciences*, vol. 114, pp. 3305–3310, 2017. DOI: 10.1073/pnas.1618020114.

[105] M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*, 2022. DOI: 10.48550/arXiv.2211.07629. arXiv: 2211.07629 [quant-ph]. [Online]. Available: https://arxiv.org/abs/2211.07629.

[106] C. Ryan-Anderson et al., "Realization of real-time fault-tolerant quantum error correction," *Physical Review X*, vol. 11, p. 041 058, 2021. DOI: 10.1103/PhysRevX.11.041058.

[107] J. M. Pino et al., "Demonstration of the trapped-ion quantum CCD computer architecture," *Nature*, vol. 592, pp. 209–213, 2021. DOI: 10.1038/s41586-021-03318-4.

[108] D. Bluvstein et al., "Logical quantum processor based on reconfigurable atom arrays," *Nature*, vol. 626, pp. 58–65, 2024. DOI: 10.1038/s41586-023-06927-3.

[109] D. Bluvstein et al., "A quantum processor based on coherent transport of entangled atom arrays," *Nature*, vol. 604, pp. 451–456, 2022. DOI: 10.1038/s41586-022-04592-6.

[110] X. Xue et al., "Quantum logic with spin qubits crossing the surface code threshold," *Nature*, vol. 601, pp. 343–347, 2022. DOI: 10.1038/s41586-021-04273-w.

[111] V. V. Sivak et al., "Real-time quantum error correction beyond break-even," *Nature*, vol. 616, pp. 50–55, 2023. DOI: 10.1038/s41586-023-05782-6.

[112] M. H. Abobeih et al., "Fault-tolerant operation of a logical qubit in a diamond quantum processor," *Nature*, vol. 606, pp. 884–889, 2022. DOI: 10.1038/s41586-022-04819-6.

[113] R. A. Howard, "Information value theory," *IEEE Transactions on Systems Science and Cybernetics*, vol. 2, no. 1, pp. 22–26, 1966. DOI: 10.1109/TSSC.1966.300074.

[114] A. J. Daley, I. Bloch, C. Kokail, S. Flannigan, N. Pearson, M. Troyer, and P. Zoller, "Practical quantum advantage in quantum simulation," *Nature*, vol. 607, pp. 667–676, 2022. DOI: 10.1038/s41586-022-04940-6.

[115] S. McArdle, S. Endo, A. Aspuru-Guzik, S. C. Benjamin, and X. Yuan, "Quantum computational chemistry," *Reviews of Modern Physics*, vol. 92, p. 015 003, 2020. DOI: 10.1103/RevModPhys.92.015003.

[116] T. E. Baker and D. Poulin, "Density functionals and Kohn–Sham potentials with minimal wavefunction preparations on a quantum computer," *Physical Review Research*, vol. 2, p. 043 238, 2020. DOI: 10.1103/PhysRevResearch.2.043238.

[117] Y. Cao et al., "Quantum chemistry in the age of quantum computing," *Chemical Reviews*, vol. 119, no. 19, pp. 10 856–10 915, 2019. DOI: 10.1021/acs.chemrev.8b00803.

[118] V. von Burg et al., "Quantum computing enhanced computational catalysis," *Physical Review Research*, vol. 3, p. 033 055, 2021. DOI: 10.1103/PhysRevResearch.3.033055.

[119] S. Jahangiri, J. M. Arrazola, and A. Delgado, "Quantum algorithm for simulating single-molecule electron transport," *The Journal of Physical Chemistry Letters*, vol. 12, no. 4, pp. 1256–1261, 2021. DOI: 10.1021/acs.jpclett.0c03724.

[120] A. D. King et al., "Quantum critical dynamics in a 5,000-qubit programmable spin glass," *Nature*, vol. 617, pp. 61–66, 2023. DOI: 10.1038/s41586-023-05867-2.

[121] E. Altman et al., "Quantum simulators: Architectures and opportunities," *PRX Quantum*, vol. 2, p. 017 003, 2021. DOI: 10.1103/PRXQuantum.2.017003.

[122] L. Postler et al., "Demonstration of fault-tolerant universal quantum gate operations," *Nature*, vol. 605, pp. 675–680, 2022. DOI: 10.1038/s41586-022-04721-1.

[123] S. G. J. Philips et al., "Universal control of a six-qubit quantum processor in silicon," *Nature*, vol. 609, pp. 919–924, 2022. DOI: 10.1038/s41586-022-05117-x.

[124] L. M. K. Vandersypen et al., "Experimental realization of Shor's quantum factoring algorithm using nuclear magnetic resonance," *Nature*, vol. 414, pp. 883–887, 2001. DOI: 10.1038/414883a.

[125] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. D. Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics*, vol. 80, no. 3, pp. 1083–1159, 2008. DOI: 10.1103/RevModPhys.80.1083.

License